NeuroTranslation: A New Paradigm In Brain-Body Mapping.

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Abstract

Both history-reading candidates lowered average error. Neither passed. In the one-macaque LINK comparison, a better average could not satisfy the original conjunction: no evaluated output may be harmed, and neural information must be positive. All five separate benefit claims fail too. LINK compares two normalized finger-group positions across four training and two development-validation sessions, using weighted-sum and history-reading controls and three fixed random starting conditions. Its 18 runs and 720 saved epochs admit neither model. These development results do not independently verify the shared optimizer or establish performance on the reserved confirmation recordings.

That failure gives NeuroTranslation's question a concrete test: can a model predict a specified body target from identified neural observations, and what would that prediction establish? NeuroTranslator addresses it with bounded, recording-specific models, replay and body interfaces. The broader NeuroTranslation Network remains proposed. LINK and the earlier reaching, sensory, attempted-finger and simulated-hand studies address different links between a recording and an outcome; each link needs its own test.

A simulated hand may execute an issued command even when the neural decision behind it is wrong. A goal-driven hand may complete a phrase without recorded brain activity driving that policy. The proposed paired experiment would compare brain and muscle observations with independently measured movement, after resolving physical calibration, clocks, contact identities and participant registration. Anatomical and functional comparisons, final-checkpoint and update verification, and complete GUI and offline checks remain unfinished. Predicting movement, delivering assistance and retaining a biological skill require different evidence. Biological teaching and universal skill transfer remain unestablished.

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1 Neural Signals and Physical Skills

A hand can follow the wrong command perfectly. That difference between an issued command and an intended action is where NeuroTranslation begins: relate an identified neural recording to a specified body quantity, then check what happened. Predicting movement does not by itself identify the mechanism that produced it, and executing a command does not establish learning. The destination is faster physical learning that remains after teaching stops. The present evidence is bounded recording-specific software, separately calibrated simulations and a completed neural-decoding comparison whose improvement requirements failed. Decoding here means predicting a specified target from recorded neural input.

The LINK archive comparison makes the problem concrete. Two history-reading models lower average development error, but both fail the requirements that prohibit a harmed recording-day/finger-group and require positive neural information. All five separately defined benefit claims fail too. Source and scalar-calculation checks support that negative report; they do not qualify either model for use. The historical reaching, muscle-signal, attempted-finger and simulated-keyboard studies answer other questions. They are separate accounts, not replications that can repair LINK's failed comparisons.

NeuroTranslation: A New Paradigm In Brain-Body Mapping. names the proposed approach, not an established priority claim or a new biological mechanism. Its models and simulators are not newly invented by being used here. *NeuroTranslation* is the proposed field; *NeuroTranslator* is the implemented software joining particular recordings, prediction models, playback and computer-body checks. The *NeuroTranslation Network* is the proposed architecture for connecting more components while keeping each value's origin and meaning clear. *Brain2Body*, with the numeral meaning "to", is the wider programme. Its companion article introduces the biological-interface proposal independently; neither manuscript uses a presentation or demonstration as scientific authority.[1]

Each study keeps its historical identity and terminology. The definition supplement identifies this edition's shared registry version and fingerprint; a changed public label cannot retrospectively change the registry an earlier study used. The following comparisons ask what each recording can support, which outcome checks it, and where the connection fails.

2 Measure Brain–Body Relationships

The programme's starting point is that neural structure constrains function without determining it. A connection map describes possible paths. It does not supply every interaction strength, cell state, delay or learning rule. FlyWire's adult female fly reconstruction is anatomical evidence with stated coverage and reconstruction limits, not a complete behaving brain and body.[2]

An electrode observes a system through an instrument. Scalp brain, spinal surface, peripheral nerve and muscle electrical recordings have different source meanings: EEG, ESG, ENG and EMG respectively. EMG records muscle activity, not peripheral nerve activity directly. Functional magnetic-resonance imaging, or fMRI, observes an indirect

blood-oxygen-related signal. In the classic simultaneous-measurement study, its relation to local electrical activity depended on the measurement and site. It is not a photograph of thought.[3]

Name the source and target of every mapping, along with the subject, task, recording method, time support, units and tested domain. An electrode's contact identifier belongs to a declared source inventory; it does not establish that the same neurons were isolated on two dates. A neural feature is a specified numerical summary of a recorded signal.

Physical size, organizational level, computational detail and statistical averaging answer different questions. The 192 LINK input features are two measurements from 96 physical contacts, not 192 verified cells or 192 separate anatomical sites. Averaging four scoring cells equally defines a scoring rule, not a physical scale.

Perception is subjective. Movement is independently measurable through specified positions, trajectories, timing, contacts and forces, each with calibration and measurement limits. Sensory input guides action; action changes the body and environment; feedback informs the next action. Feedback-control theory provides a framework for testing that loop under uncertainty.[4] The present offline neural comparisons and simulated feedback loops remain distinct from a measured living loop. A biological causal claim needs appropriate intervention and outcome evidence in addition to predictive association.

Context also limits language-like reconstruction. The published semantic reconstruction depended on the participant and cooperation; it did not establish unique universal access to subjective content.[5]

The quantum-measurement analogy concerns observation and interaction. Quantum backaction means that measurement changes the state of the quantum system being measured. Quarks are subatomic particles. Neither backaction nor quark-scale measurement is offered here as an explanation of neural decoding limits at much larger physical scales.[6] This programme claims no one-to-one decoding of private subjective experience.

Accuracy means agreement with an independent reference; precision means agreement across repeated measurements under specified conditions. A repeatable wrong output can be precise and inaccurate. A reference is not assumed error-free: its acquisition, sampling, calibration and uncertainty remain part of the test.[7] A correct guess is not a validated model. A disclosed approximation may earn useful evidence in its tested domain. Invented biological content presented as faithful reproduction is hallucination. "Biological virtualization" would require a digital model independently shown to reproduce specified behaviour and causal dynamics within stated error bounds. Neither a realistic surface nor a high simulator reward establishes that claim.

3 Recording, Prediction and Skill Transfer

A map is useful only if its inputs can be observed, its outputs can be checked, and its meaning survives the conditions under which it will be used. We therefore ask three separate questions. Can we observe useful signals? Can we connect them to a particular body and a measured task? Does the connection remain useful as recording condi-

tions, bodies and tasks change? Each question requires its own measurement. A clear drawing or a stable replay file cannot answer all three.

The software reported in the original evidence snapshot addresses selected parts of those questions. It uses available macaque and human recordings, a human arm simulation and a separate macaque arm reference. It does not establish a lasting implanted interface or teach a living nervous system a skill.

A future experiment should begin with ordinary behavior and a defined baseline. If a human-related biological feature or a tissue connection is studied, the feature and comparison must be specified rather than treating a broad label as an explanation. Experienced partners, computational tutors and disease-specific conditions are possible future comparison families. They are proposals, not outcomes of this computational work.

Task difficulty and assistance need controls. Simple straight routes and routes with changing headings should be compared separately. Useful competitors include movement-only predictors, signals shuffled in time, matched assistance and sensing delay, unfamiliar sessions and animals, and performance after tutor instructions are withdrawn. A convincing animation cannot replace those comparisons.

If a learner needs an assistive bridge to express movement, that bridge should remain standardized while tutor instructions are removed. Switching off the whole bridge may prevent movement regardless of what the learner retained. Assistance, retained learning and tissue repair therefore need separate outcome measures. A signal that remains useful in a saved file does not demonstrate that a living recording interface remains useful over time.

For a proposed biological tutor, compare an experienced partner, a novice partner, individual practice and a matched no-link condition, then test the pupil alone. Replay comparisons should distinguish correctly aligned signals from stale or unmatched signals and an inactive comparison. Unfamiliar tasks and bodies must stay outside fitting. A new-looking maze or an equal target score does not by itself establish equal difficulty or retained learning.

Body observation also needs independent checks. For the proposed hand–arm task, synchronize the left and right tripod views and compare what each reveals or misses. A separate perpendicular depth-sensing view could be compared for coverage. This software did not acquire or calibrate either arrangement. Transparent layers and cutaways explain a model; they do not turn hidden nerve activity into a measurement.

4 Recording-Specific Models

The earlier computational studies explain how the recording-specific software works and where it fails. The current LINK cohort and its completed negative comparison follow. Each older study keeps its original tested scope; support already inspected cannot become fresh confirmation.

The earlier software snapshot had seven executable translation routes. Each route belongs to an identified recording and a particular signal quantity. A packet carries recent values, their recording identity, channel order, quantity, units, body frame where relevant, and optional sample times. The translator returns a named estimate with an

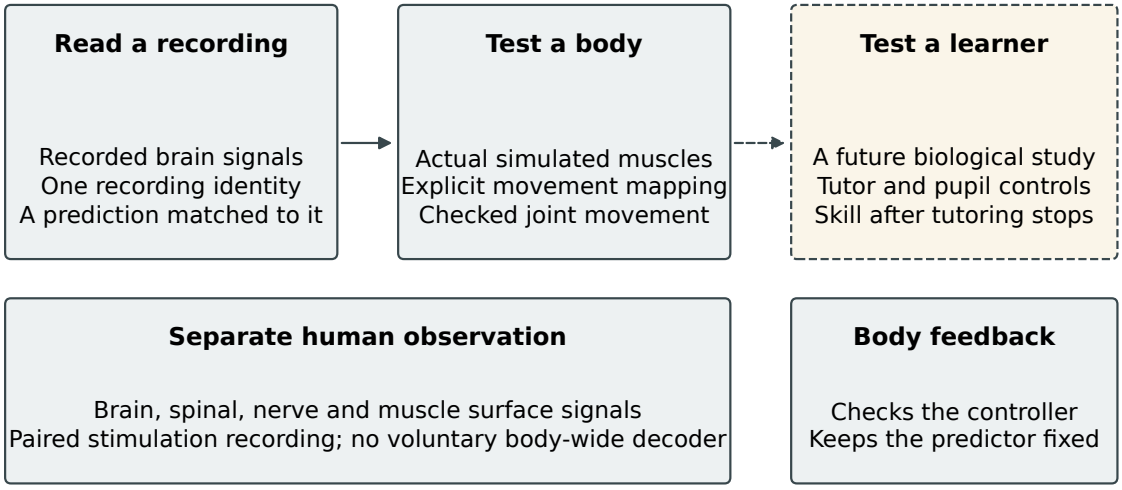


Figure 1: Historical diagram: the implemented chain and the proposed next stage. Independent recordings keep separate identities and clocks. Body feedback checks simulated movement; it does not reveal hidden nerve activity.

explicit time and source. It never silently inserts missing spinal or peripheral motor signals.

One route estimates fingertip velocity from a macaque movement-brain recording. A second estimates 16 normalized muscle-activity channels from a different macaque recording. A human route classifies attempted finger identity from movement and parietal brain areas. Three touch-brain routes estimate earlier mechanics or expected brain activity from mechanics or joint history. The seventh estimates velocity on a genuinely different macaque day after separate calibration. These routes share a software interface. They are not one simultaneously observed brain, spinal cord, nerves and body.

The seven routes give the software a common way to handle different recordings. Their separation is essential: combining the interface cannot supply simultaneous observations that the recordings lack. This recording-specific implementation, defined in “Neural Signals and Physical Skills”, establishes neither a universal trained decoder nor the proposed larger network or biological teaching. Older technical names remain glossary aliases.

The scale distinctions in “Measure Brain–Body Relationships” matter here. A small prediction model does not imply a small measured biological structure. Physical size, organizational level, computational detail and statistical averaging remain different axes; averaging output groups cannot supply an anatomical measurement.

5 The current LINK development cohort

What does recent neural history add when every model must answer the same position question? The LINK study uses DANDI 001201, version 0.251023.2336, Monkey N in the centre-out finger task: one rhesus macaque, six recording sessions, 375 trials per session.[8] Four dates are training data (January 27 and 30, February 4 and 5, 2020); two later dates are development validation (February 6 and 11). Two further February dates and two June 2023 drift recordings remain reserved and closed. These are distinct future confirmation supports; they are not part of the reported development results.

Ninety-six physical contacts in hand primary motor cortex provide 96 threshold-crossing features and 96 spiking-band-power features. Threshold crossing counts detected events crossing a chosen threshold; spiking-band power summarizes signal power in the source's selected high-frequency band. The counts use $\log(1 + x)$, a transformation that compresses large counts while retaining zero. The archived power features keep their supplied scale. Absolute calibration is unverified, and an archive voltage label conflicts with the publisher's threshold-count description. File labels do not resolve that conflict.

The outputs are two archived normalized sensor positions: index finger and the combined middle, ring and small-finger group. We call the latter the combined finger group rather than implying three independently measured trajectories. These targets are not radians, newtons, individual joint angles, muscle force, EMG, peripheral nerve activity or spinal activity. Conversion to a body model requires separate physical calibration and an explicitly tested interface.

That comparison starts before fitting a predictor: even the input scale must come from training alone. The scaler's input means and population standard deviations were fitted to 107,605 unique trial-owned training frames. Population standard deviation measures spread around the mean using the count of all fitting values as the divisor. Validation supplied no scaler statistics.

Every candidate uses the same eleven-frame history inside one trial, with the following neural frame added where the auxiliary target requires it. The position label belongs to the final input timestamp. This gives 91,105 training endpoints and 45,757 development-validation endpoints. The eleven timestamps span about 199–201 milliseconds. All six series in each file share the recorded clock, but unresolved bin anchoring and filtering prevent that span from being treated as physiological input support or verified online delay.

A saved independent source audit reconstructed the six sources and the prepared packet values. Its scaler difference was at most 4.54×10^{-13} within the frozen criterion. That evidence authenticates the named preparation. It does not demonstrate accurate decoding, causal physiology or independence of the common learning engine. The separate human EEG/ESG/ENG/EMG replay is an evoked stimulation source on its own clock; it is not input driving this macaque study or the historical macaque-to-human-arm simulation.

Date	Role	Trials	Source frames	Eligible endpoints
2020-01-27	Train	375	27,649	23,524
2020-01-30	Train	375	26,944	22,819
2020-02-04	Train	375	26,441	22,316
2020-02-05	Train	375	26,571	22,446
2020-02-06	Development	375	26,957	22,832
2020-02-11	Development	375	27,050	22,925

Table 1: The current one-animal cohort. Each session has 4,125 frames that are not prediction endpoints but can supply declared history, auxiliary labels or training scaler statistics. Endpoint counts enumerate archive support; they are not independent subjects, independent windows or confidence intervals.

6 Recorded Signals and Missing Measurements

The earlier sources extend the range of targets, not the measurements available in LINK. The public reaching archive pairs macaque movement-brain signals with fingertip position. The original task release supplies a processed fingertip-velocity target; the independently acquired day supplies recorded positions from which we derive fingertip velocity. The touch-brain study adds hand motion, handle force and turning force, with joint angles and muscle lengths obtained through tracking and body analysis. Those modeled lengths are not measured living muscle fibers. [9–12]

The human finger recording contains cues and attempted finger labels. It does not provide executed finger trajectories, sewing, cube manipulation or measured muscle forces. A separate human source contains simultaneous scalp brain, surface spinal, peripheral nerve and muscle signals during nerve stimulation. Its companion lacks muscle channels. These surface signals are useful paired observations, but they are not individual spinal-cell spike trains or evidence for voluntary whole-body control.[13–15]

The historical curated catalog had 37 resource records. A catalog entry does not mean that we acquired every file or trained on it. It includes functional recordings, movement data and anatomical connection maps with different licenses. An anatomical map describes connections; it cannot supply unrecorded activity over time. Sources remain identified by release, asset, bytes and checksum.

The historical second reaching day was chosen before this study inspected its raw data or decoder results: the closest eligible published Indy day within the initial download limit, excluding the existing day. Its 208,792,806-byte original file was checked against the publisher's fingerprint and the stronger local fingerprint. It contains 171 nonempty sorted signal features; the unsorted residual channel is excluded. Channel numbers do not establish that the same cells were recorded across days. Because the selected day precedes the original model-training day, this is a look-back comparison, not a future-day test.

We divide the recording into short completed intervals, called bins. Every completed 20-millisecond bin uses its last available 250-times-per-second fingertip sample. We subtract successive positions and divide by the elapsed time to obtain velocity. The actual position sample ends four milliseconds before the neural bin ends. That offset comes from sampling; it is not a measured nerve-conduction delay. Cursor and fingertip positions can differ by 11.9 millimeters and remain distinct.

7 Compare Against Simple Models

If the history helps, its advantage should survive comparison with a single sample and with a weighted sum of the same history. The frozen LINK comparison includes a constant training-target mean, a linear model using the last sample, a linear model using the eleven-sample history, and two matched gated recurrent-unit models. A linear model computes weighted sums; ridge fitting adds a penalty on large weights. Its five predeclared penalties are 0.01, 0.1, 1, 10 and 100, with an unpenalized constant term. Both linear models select 100 using the same two development dates.

A gated recurrent unit is an established sequence model that carries a numerical state from one input step to the next; learned gates control how that state is retained or replaced. This is memory in the software, not evidence of biological memory. Here it has 192 inputs, one memory layer with 128 values, a two-position output and an additional next-neural output. The two candidates retain the same full 169,090 parameters, although only 144,322 are active in the position-only objective. The auxiliary candidate adds 0.1 times next-neural mean squared error to position mean squared error. An objective is the numerical loss minimized by software fitting. It is not a learning rule shown to operate in the animal.

Both candidates use three fixed seeds, 20261007, 20261008 and 20261009, with matched initialization and sample order. Every run retains all 40 epochs, each a pass through the supported training windows. Batch size is 256, and the final partial batch is retained.

The established Adam update uses learning rate 0.001, averaging constants 0.9 and 0.999, numerical stabilizer 10^{-8} and no weight decay. Weight decay pushes model weights toward zero. Dropout randomly omits internal model units during training. A gradient measures how a small weight change affects the loss; gradient clipping limits the size of the update-direction vector. Target clipping replaces out-of-range targets with chosen bounds. None of these four controls is applied.

The reported study configuration specifies four mathematical worker threads and one framework scheduling thread.

Each seed selects its lowest equal-date/group validation error, taking the first epoch on an exact tie. The final predictions average all three selected seed models equally; a favourable seed is never substituted for that ensemble. The measured input condition and matched association and temporal-order controls are distinct. Association controls break the relevant pairing; order controls challenge use of input sequence order. Neither control enters the best-model selection contest. Two candidates, three seeds and three conditions give 18 runs and 720 saved epochs.

The primary error averages mean squared error for two groups on two dates: four equally weighted scoring cells. Mean squared error averages squared prediction-minus-reference differences, in squared source-normalized position units.

The whole-trial paired bootstrap repeats the error comparison 1,000 times. It resamples complete trials within each date and shares the draws across models and groups, preserving correlated frames inside a sampled trial. A percentile is a cut point below which a stated fraction of values lies. The resulting percentile interval is a descriptive 95-percent interval conditional on these two dates and the validation-selected models. It does not confirm performance on a new animal or recording.

A development improvement must meet *all* original conditions: lower aggregate error than the selected ridge; no higher error in any date/group; a negative upper interval bound for candidate-minus-ridge error; and positive, defined ordinary R^2 in every cell that also exceeds its matched association control. R^2 compares squared errors with a prediction of that evaluation cell's observed mean. The separately reported training-mean skill score uses a different constant reference and is not interchangeable with ordinary R^2. History, order-use and auxiliary-benefit claims retain their own conjunctions. An improved average cannot override a failed cell or rewrite those criteria.

8 Why the earlier implementation used small models

A larger model would change the cost of these comparisons; it would still have to pass them. The earlier environment receipt identified a Ryzen AI Max+395 processor and roughly 125 gibibytes of memory. Those training runs used four processor threads; graphics-processor training was unverified, and no paid cloud training was used. The current LINK study retains its specified CPU route. Its reported bounded GPU prefix took 4.565 seconds end to end, versus 2.427 seconds on CPU, and remains unpromoted. That prefix qualifies neither full training nor energy or cost savings. Compact models made training runs, controls and replay possible in that environment.

The first movement model has 232,388 adjustable numbers; the muscle-activity model has 195,216. Both first learn to predict the next signal bin from recent signal history, then learn the paired movement or muscle target. This is real training followed by fine-tuning. It is not a claim that a public pretrained brain foundation model was downloaded and adapted. The new-recording model trained from scratch has 256,045 adjustable numbers.

Large-scale neural models already exist. Some read spikes, others scalp electrical signals or magnetic-resonance measurements. Their inputs differ, and a larger model does not make those measurements interchangeable. [16–19] The question for a larger shared model is whether it beats well-controlled specialists on unfamiliar recordings and bodies. Our first shared movement and muscle candidate failed its validation comparison and remains rejected.

9 How the earlier predictions were evaluated

Validation chooses the model; testing checks the result of that choice. Training examples adjust its parameters. Validation examples choose between candidates and decide when training stops. Test examples describe performance after that choice. The continuous movement split is 70/15/15 percent over time, with a one-second gap at boundaries and complete signal histories. The muscle and touch-brain splits keep whole trials together. Our normalization is fitted only on training examples. Source-provided filtering or normalization may already have used later data, so our split does not certify the entire upstream recording pipeline as causal.

The prediction score R^2 compares a model with predicting the evaluation-set mean. One means perfect agreement; zero matches that reference; a negative value is worse.

Scores are averaged across named output channels or defined groups. They are neither percentages of learned skills nor official benchmark rankings.

Shuffled inputs and simple linear predictors supply additional comparisons. A linear predictor is a fitted weighted sum of input values. “Typical error” below squares the errors, averages those squares and takes the square root. In the two-joint tracking case, each frame combines both joint-angle differences before that summary is calculated. Radians measure angle; meters and millimeters measure length.

The touch-brain reconstruction compared zero, 40 and 80 milliseconds of alignment on common usable examples. Validation selected 40 milliseconds: brain activity reconstructs *earlier* mechanics. It is not a future movement command or proof of biological conduction time. Its separate reverse model predicts positive expected count means. Those means are statistical estimates, not new recorded spikes or stimulation instructions.

The different-day study fixes two supervision amounts before outcomes are known. The small amount uses 1,425 new training labels, or 28.5 seconds, *plus* 6,000 validation labels, or 120 seconds. All methods can use 570.14 seconds of new training-brain signal history for signal-only learning. The original model also carries 22,598 paired training and 4,723 validation examples from its earlier study. Thus “first 5 percent” is not five percent of all supervision or evidence of biological mastery in 28.5 seconds.

Four candidates receive the same new paired examples in each supervision amount: an input adapter into the old frozen model, the same-sized adapter into a randomly initialized frozen model, a newly trained model and a simple linear predictor. Frozen transfer is accepted only if its validation score exceeds every control by at least 0.01. The full amount is the interface default; both evaluated amounts were fixed before outcomes. All eight outcomes share the same 6,050 local test windows, now inspected. They are not eight independent prospective experiments.

10 LINK Results: Neither Model Passed

Across 18 runs and 720 saved epochs, lower mean error was not enough: neither candidate passes the frozen improvement conjunction. The postfit audit checks the named sources and scalar calculations in this negative report. Postfit means checking reported outputs after fitting; it does not independently certify final-checkpoint or full optimizer arithmetic.

For the position-only ensemble, the difference from ridge is –0.0002678134, with conditional 95-percent interval [–0.0014148839, 0.0009468841]. For the auxiliary ensemble, the difference is –0.0015373207, with interval [–0.0027586737, –0.0003470099]. Its aggregate interval excludes zero. Its no-harm and neural-information conditions still fail. The aggregate result and the failed conditions must be read together.

On February 11, position-only error exceeds ridge by 0.0052471986 for index and 0.0061405635 for the combined group. Auxiliary error exceeds ridge by 0.0043574767 for index. The position-only ordinary R^2 values on that day are approximately –0.099742 and –0.353799; auxiliary values are –0.071162 and –0.116390. The corresponding association-control values stay visible in the ledger. Exceeding a negative

Model	Development mean squared error	Outcome
Training-target mean	0.0317577871	Reference
Last-sample ridge	0.0314982382	Selected reference
Eleven-sample ridge	0.0383765124	No history advantage
Position recurrent ensemble	0.0312304247	FAIL
Position plus next-neural ensemble	0.0299609175	FAIL

Table 2: Saved reported scalar summaries from one macaque and two development dates. Lower aggregate error alone is insufficient. Both recurrent candidates fail the complete original development gate, and no model is promoted. These figures were transcribed from accepted source-bound reports, not recalculated.

control in one comparison cannot satisfy a requirement for positive information in every cell.

All five additional claims fail: history benefit for position-only, order-use for position-only, history benefit for auxiliary, order-use for auxiliary, and auxiliary-objective benefit over position-only. The original selected linear models also retain negative ordinary R^2 cells. Neither averaged performance nor one agreeable trajectory establishes reliable coverage of all evaluated support. These are development observations, not population estimates, cross-species translation or biological teaching.

The auxiliary ensemble exposes what the average concealed. Its aggregate interval excludes zero, yet its average improvement coexists with a harmed output and negative information scores. Under the unchanged requirements, that is a failed development comparison. The four reserved recordings remain closed, so this result supplies no confirmation beyond the development support.

The negative result leaves a separate implementation question open: were the full learning calculations independently verified? Both routes share Torch automatic differentiation, which computes changes in the objective, and Adam, which uses those changes to update fitted state. Source review and saved numerical checks of the report do not independently verify that common learning engine. An independent numerical route must still check the full learning calculation: complete NumPy calculations, restored random-generator states, reload equivalence, final-checkpoint forward calculations, derivatives, Adam states and complete update trajectories. NumPy is a numerical-array library. Those checks remain outside admitted scope; the final-checkpoint/update gate and current fitting remain held. No model or GPU route is promoted. The documentary details are retained in “Frozen LINK Development Methods”.

11 Failure Analysis

Where did the auxiliary ensemble lose its advantage? The saved residual diagnosis decomposes the auxiliary model’s second-day index excess mean squared error. About 78 percent is the difference in squared mean error and about 22 percent is the difference in centred residual variance. Its mean prediction is 0.058645 below the target in

the source-normalized sensor units. This describes the miss; it does not establish a cause. Trial composition, state-dependent offsets and recording/calibration changes remain competing explanations. Centring predictions on validation answers would fit those answers and invalidate the intended comparison.

A separate training-only context analysis uses the mean of ten preceding neural bins in each existing eleven-bin window. Across four dates, 1,500 trials and 91,105 supported endpoints, summed threshold-feature variance is partitioned into 71.56 percent within trial, 18.85 percent between trials within date and 9.59 percent between dates. The corresponding power-feature shares are 51.28, 13.55 and 35.17 percent. These ratios weight features by variance and are descriptive archive summaries, not uncertainty intervals or proof of biological state or neural drift. Equal date/trial/frame-within-trial weighting gives similar fractions. The existing recurrent model already sees these bins; variation alone cannot demonstrate the usefulness of an added branch.

Joining the exact source indices, times and trials leaves 89,605 endpoints with one supported prior sensor sample and 88,105 with two. Trial beginnings without labels remain missing; no labels are borrowed across trials or interpolated. Each training date supports all sixteen joint prior-position quartile cells. A quartile is one of four divisions of an ordered numerical distribution.

Correlated frames are not independent samples. Body-position strata are groups of body positions used as diagnostic labels. They can locate a neural-only predictor's failures; supplying those labels to the predictor would change its input condition, so they cannot enter it.

The first descriptive calculation failed two skewness checks at the unchanged absolute and relative 10^{-12} tolerance. Skewness measures asymmetry of a numerical distribution. The corrected second version changes only the per-date accumulation of that quantity; 768 values pass the named independent checks and all 119,848 other numerical leaves are unchanged. The rejected version remains in the evidence record. Acceptance concerns these descriptive calculations, not model improvement or the final-checkpoint gate.

The proposed next readout must obtain its context from permitted neural history. It would let past neural context change the decoder's output through 780 additional parameters. Its rank-four branch uses four learned context summaries to modulate that output. Rank describes the restricted numerical representation, not anatomical scale. The branch needs constant-context, association and order controls.

Selection must stay strictly inside training, using independently verified whole-trial inner folds: training trials divided into internal fitting and checking groups. Scaler statistics must come from unique inner-fit bins. Renormalizing rounded packets that were already normalized across training cannot establish that independence.

The branch is unexecuted, and current fitting is unadmitted. A prospective test must preserve initialization, random seeds, selection, uncertainty, complete interface/derivative/update evidence and the four closed reserved dates. Body feedback, reward-driven learning and broader extensions remain proposals, not LINK outcomes.

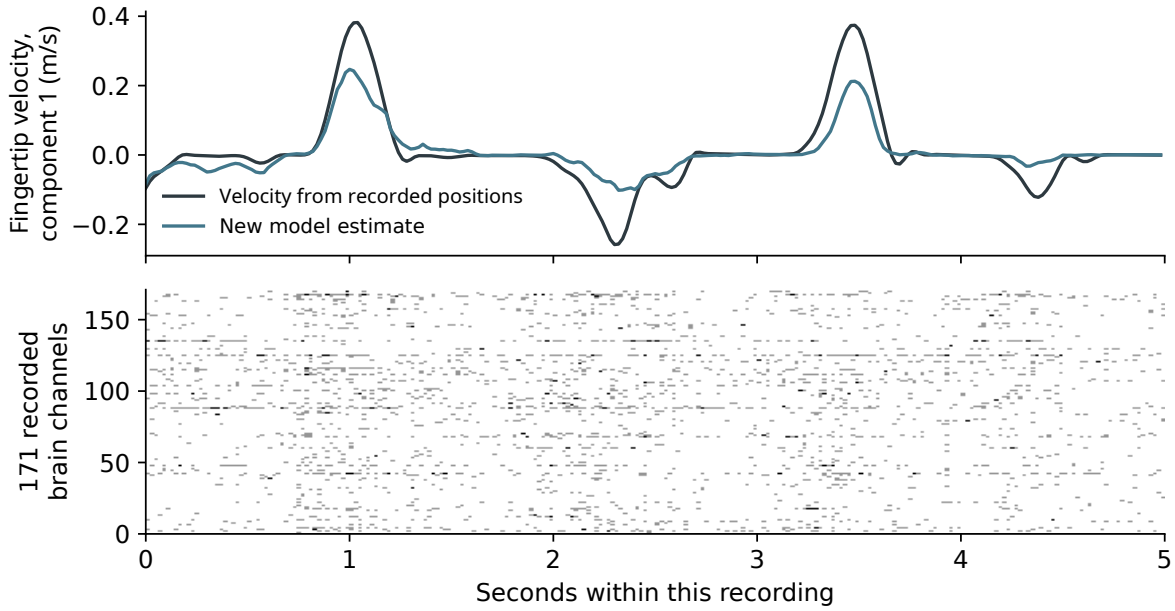


Figure 2: Historical study: actual different-day brain counts and a fingertip-velocity component predicted by the selected new model. The reference component is calculated from recorded fingertip positions. The rows belong to this recording; they are not matched cells across days.

12 Earlier prediction and transfer results

The earlier studies answer narrower prediction questions, with different targets and recordings. The original movement specialist scores 0.729 against 0.559 for its linear control. The corrected 16-channel muscle specialist scores 0.704 against 0.595; its weakest channel scores about 0.088. Source muscle activity has already been normalized. A stale source file label saying millivolts does not turn those dimensionless values into physical amplitude.

The human selected classifier correctly identifies 45 of 50 attempted-finger trials, or 90 percent; its recent-signal model scores 68 percent. That evaluation run was inspected in an earlier pilot and is development evidence. A separate uncalibrated day comparison scores 10 percent, matching ten-class chance. It does not establish executed finger control or cross-day skill teaching.

The touch-brain decoder scores 0.710 over five named mechanics groups. The reverse mechanics-to-brain model gains 0.177 bits per spike over a training-mean reference, and the seven-joint-only version gains 0.119. This information score compares how well expected counts explain recorded counts. It does not prove that the model can generate a complete biological spike process.

The old model carries useful information compared with a random frozen model, but training a new model wins at both supervision amounts. Both frozen-transfer candidates remain rejected. After local calibration, the selected full-amount new model scores 0.747. That result supports a recording-specific specialist, not zero-shot trans-

New examples	Method	Selection score	Local test score
First 5%	Use the old fixed model	0.393	0.425
First 5%	Use a random fixed model	0.312	0.331
First 5%	Train a new model	0.533	0.520
First 5%	Simple fitted formula	0.276	0.220
Full	Use the old fixed model	0.683	0.704
Full	Use a random fixed model	0.631	0.646
Full	Train a new model	0.724	0.747
Full	Simple fitted formula	0.550	0.548

Table 3: Historical study: all eight different-day outcomes. The prediction score is defined in the text. Both supervision amounts also require validation examples, recent brain-signal history and, for the transferred model, prior training. The local test examples are shared across methods.

fer or shared neuron identity. Different recordings and velocity preprocessing prevent treating it as an improvement over the original 0.729 case.

13 Separate human intention and simulated-hand studies

The hand studies let us separate two questions that a successful animation can blur: did the neural decision identify the attempted finger, and did the body execute the command it received? These October 4 studies are separate from LINK and from the earlier reaching tests. Their saved acceptance scopes do not change the later LINK failures.

13.1 Recorded brain signals select an attempted finger

The later study used public recordings from one participant with tetraplegia, a condition affecting movement in all four limbs. Its labels identify the finger the participant was cued to attempt. They are not measured finger movements. The three held recording dates were May 16, May 20 and May 23, 2016. Those dates contain posterior parietal cortex recordings, from a named brain area. They do not supply paired motor-cortex, spinal, peripheral-nerve or muscle recordings.

The model used eighteen completed 50-millisecond bins, covering 0.1 to 1.0 seconds after the source Go signal. A decision could first be released one second after Go. Fifty early labeled examples from each new date supplied local calibration: examples used to adapt the model to that date. The same participant supplied all three dates; this is not a study of 150 people or an uncalibrated new person. Recorded unit identities remained local to each date. Missing observations were kept unavailable rather than filled with invented measured zeros.

Seventeen fixed model candidates were compared. Selection used earlier validation dates before the later held examples were opened. The selected simple classifier correctly identified 116 of 150 later attempted-finger labels: 35, 44 and 37 of 50 on the three dates. A population model without those local labeled examples correctly

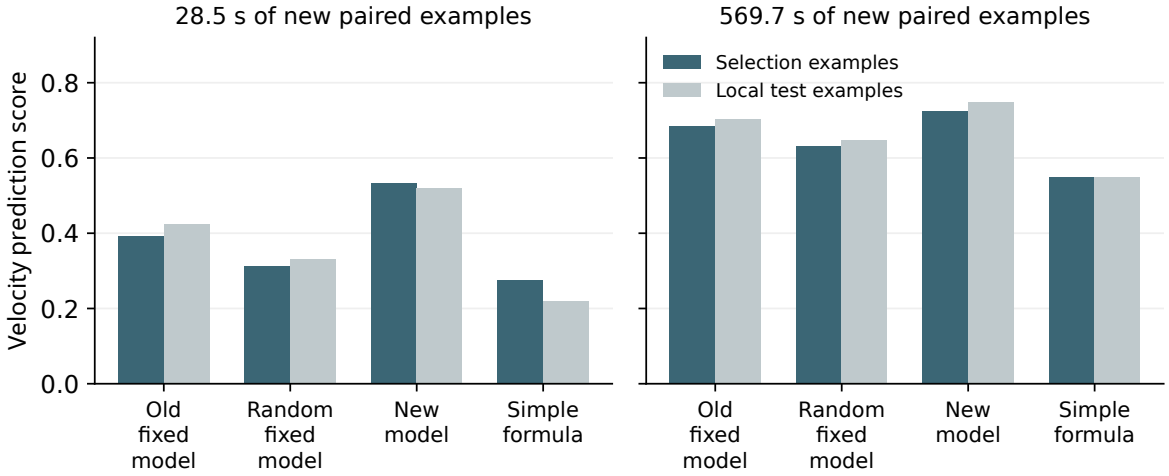


Figure 3: Historical study: transferred information beats its random control but does not beat the newly trained model. These are point estimates from one fixed experiment, not uncertainty intervals across repeated random seeds.

identified 11 of 150. These results do not show that a larger model is better, that the same cells were matched across dates, or that the participant learned a physical skill through replay.

13.2 Neural Decision Errors and Muscle Responses

The actual classifier decision, including errors and unsupported fingers, set the body command. The system did not choose the best right-finger class after seeing the answer. Four right fingers could issue goals to four physical keys; left fingers and the right thumb remained unsupported. Reference labels scored the learned replay and did not repair its predictions.

A saved learned policy supplied 39 muscle excitation commands to an openly available, generic human-hand simulation. The wrist, thumb and finger abduction were braced. The goals used fixed fingertip press targets and a 0.5 second pulse. Body positions and velocities were initialized once per recording, then advanced through physical integration without pose or velocity overwrites. Contacts, key travel and release were measured in the simulator.

All 54 issued supported goals were physically executed. However, only 41 of the 60 supported right-finger source intentions produced the correct contact, and 13 unintended presses remained. The two fractions answer different questions: 54/54 measures whether the body followed the issued command; 41/60 measures whether the recorded attempt led to the right physical result.

The fixed controls retained all source trials. Zero muscle excitation produced 0/60 source-intention contacts. A reference control given the true source labels produced 60/60. Deliberately reassociating neural decisions with different trials produced 4/60, despite executing all 54 issued supported goals. This reassociation is an offline control, not a new chronological neural recording.

The probabilities were computed offline and released only after their original source windows completed. The result is paced recorded replay, not live brain acquisition. The participant did not perform the simulated motion or play music. These results do not establish accurate hidden nerve activity, a learned piano skill, biological teaching or a universal brain–body map.

13.3 A separate simulated hand learned phrases and chords

That replay asks whether a recorded decision can drive a press. The separate phrase study asks whether a goal-driven muscle policy can complete a requested sequence. It trained on requested musical phrases in a generic four-key simulated hand with braced wrist, thumb and finger abduction (sideways spreading), driven by scripted smooth fingertip goals. Recorded neural clips did not drive this policy.

The 32 episodes combine eight phrase orders, two tempos and two starting poses; they are not 32 subjects. The policy received requested notes, timing and fixed smooth fingertip goals; it received no teacher actions during held evaluation. New phrase orders, rhythms, two-finger chords, tempos and starting poses were reserved before tuning.

The selected corrected policy completed all 32 held phrases: 704 intended strikes, 192 chords and timely releases, with no extra presses. The starting policy completed 0/32, the first new fit completed 20/32, the named reference controller completed 24/32 and zero excitation completed 0/32. Independent numerical replay verified the saved results. These are constrained simulation results. They do not show that a living hand learned music or that brain signals encoded those phrases.

The original evaluator exited with code 143 after the closed results and complete final output had been saved. The termination's cause remains unknown. The independent full replay verifier exited 0; that numerical check does not make the simulator a biological reference. Spring-key limits are soft numerical constraints, so travel can exceed the nominal 6 millimetres. Saved poses and travel were neither clamped nor fabricated for display.

Local training of that muscle-policy study took about 2 minutes 36 seconds. The later neural candidate training and validation took about 24.7 seconds. Those are measured software runtimes; acquiring recordings, supplying labeled examples and verifying physics require additional time. No completion time for biological teaching can be inferred from them.

14 The earlier muscle-driven arm experiments

The arm study tests another link: can decoded movement goals produce simulated muscle-driven motion? The working human arm uses 63 simulated muscles that produce force. Their activation changes over time rather than jumping instantly to the command, with 500 simulation steps per second. Two predicted fingertip-velocity components are explicitly mapped to shoulder and elbow movement goals; other independent joints are braced. The simulator calculates each next body state from muscle

forces and the current state. Joint poses are not overwritten, and extra artificial joint-turning forces do not replace the muscles. [20–22]

Feedback correction reads simulated body state. Recorded-position-derived movement is reserved for scoring. In the bounded 500-frame new-day replay, typical corrected tracking error is 0.0374 radians against 1.5803 for raw control. This comparison follows each branch's own command target, which can change under the correction's bounds. Against a common scoring reference mapped from recorded positions, the errors are 0.26302 and 1.49325 radians. These are numerical body-control results, not neural accuracy or measured macaque muscle behavior.

The corrected arm's largest movement beyond a declared joint limit is 0.000406 radians, below the declared 0.001-radian numerical tolerance. Raw replay violates that tolerance in 75 of 500 frames. Passing the tolerance means passing this solver check. It does not establish biological safety or physiological fidelity.

The separate macaque model has seven coordinates, 39 muscle paths and a rigid hand. Its retrieved source revision postdates the touch-brain paper, so it belongs to the same source family rather than being a certified copy of that study's specimen model. The original muscle-rate parameters contain zeros. A separately named copy changes 117 muscle settings to the software's official defaults. That documented assumption does not establish living-muscle behaviour.[23, 24]

Three original-model muscle controllers remain rejected. The force-feedback case has per-coordinate typical error 0.05505 radians versus 0.04512 for a constant muscle command. This error squares all joint-angle errors, averages them across coordinates and time, then takes the square root. An initial-speed protocol correction is retained, and this already inspected segment is not called a fresh test. The later controller plans short movements in isolated copies of the simulated body state. It loses 0.05971 to 0.03263 radians despite improvement on constructed simulation targets. Planning takes about 4.52 seconds for a 20-millisecond interval and is offline. Its cached replay can play at normal speed; that is not real-time planning.

15 Anatomical Templates and Participant Registration

Before a neural overlay can identify a participant's anatomy, its template must be checked and registered to that participant. The retained reconstruction audit reproduced 122,284 template vertices, 244,760 triangles and 81,924 atlas values, with three source MRI planes. Vertices are mesh points, triangles connect them into a surface, and the atlas assigns template regions. Both mid-cortical surfaces passed the specified closed-surface checks, including Euler characteristic +2, a connectivity summary appropriate to those tested surfaces.

In that audit, the optional region boundary had 432 open edges, 838 winding inconsistencies—inconsistent triangle orientations—and six invalid vertex fans, defects in neighbourhood connectivity. A second reader using the established nibabel imaging library was unavailable. Successful counts elsewhere do not repair these defects.

Those checks establish named template/source correspondence. The recorded subject still needs independent spatial registration; the checks neither localize individual neurons nor demonstrate physiological function. Registration means independently

establishing spatial correspondence with the actual subject. A group template may contextualize “hand motor cortex” while an electrode’s local identity remains local. A region colour cannot claim that an absent peripheral pathway was measured.

A template cannot fill those measurement gaps by becoming more lifelike. Every visual body and neural map still needs a lifelike, tangible three-dimensional representation refined against a *series* of identifiable real-world references. Those comparisons must cover recording site, time, pose, physical scale and anatomy, with independent review of both the parts and their assembly. Biological identification colours, cutaways and useful views can explain those relationships. Functional overlays must also preserve the displayed quantity, clock, availability and measured, predicted or simulated origin; otherwise the view changes what it appears to claim.

The required held-out trajectories, joint and force measurements and complete anatomical and functional validation remain missing. The four historical figures are evidence diagrams and plots. Retaining them preserves those earlier comparisons; it supplies neither newly graded 3D biological models nor a fresh visual review.

16 Earlier source and body defects remain evidence

The earlier source audit shows why a plausible score cannot authenticate its input. It found that 33 of 64 brain-event streams in the muscle study were not globally sorted. Counting them as though they were sorted corrupted 866,594 entries in the table of channel-by-time counts. These are count-table entries, not 866,594 living cells. Earlier muscle, shared-model and uncertainty results were withdrawn. The corrected historical streams were sorted, checked against independent whole-source counts and retrained with the unchanged seeds and split rules. Initial green tests had not proved source correctness.

The same demand for source checks reaches into the body geometry. A separate geometry audit found an implausible native triceps detour around the elbow. A rule based on the original attachment locations restricts three wrapping sections to the part of each muscle path between the upper-arm bone and the ulna, one of the two forearm bones. It removes 14 off-surface contacts in seven inspected anomalous frames, without deleting muscles, moving attachment points or fitting anatomy to those frames.

A global size factor is then fitted only on 128 training poses:

$$a = \frac{\sum_{t,i} L_{t,i}^{\text{model}} L_{t,i}^{\text{reference}}}{\sum_{t,i} (L_{t,i}^{\text{model}})^2}. \tag{1}$$

Here a is one common multiplier; t identifies a training pose; i identifies one of the 39 muscle paths; and L is its length in meters. The equation chooses the multiplier that minimizes squared length error on those training poses. The reference lengths were generated by the recording’s tracking/body analysis, not measured directly from living fibers.

The fitted factor is 1.2413815. Length error drops from about 27.75 to 0.552 millimeters on separate validation/test poses. A later, separately named model applies that



Figure 4: Historical study: one size factor fitted on training poses reduces length error against tracking/model-derived references. These reported values remain those of the earlier study. Agreement with those reference lengths does not identify specimen anatomy or establish living muscle behavior.

factor through official body-model scaling rather than multiplying displayed paths after export. Its native lengths reproduce the scalar predictions within 6.13 micrometers. The modeled masses are retained; the geometric changes to inertia—resistance to a change in turning motion— and muscle lengths remain assumptions. Dynamics are blocked on that scaled viewing candidate.

The 400 displayed observed and decoded poses are reused diagnostics. Their small observed-pose length error is not evidence for accurate decoded movement. The known wrap anomaly also occurs later than the evaluated lookahead-controller window, so it is not proved to explain that controller's rejection.

17 Earlier interface checks and their bounded scope

An interface can reject a malformed packet without knowing where its values physically came from. The interface checks recording identity, signal shape and quantity, channel order and, for mechanics, names, units and body frame. When times are supplied, it checks complete-bin cadence and the last sample time. Without them, nominal cadence is assumed and reported. Integer spike counts are checked before conversion to the model's lower-precision numbers. These rules validate the caller's declared contract; they do not independently authenticate the physical origin of supplied values.

The model outputs remain labeled as estimates. Expected counts cannot silently become measured spikes. Missing motor-spinal and peripheral-motor observations remain unavailable. Controller correction acts on simulated movement, not on recorded nerve counts. The saved different-day cache contains actual body positions, activations, geometry and muscle paths. It contains no saved joint speed array; no such array is fabricated from differences in position.

In the historical verification, all seven routes also ran through exported NumPy weights. NumPy is a numerical-array library; this route calculated outputs without importing the main training library. A separate historical live verifier checked all seven examples within numerical tolerance 0.000002, preserving exact response identity, units, times and support descriptions. The checked route rejected mismatched requests. These checks establish replay and interface consistency, not a universal neural language.

The earlier dashboard snapshot showed actual recorded counts, estimates, model choices, body state, activation and playback. Its natural-language task interface supported explicit reaching lessons and measured reach-to-grasp replay. The practice software example adjusted two simulated controller gains and displays the measured simulation errors and rewards. Sewing, cube solving and breakdancing lack supporting recordings and task physics here. They remain future composite tasks.

18 Current Viewer and Missing Measurements

Seek to another moment in a replay and the displayed quantity must still mean the same thing. In the retained cached-hand component check, the graphical user interface (GUI) loaded an identified saved replay with 336 frames at a 20-millisecond source interval. Play, pause, seek, stepping and restoration preserved that source. An uncached selection triggered no automatic replay or run request. Missing values appeared as dashes, and unavailable or failed sources cleared stale replay and geometry. These behaviours matter because an old hand pose beside a newly selected recording would describe a relationship the source never supplied.

The saved development example contains 22 of 22 strikes, 6 of 6 chords and 22 of 22 timely releases, with a 34-millisecond worst timing error and no extra strike. It is one constrained simulated example. Separate human neural clips did not drive that goal-driven hand policy. Its snapshot-level check does not establish current operation of every GUI route.

The original-control checks cover bounded source switching, gaps, replay controls and camera/restoration behaviour. Complete validation still requires the original source-time, geometry, materials, renderer-lifetime, functional-map, anatomical-reference, device and offline checks. No whole-GUI acceptance or performance improvement follows from the checked component.

Opening a package is a separate question from inspecting a replay. The online candidate needs its whole directory and assets; the original-source static offline export is a distinct artifact. Views that require the application’s service interface (API) are unavailable offline. The recorded review could not use the candidate’s browser route, so the candidate remains unpromoted and has no newer browser qualification. The exact four byte counts are retained in “Evidence, scope and edition status”. They describe file packaging, with no measured interaction, loading-latency, memory, crash or GPU benefit.

The same separation applies to the observations on screen. Human-arm feedback is simulated. The human EEG/ESG/ENG/EMG preview comes from a separate evoked experiment on its own clock. Anatomical site, time, modality and missing pathways must

remain visible so that it cannot appear to be simultaneous voluntary cortical-to-spinal-to-nerve control of the macaque-derived human-arm simulation.

19 Alternative Explanations

The distinction between a plausible display and a measured mechanism also changes how outside examples should be read. Useful approximations can succeed on some predictions and fail on others. Shiu and colleagues tested connectome-derived predictions through perturbations and behaviour. Specific successes coexist with missed Phantom and Usnea effects and incorrect G2N-1/Usnea water-silencing predictions. Missing basal activity and neuropeptide mechanisms—neuropeptides are small chains of amino acids used as signals—are possible limitations, not uniquely established causes. The evidence supports specified circuit predictions, not complete whole-fly fidelity.[25]

The official March 2026 Eon account describes hand-chosen brain–body mappings, heavy dependence on existing body controllers, missing state/plasticity and unvalidated internal dynamics. It demonstrates an engineered integration under its reported conditions, not independently recovered complete biological function.[26] Pinned fly-chess and DoomFly sources add explicit encoders, readouts or injected game-engine information; those software capabilities cannot be attributed to living fly biology.[27, 28] These are source inspections and author disclosures, not reproduced gameplay strength or proof of a lifetime-impossible biological ability.

Physics also needs independent checking. NeuroMechFly v2 reports slightly below-ground simulated leg tips and a 0.05-millimetre margin used to prevent false over-stretch detection. It does not establish dramatic tunnelling or an impossible FlyWire-brain capability.[29] Published first-hand simulation accounts describe numerical or contact errors supplying artificial energy to virtual creatures. A higher reward can likewise satisfy the wrong objective, as in the reported CoastRunners case of circling reward targets instead of finishing the race.[30, 31] These are different failure modes, not one invented universal error rate.

The same distinction matters when the output is a structure rather than movement. The original AlphaFold3 structural benchmark reports 62.9 percent success over 65 interface clusters from 71 complexes at DockQ greater than 0.23, with a 95-percent interval of 51.4–73.5 percent. DockQ measures reported structural-interface agreement.

Predictions were confidence-ranked from 1,000 seeds times five diffusion samples per target. Diffusion sampling generates a candidate structure by refining noise in stages; these are model samples, not biological replicates. The 37.1 percent complement belongs to that filtered benchmark and confidence-ranked sampling process. It is not a percentage of all proteins failing in real-world function. A structural interface score is not a functional assay.[32]

Crystallization limits some structure measurements, but cryo-electron microscopy and nuclear magnetic resonance provide other approaches. Crystallization alone supports no majority-unknown claim.[33] Pretraining on measured motion alone and declared unpaired transfer between homologous or mechanically analogous features with roughly similar biomechanics remain candidate training routes. Simulated or generated examples, including model-produced targets called pseudo-labels, may support

training without each being a simultaneously measured neural–muscle–movement triple. Those examples remain generated or derived; agreement with the model or musculoskeletal simulation that produced them is not independently measured biological validation. Synthetic probes can check specified computational behaviour, but neither probes nor generated targets can supply a missing measurement. Biological claims still need comparison with independent measurements in the claimed domain.

The positive alternative is measurement. A brain–spine interface evaluated cortical signals, spinal stimulation and actual standing/walking in one person with chronic tetraplegia. Its measured interventional outcome is valuable without becoming universal decoding or skill upload.[34] For this programme, predictive accuracy, mechanically plausible motion and causal biological reproduction each require their own references and tests.

20 Proposed Body and Tutoring Tests

A body-wide model needs a way to learn across bodies without pretending that unrelated macaque, human, fly and rodent recordings came from one nervous system. Shared functional roles—a limb, a joint, effort, contact and feedback—can support that proposed comparison, with separate verified body adapters. Homologous structures share evolutionary origin; mechanically analogous structures may serve comparable roles without that common origin. Either can motivate a transfer map when roughly similar biomechanics are specified. Anatomical similarity alone does not establish matching cells or interchangeable muscles.

Motion-only pretraining, unpaired representation learning and mechanical retargeting are candidate routes, alongside learning from paired neural and body observations. Scans can constrain geometry or test an anatomical assumption where they add information beyond the selected musculoskeletal model. For each transfer, state the source and target, joint coordinates, body frames, proportions, muscle roles, force and time scales, contact conditions, missing features and uncertainty. Measured, tracking-derived, model-estimated, retargeted, predicted and simulated quantities must keep those identities; transferred targets do not become simultaneous neural measurements.

Matched voluntary recordings, reliable clocks and body frames remain necessary for the claimed paired test across relevant levels, with missing measurements explicitly identified. Missing simultaneous observations limit that empirical claim; they do not forbid the proposed unpaired transfer routes. Compare proposed transfer with target-only training and naive or incorrect mapping under matched resources, preserving independent target evaluation and the original held-out restrictions. Virtual-rodent work offers a relevant example of combining an embodied model, movement and neural analysis, not a license to assume species transfer. [35]

Both recurrent candidates fail the original no-harm and positive-neural-information requirements. All five added benefit claims fail too. Lower average error admits neither predictor. NeuroTranslation remains a proposed field; NeuroTranslator remains recording-specific software.

The proposed loop should expose each transition: supported observations become decoded estimates, pass through body adapters and return as feedback. Simple base-

lines must face the same matched tests. Illustrative motor replay and a deliberately goal-driven simulation can serve that limited question when their goals, assistance and simulator information are disclosed. A positive neural-control claim requires a predictor that passes its unchanged admission criteria. The current failed candidates do not pass.

Check the reward, the simulator and the independently measured task outcomes separately. Physical calibration, synchronized recording chains, participant-specific registration and generalization remain unverified. A controller that improves inside a simulation earns a claim about that tested computation. Biological teaching, retention and repair need their own evidence. Those missing biological results do not prevent a clearly limited computational proposal.

A later correction-learning comparison must keep the baseline and its information resources fixed and test unfamiliar recordings where applicable, bodies, delays and contact conditions. Simulator rewards for task completion, tracking, effort and contact limits require verification against independently checked outcomes; that verification is a future gate, not a result established here.

The biological teaching claim would require a further matched experiment with unfamiliar tasks, appropriate controls and retained performance after tutor instructions are withdrawn. Assistance budget and any necessary bridge must remain controlled. Failure to beat a simple predictor, failure on a new recording, or failure to retain performance must remain genuine ways for the proposal to lose.

21 A grounded conditional learner with distinct output interfaces

Start with a movement we can measure well. For the proposed snap showcase, the arm begins extended with the palm down. The elbow and forearm swing and turn; the wrist and hand turn upward during the snap, then move down to rest. The model must account for that whole movement, including the preparation, contact and return. Finger pairing, organism-specific anatomy and timing still need a real-reference series; a human-looking animation cannot establish that a particular monkey can perform the same gesture.

The cup reach supplies a simpler precision comparison: relate recorded signals to measured movement before attempting the coordinated snap. Each task still needs its own anatomy, contact, timing and independent comparison.

That comparison connects three interacting groups: task-linked brain observations, named-muscle observations and a musculoskeletal simulation. A learner would relate the observations to the mechanics; independent movement and contact measurements would test its predictions. EEG records scalp electrical activity and surface EMG records muscle electrical activity. A separately recorded nerve or spinal signal needs its own declared channel. The proposed apparatus uses two arm sleeves, scalp EEG and synchronized, calibrated left and right tripod views. Vendor, channel, bandwidth and clock qualification remain open.

Intent to action is the target. By understanding and measuring the intermediary, we aim to make that mapping as close to one-to-one as possible within a declared task.

Specify what counts as the intended action, the participant and session, state and context, permitted history, tolerated error and independent reference. The test then asks whether observing the middle reduces ambiguity and improves prediction or qualified control. It does not assume a unique hidden intention or claim that the best possible mapping has already been achieved.

These observations define what the learner would have to explain: recorded brain and muscle activity, where the hand moved and when. Delivering sensory feedback and producing an action would require further tests of separate outputs; neither follows from a successful prediction. No paired trial, trained model, delivered sensory output or physical actuation is established for this proposal. Sources and permissions, physical and computational behaviour, clinical interfaces, reference comparisons and held-out tests still need independent assessment.

The retained LINK comparison answers a different question. Its 18 runs and 720 saved epochs are not the proposed paired EEG/EMG/movement experiment. Both model-admission gates and all five benefit claims failed. Recording, prediction, simulated movement, sensory delivery and physical action require different evidence. EEG and EMG recorders alone cannot write sensation. The prospective methods below state what the next experiment must measure.

21.1 Form, function and action need a measured middle

A structural map constrains what a system could do. A NeuroConnectome maps specified neural connections; its organism, coverage, resolution and source would have to be declared. The signals passing through those connections still have to be observed or inferred, with the difference made explicit. To predict the next action, we need the processes that turn structural constraints into an outcome. The proposed programme follows form to function to action, while intent to action names the task-level translation we want to improve.

DNA to RNA and RNA to protein put a process between a starting constraint and its downstream product. Regulatory context, RNA processing and cell state change what that process produces; protein structure alone does not determine the behaviour of a whole cell. AlphaGenome, RNA or cell-state models and AlphaFold illustrate these different modelling jobs. AlphaTranscriptome names a desired predictive capability here. Connecting the predictors would still leave system dynamics and the environment to account for, with experimental outcomes needed to test the combined prediction.

AlphaGenome regulatory prediction [36]; scGPT cell-state modelling [37]; AlphaFold 3 structure prediction [32]. These provide modelling examples, not evidence of an integrated cell programmer. Our account of AlphaGenome and scGPT is limited to their reported modelling roles; their complete methods have not been independently assessed here.

In the biological pathway, Brain2Spine and Spine2Body name proposed links to test; their composition is the Brain2Body aim. The spinal cord integrates descending and sensory signals and contributes local coordination, reflex regulation and adaptation. This is a different mechanism from RNA processing, but it serves the same explanatory purpose: account for the context-dependent process between an initial constraint and its outcome. Supported spinal observations would test that account. Where they are

absent, the internal states remain inferred or simulated. Physical spinal work remains read-only.

Spinal sensorimotor adaptation [38] is a constrained example, not evidence of unrestricted skill transfer or validation of our proposed pathway. Our comparison draws on the abstract and relevant author-manuscript passages; equivalence to the final published edition and the complete methods and statistics remain unverified here.

Thought2Action relates a declared intention to an independently measured action. Action2World follows the action to its consequence in the task environment; Thought2World names the proposed composition. Body2World drew attention to the consequence outside the body; Body2Action focuses on body mechanics and a specified task outcome, while Brain2Motion focuses on the movement pathway. These working labels identify related scopes for a test. They do not establish interchangeable or working controllers.

The engineering route connects neural activity to musculoskeletal simulation and then to a programmable model, toward NeuroTranslation. The proposed movement-focused learner is a large movement model. System 3 thinking names the proposed human–AI supervisory partnership around it. The supervisory model would compare task strategies and diagnose where neural or mechanical predictions miss independent measurements, then propose training or calibration comparisons for the signal mapping and virtual body. The task learner would perform the declared prediction or control; the supervisor would help choose what to test. Neither role replaces the biological spinal intermediary or a physical interface. Final held-out outcomes must remain outside rewards and model selection.

Shaw and Nave's SSRN preprint, *Thinking—Fast, Slow, and Artificial*, and its institutional discussion provide conceptual context for this proposed human–AI role. [39] The comparison is limited to that framing; the complete preprint and institutional article have not been reviewed here. Neither establishes a third anatomical brain system or validates this programme's proposed learner or supervisor. Expert footage may supply movement and tactical examples, with rights and camera uncertainty accounted for; it supplies no measured expert EEG, EMG, muscle force or stimulation plan.

Action changes the outcome. Its sensory consequences inform perception and the next intention. Independently measured movement would anchor the action part of that loop; it would not by itself decode a unique sensation or perception. Known stimuli, measured feedback and appropriate recordings or reports would have to test those further mappings. The return path must therefore be measured rather than drawn into existence. The full Interoceptomics aim includes internal-body sensing and connectivity; external touch and proprioceptive tasks contribute evidence but do not by themselves solve that wider problem. A prediction, assisted execution and retained skill without assistance answer different questions. Each wider claim must earn its own independent comparison.

21.2 Context connects. Grounding checks

Within a reach, context has a concrete job: relate brain activity to muscle activity in this task and at this time. Attention offers a candidate computation for those relationships, following Vaswani et al. (2017) [40] and Tsai et al. (2019) [41]. The model compares a

query with keys and uses the resulting weights to combine their values. Masks keep unavailable observations and future samples out. The calculation weights information; it does not establish that attention understands all context.

A pattern can fit inside the model and still describe the wrong movement. Grounding therefore means testing the declared prediction against qualified observations under the same conditions. Independently measured movement would check the outcome, while muscle activity would add an observation that constrains the brain model. Even a matching reach cannot identify a unique private experience. No acquired paired data or trained model is established for this proposed loop; those comparisons still have to be made.

21.3 Muscle activity tests the brain model

Record brain and named-muscle activity during the same reach. In the proposed paired trial, EEG would sample scalp electrical activity and surface EMG would sample activity associated with the named muscles. Paired training could use the muscle observation to constrain their relationship. Then withhold the muscle target and movement reference from the EEG-only predictor: can the permitted brain observations predict a specified muscle-activity observable or movement?

Pairing and calibration remain proposed. Shared cues and movement artefacts could give that predictor a shortcut, so the comparison needs controls for both. The muscle observations would cover only the named muscles, and EMG supplies neither afferent nerve activity nor direct force. Sensory claims still need known stimuli and measured feedback. Several neural states could remain compatible with even a well-matched pair; the pair could narrow a task-specific explanation without revealing the complete hidden brain. Checking the movement itself requires the next independent observation.

The proposed pairing still needs its own evidence. The candidate reference at https://www.nature.com/articles/sdata2018133 has not yet been checked for exact support or reuse rights.

21.4 Measured motion tests the virtual body

Measure the same hand and arm from the aligned left and right tripod views. The proposed motion-capture relay-composite would align the two point maps in a common clock and coordinate frame, then join them into one composite while retaining each point's origin. Before that joined map can check a prediction, its calibration, alignment and capture precision would have to be independently verified.

The movement map would constrain the external movement model: surface motion and pose in the task's physical frame. To explain that motion mechanically, the internal musculoskeletal model would describe specified bones, joints and soft tissues. Organism, developmental stage and task would determine the appropriate anatomical sources. X-ray projections could inform bone and joint relationships; appropriate MRI references could inform specified soft tissues. Registration, rights and source-specific limits would need independent checks. No scan set is selected for this proposed comparison. Anatomy still leaves activation, forces and neural function to test.

With a qualified composite, the musculoskeletal-simulation physics-engine's joints, muscles, contacts and physical limits would face a measured comparison. Fit the model on training data, then compare its predicted reach with held-out measured movement. A force claim would require an additional force measurement. The body's form limits possible motion; measured function would test the mechanics and show where they need refinement.

Seth et al. (2018) [23]; MyoSuite documentation [42]. The simulation tools supply methods for that comparison; the proposed capture and body models still need its independent validation.

A lifelike hand makes that comparison visible. It cannot establish biological fidelity. Capture alignment, precision, mechanics and biological fidelity have not been demonstrated by this proposal, and captured geometry cannot establish hidden neural state, internal forces or unique subjective experience. The measured reach would check a movement prediction within those limits.

21.5 Two pairs. One conditional learner

The two comparisons would meet in the proposed large movement model: EEG observations paired with muscle activity, and measured movement compared with predicted mechanics. Large movement model names the proposed family; deep neural network describes a possible implementation. Large refers to the intended scope, rather than a demonstrated parameter count. Encoders would receive only the observations permitted for the named test. Two branches could use the attention and multimodal computations of Vaswani et al. (2017) [40] and Tsai et al. (2019) [41] to summarize those relationships in a shared task representation; separate output parts would predict each direction's targets.

Signal features belong to branches, and branches belong to the learner. That hierarchy describes the computation, not two anatomical thinking systems. A common clock, task and physical frame would connect the branches; independent measurements would check their errors. Form constrains the prediction, and checked function would guide refinement. Several hidden states could still fit the same reach, so neither a shared representation nor conditional predictions would establish an exact inverse nervous system. The architecture has not been selected, implemented, trained or independently accepted. To choose it, we would have to compare it with simpler baselines on appropriately reserved observations.

21.6 Learn the movement, then test what transfers

A reach can provide a training target even when no EEG accompanies it. Earlier movement supplies the input; a later observation supplies the target. A separate proposed comparison would ask whether learning that relationship in one person helps forecast another person's grasp-and-lift movement. The WAY-EEG-GAL descriptor provides a relevant continuous-recording candidate, alongside its distinct windowed trials and derived lift summaries.[43] This movement-only question needs no AllLifts Run/trial join or invented cross-person neural pairing. The original condition-specific five-weight study remains a separate design.

The mapping has to explain what two bodies share. Homologous structures share evolutionary anatomical origin; mechanically analogous structures can serve comparable roles without that origin. A human digit aperture and a robot gripper opening could support a task-space comparison after separate mechanical qualification; this human-only proposal would not test that robotic extension. Joint degrees of freedom, axes and signs, proportions, muscle–tendon roles, loads, inertia, contact and time scales remain body-specific. Similar geometry cannot establish equivalent dynamics or neural coding. Retargeted motion remains derived; tracking, estimation, prediction and simulation keep their different evidential roles.

For the proposed two-person comparison, the representation contains index-, thumb- and object-sensor positions relative to the wrist, digit aperture and causal wrist velocity: thirteen features, with thirteen availability masks. Each participant needs qualified position units, a proper wrist-frame rotation, orientation convention and sensor-attachment calibration. A positive median wrist-to-index-sensor distance, expressed in qualified metres, supplies the characteristic length from assigned training support only; it is not a bone measurement. Multiplying normalized velocity by the same fixed 0.1-second reference interval makes all thirteen features dimensionless. Physical velocity errors invert that scaling with length divided by 0.1 seconds. Missing features receive no target loss; an unavailable observation is not a measured zero. Full finger joints, individual muscle forces and subject-specific inertia are not supplied by these sensor features.

The continuous target would be the same observed feature set 100 milliseconds beyond the latest actually available filtered input time. The proposal retains fifty frames at 100 Hz and a causal 51-tap filter before decimating qualified 500-Hz kinematics by five. Its 50-millisecond linear-phase delay must remain in the physical timing account; shifting information backward would change the prediction claim. Coefficients, passband, timestamp convention, gaps and boundary handling still require qualification. Only named kinematic columns may enter: the documented kinematics matrix also contains force-plate signals.

Identity and independence answer different questions. Trustworthy versioned participant-release attribution, an immutable selected archive entry and a qualified capture can identify the participant record and input object without requiring one particular internal ID-field spelling. They cannot establish biological identity or independent original acquisitions. Shared samples, trims, derivatives and overlapping recordings must form indivisible acquisition groups before filtering, windows, normalization or role assignment. Different names, checksums or series numbers do not prove separation. The complete public non-ST candidate roster and these relations still need establishing; all 294 rows of the prior AllLifts table remain excluded from training, evaluation and assigning trials or acquisition groups, and ST, competition storage and the reserved LINK recordings remain unavailable.

Qualified P1 groups would supply source pretraining; P2 would supply target training and development. With $G \geq 2$ independently established P2 acquisition groups, the fixed identifier-hash ranking assigns the first $\lfloor 3G/5 \rfloor$ to training and the remainder to development. Canonical original-parent identifiers, their serialization and the roster must be fixed before ranking or examining outcomes. This is a nominal 60/40 split, not chronological ordering or a promise of condition balance. If unresolved ancestry merges P2 into one group, the comparison stops; splitting its windows cannot manu-

facture an independent development group. No group or segment has been assigned by the proposed identity method.

Three matched arms would test the transfer. B0 pretrains on correctly mapped P2 training movement. B1 and B2 use identical P1 windows and order: B1 sets the index-role input to the thumb sensor, the thumb-role input to the object sensor and the object-role input to the index sensor, while B2 retains the correct correspondence. Both share the correct source calibration and scale before that corruption, keep the wrist anchor, and use the same correct target adapter and ordered target batches. Parent-balanced sampling must record unique windows and repetitions. The false mapping can be relearned or can alter feature availability; matched update counts do not match source diversity. These are limits on attributing any difference to anatomical correspondence, not reasons to replace the control after seeing its result.

The proposed learner is one GRU layer of hidden width 32, with 26 inputs and thirteen outputs. Each arm and fixed seed—17, 31 and 73—receives 200 pretraining and 100 target updates, batch size 16, identical per-seed initialization and the fixed last checkpoint. Adam uses learning rate 0.001, betas 0.9 and 0.999, epsilon 10^{-8} and zero weight decay, without clipping, scheduling or early stopping. Training minimizes mean squared residuals over observed target features; a batch with no observed targets is rejected. The resulting 2,700 updates are planned, not performed. No external checkpoint is selected; numerical updates, filtering, precision and actual workload limits remain to be checked independently.

For each development parent and seed, the proposed score takes the square root of summed squared errors divided by the count of observed target features, then averages parent scores without weighting by parent duration. A zero denominator rejects the comparison. Report feature counts, missingness, physical-unit errors, uncertainty and persistence at the same information endpoint, including every worse parent or feature. B2 would have to lower normalized error against both B0 and B1 in every fixed seed, with no applicable original criterion failing, to justify another investigation. This remains a two-person development comparison, not independent final confirmation, population transfer or biological teaching.

Broader motion collections could supply additional movement examples only after their exact assets, rights, task coverage and original-recording overlap are qualified. In AddBiomechanics, measured markers and ground reactions must remain distinct from fitted kinematics and estimated inverse-dynamics torques. AMASS requires checking the selected asset's research-use, non-commercial and redistribution restrictions. Copies across collections stay with their original acquisition. LaBraM's WAY source-family exposure is flagged, while exact checkpoint/participant/trial exposure remains unknown; reserving a recording later cannot erase earlier pretraining exposure. No such checkpoint is selected for this comparison.

Before adding X-ray/CT or MRI, inspect the anatomical sources already used by the selected musculoskeletal model. A scan is useful when it constrains a stated gap or supplies an independent check beyond that model. Its person or species, scale, registration, rights and uncertainty still matter. Another person's scan remains an external anatomical reference; a generic model does not implicitly measure this participant's geometry, activation or forces. No scan set is selected here.

Actual source eligibility, acquisition groups, units, frames, clock and calibration remain unresolved for this continuous comparison, so no new fit or successful transfer is re-

ported. A useful movement prior could later be tested with actual paired neural/body observations and the permitted input masks. Its transferred targets would still not become simultaneous measurements. The next result must identify where the mechanical forecast improves, where it worsens and which measurement or assumption accounts for the difference; neural equivalence, causal physiology, sensory delivery and retained skill would require further experiments.

21.7 Pretrain. Pair. Verify. Refine

Before a paired example can teach the mapping, we have to know whether we may use it and what it measures. Data rights, calibration and the held-out subjects, sessions and tasks determine which observations may enter; resolve those conditions before processing data or choosing a model. Open-source code supplies no blanket permission for datasets, checkpoints or clinical interfaces. Qualified data could first train a representation for each recording type, then synchronized paired observations could train the declared mapping from permitted inputs. LaBraM is one candidate EEG encoder; it does not supply a complete brain-body writer.

Paired prediction would give later refinement a baseline to beat. Before optional reinforcement learning, post-training or task-specific fine-tuning, the physical predictions and reward need checking against independent outcomes. Offline reinforcement learning needs qualified coverage of actions, transitions and rewards: labels alone cannot tell us what an untried action would do. Backpropagation can update differentiable neural components, but it cannot automatically pass through every physics engine; universal differentiability is not established. Simpler baselines and frozen final tests remain through every phase.

Jiang, Zhao and Lu (2024) [44]; Kumar et al. (2020) [45]; Seth et al. (2018) [23]; MyoSuite documentation [42]. These sources offer candidate methods for the sequence; validation of the proposed paired system still requires the experiment below.

No data acquisition, training, fine-tuning, RL rollout or accepted checkpoint is established for this proposed loop. The next experiment must test what the prediction and reward represent against independent outcomes. A reward that the computer can calculate supplies no biological proof on its own.

21.8 Respond quickly. Improve carefully

A response should not become a new experiment every time it is issued. The proposed response path would use a frozen accepted model to predict from permitted past and present observations and prepare encoded outputs for a qualified interface. A separate path would inspect errors, calibrate and propose a candidate update. Independent evaluation and a restorable checkpoint would precede promotion. Updating an estimate of the current state would therefore remain distinct from changing learned weights; examining errors would not silently change the running model.

This fast-and-slow comparison is a Kahneman-like analogy for the proposed division of work. It does not identify two biological systems or assign inherent thinking speeds to EEG and spatial data. Real-time performance would have to be measured from acquisition through filtering, alignment, inference, output, feedback and rendering, including

delay tails and jitter. No accepted model, real-time demonstration or running system that adapts under these rules is established here. The loop has no measured end-to-end latency or accuracy guarantee.

21.9 Write sensation. Enact action

The prediction ends at the computer. A separate interface would have to turn it into sensation or action. Input and output describe direction relative to a computer; afferent and efferent describe physiological direction relative to the central nervous system. A computer output could feed a sensory transducer, or a controller and actuator for a virtual, robotic or later qualified biological body.

A display or haptic interface supplies sensory feedback, which differs from direct neural writing. Direct cortical writing could have a sensory role without being a peripheral afferent signal, just as a robot command is not an anatomical efferent nerve. EEG and EMG recorders do not write, and predicting an EEG-like pattern does not deliver sensation. Each proposed output needs an independently qualified interface, permission for its use and a measurement of what it delivers. No such physical interface, stimulation, acquisition or actuation is established by this proposal.

Comparing a recorder with a sensory or action device requires evidence about what each measures or delivers. The clinical source could not be read, so it supplies no methods evidence here. Implanted precedents cannot establish that EEG or EMG recorders are equivalent to devices that deliver sensation or produce an action. Those physical additions remain unpriced and outside the acquisition and virtual-modelling scope.

21.10 Close the loop with evidence

Return to the cup and test each prediction against the observation it was asked to explain. The brain-to-body prediction would face withheld observations; the body-to-recorded-neural estimate would face its own recorded target. A sensory or action interface would need a further test of what it delivered, under which conditions, with what error and delay. Two models can agree and still be wrong; a high reward can have the same problem. Neither agreement nor a successful prediction would, by itself, identify a unique inverse brain or establish someone's private experience.

The programme's distinction remains that approximation gets us towards plausibility and virtualization gets us to precision. Independent measurements, repeatability, causal checks and stated error bounds must earn that precision. It is not an automatic result, and the proposal establishes no perfect biology. Keep the original held-out, real-reference and retained-learning controls, including the distinction between useful guidance and learning that remains after guidance ends.

Acquisition and virtual-modelling costs do not establish the cost of delivering sensation or action. Physical sensory writers, transducers and actuators remain unpriced additions. No loop, training or physical-writing result is claimed here. The next experiment must report where each declared prediction works and where it fails against its independent measured target.

21.11 Technical representation and mask

Learned query, key and value representations give the proposed attention calculation a precise form. An admitted mask would restrict which observations may contribute; biological grounding still requires the independent comparison.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^{\top}/\sqrt{d_k} + M\right)V.$$

Here Q, K and V are queries, keys and values; M expresses the admitted mask. The dimension d_k is the common feature width of each query and key vector. The equation is methodological context from the supplied technical proposal. Modality, channel, task and time are part of the observation contract. No weights, EEG encoder or gradient route are selected or executed.

21.12 Permitted observations and causal modality routes

Declare B=EEG, E=EMG, Y=independently measured movement, C=task/stimulus/context, and P=the source-bound physics model. Training may relate qualified paired observations. Inference uses only its explicitly admitted subset:

- EEG-only: permitted past/present B and declared context predict a specified EMG/movement target. Withheld E, Y and future samples cannot enter the predictor.
- Body-only: permitted E and/or Y predict specified recorded neural features as a conditional distribution with uncertainty. This is not a unique inverse brain.
- Hybrid: additional modalities require a separately named route and evaluation; a hybrid result cannot be reported as EEG-only.
- Simulation-derived inputs stay simulated. They do not replace independent observations or inherit the biological validity of a rendered body.

Separate training signals from evaluation references. Cycle consistency can make two models agree while both are wrong. Shared cues, movement artefacts, subject identity, session overlap and preprocessing leakage need the original controls. Do not expose final held-out targets to model, physics or reward selection.

21.13 Representation, supervised learning and policy refinement

0. Establish source rights, subjects/tasks, measurement units, calibration, uncertainty, clock alignment and frozen subject/session/task splits. Audit pretraining overlap before preprocessing and model selection.
1. Pretrain modality representations on the qualified inventory. LaBraM is one candidate EEG encoder: neural-tokenizer training, masked neural-code prediction and downstream fine-tuning. It is not a complete brain-body writer. Code, checkpoint, dataset and individual-use rights remain separate qualifications.

2. Learn paired supervised mappings with named targets and explicit modality masks. Compare appropriate simpler, cue-only and artefact baselines. Keep raw, processed and model-derived values distinct.

3. Qualify physical predictions and the reward verifier against independent measurements. A successful task, smaller calibrated movement error and valid physical states are prospective reward ingredients, not completed results. Declare loss/reward weights on training/development evidence. A computable objective or high reward does not confer biological ground truth.

4. Where admitted action/transition/reward coverage supports it, evaluate optional RL and post-training. Offline trajectory labels alone do not establish the consequences of actions never supported by the data. Conservative Q-Learning is a methodological candidate, not a safety or performance guarantee here.

5. Fine-tune for the declared task and qualify both directions on frozen final tests, uncertainty, baselines and every original threshold. Keep failures.

6. Serve a frozen accepted version. Propose later updates separately and promote only after independent acceptance, restoration proof and applicable authority.

An illustrative supervised objective would combine declared errors for specified EMG, movement and recorded-neural targets with applicable physical constraints and optional consistency terms. Each term needs its units, measurement uncertainty and target definition before a weight is chosen. Consistency can check whether the outputs fit together; an independent observation checks whether they fit the measured outcome.

Backpropagation changes differentiable components of the computer model; biological feedback is a different process. A physics engine in the path does not automatically give the learner a gradient. Qualify the actual OpenSim/MyoSuite rollout and optimization route, or a separately validated surrogate, rather than assuming gradients through every route. No training, RL rollout or task checkpoint is created or certified here.

21.14 Frozen response and independently accepted adaptation

What must the response path preserve when an observation arrives late? The frozen-weight distinction in “Respond quickly. Improve carefully” means that the state estimate may change as observations arrive, while learned weights change only after a separately tested candidate is accepted. Each response needs its causal time window, permitted observations, target and output preparation. Future observations cannot repair a missed deadline, and an unaccepted candidate update cannot become the running model.

The timing test must follow that response through the whole chain specified there, including transduction or actuation where applicable. Delay tails, jitter, missed deadlines, uncertainty and coexistence with healthy workloads determine whether the chain can meet its declared task; component inference time cannot settle that question. The Kahneman-like analogy describes this division of computational work. It establishes neither two anatomical systems, inherently fast or slow sensor modalities nor real-time performance.

21.15 Distinct sensory and action interfaces

The output tests follow the two routes in “Write sensation. Enact action”. A visual, haptic or other sensory transducer would be tested by stating the sensory content, interface and authority, then measuring the actual response. Direct neural writing remains a separate later research interface; a sensory role would not automatically make it a peripheral afferent signal. The boundary matters: a recording related to the body’s motor output can enter a computer as input. Computer direction and direction relative to the central nervous system must therefore both be stated.

For the action route, the test would identify the control delivered to the virtual body, robotic actuator or later qualified biological interface and measure the resulting action. The robot command remains a computer output, not an anatomical efferent nerve. That distinction determines the evidence needed for simulated action, physical actuation, retained learning and biological causation. Predicting an EEG-like pattern delivers no sensation, and EEG and EMG recorders supply neither of these writing functions.

The output design therefore needs explicit subinterfaces within the unchanged five programme pillars: sensory encoders and transducers for sensation, and action encoders, controllers and actuators for enaction. This document gives no stimulation dose, implantation, electrode-placement, acquisition or actuation procedure, and grants no participant, clinical, device, purchase or deployment permission.

22 From contact to coordinated action

A ball leaves one hand and reaches the other. A prediction can follow both hands and still miss the release, the contact or the order of events. Those differences give the proposed paired study its next questions. We want an intent-to-action mapping as close to one-to-one as the declared task permits; measuring the intermediary gives us a way to test whether the ambiguity actually falls.

The proposed sequence moves from contact and comparatively simple movements to bilateral object handling, punch discrimination and toy archery. It extends the single-hand precision example without treating every task as another copy of a reach. Each task needs its own anatomy, contact conditions, permissible context, independent reference and error specification. The full snap remains a coordinated arm, forearm, wrist and hand movement, including its preparation and return; species-specific reference evidence must precede any claim about a monkey’s ability to perform it.

22.1 Record one event across the instruments

The proposed apparatus pairs scalp EEG with two surface-EMG recording sleeves, one on each arm, and synchronized, calibrated left and right tripod views of the hands and arms. OpenBCI or an equivalent configurable system remains a candidate. Channel count, electrode sites and references, bandwidth, units, device delays and clocks must be qualified for the chosen task before the recordings can enter training. A nominal sampling rate cannot establish that the instruments observed the same event at the same time.

These left/right views are the proposed hand-task configuration of the motion-capture relay-composite. The earlier opposing top/bottom dot-map configuration describes a different proposed apparatus. Each configuration requires its own camera geometry, calibration, timing, coordinate frame, visibility and occlusion checks. Joining two maps preserves their source and uncertainty; it does not turn a model-filled gap into an observed point.

The musculoskeletal-simulation physics-engine would predict the body's response within its qualified anatomy and mechanics. Independent movement observations would test that response. Contact labels need an appropriate independent observation, and a force claim needs a force reference. Surface EMG measures muscle electrical activity: it supplies neither direct force nor sensory-nerve activity, and it is not automatically the simulator's activation variable.

22.2 Four tasks, four reasons to measure

Touch and movement. Begin with rapid hand rubbing, then record the requested knuckle movement and interlaced-hand stretch as separately identified conditions. The stretch extends the arms with the palms turned outward. Rubbing combines contact and movement; the comparison therefore needs matched conditions that can distinguish their contributions, rather than labelling every change in EEG as touch. The cold-hands reference supplies the intuitive action, not a cold-exposure result.

Transfer and juggling. Pass one juggling ball between the hands in the proposed hot-potato task, then progress to the preferred two-ball juggling task. Release, flight and catch connect the two limbs through the same object. Preserve the observed event order, each limb's trajectory and the limits of contact observation. A predictor that knows the task cue or the previous trajectory may succeed without useful neural information, so those simpler explanations remain in the comparison.

Punch discrimination and NeuroBoxer. Define punch classes from independent movement observations, then test whether the permitted recordings predict the class, trajectory and timing specified by the protocol. Boxing also involves hand and wrist stabilization. NeuroBoxer is the proposed virtual game built around these comparisons; a prime-era Mike Tyson opponent is a proposed virtual game character. It supplies no physical opponent or evidence of safety. Expert footage can inform movement and tactics within its rights and camera limits. It supplies none of the expert's EEG, EMG or muscle-force data.

Toy archery and NeuroArcher. The proposed toy task uses arrows with small, cup-shaped suction tips and an over-arm pouch with multiple arrows. Grasp, draw, release and return give the two hands different roles under opposing loads; finger and wrist changes matter as well as arm trajectories. NeuroArcher would use that coordination to act on a virtual target. Neither the target design nor a successful game action establishes measured biological mechanics.

22.3 Test the neural contribution before assigning it control

The first game milestone uses measured body motion to control the virtual body. A later EEG-only milestone would test a genuine contribution from the permitted brain observations. Its predictor must not receive withheld EMG or movement targets. Eye activity, covert muscle activity, residual movement, shared cues, trial pairing and future information need separately declared controls; apparently hands-free play cannot settle those questions by appearance.

Keep the original brain-only, body-only and paired observation comparisons, target-persistence controls and benefit criteria. Record permissible history and causal windows before fitting preprocessing or choosing a model. Reserve the required subject, session and task partitions before window construction; fit transforms on training data only. Pretraining exposure must be resolved before calling a final recording independent. Each reverse test predicts specified recorded EEG features conditionally; it cannot identify a unique hidden brain state or deliver sensation.

Qualified pretraining, paired fitting and fine-tuning would follow admitted data, rights, clock, unit and baseline checks. Optional control-policy reinforcement learning also needs declared states, actions, transitions and rewards with appropriate coverage. A simulation reward tests the declared computational objective; independent biological outcomes require their own evidence. Final held-out observations stay outside rewards, tuning and model selection, and frozen inference remains separate from independently evaluated adaptive updates.

22.4 Separate the later writer from the recording proposal

EEG and surface EMG record. Neuromuscular electrical stimulation (NMES) and functional electrical stimulation (FES) denote stimulation approaches that can produce muscle contractions or functional movement; those are separate physical output interfaces. Future sensation delivery likewise needs a qualified writer or sensory transducer. Predicting a neural feature, commanding a virtual body and producing a biological response answer different questions. Physical spinal scope remains read-only.

The proposed later blindfolded condition changes visual feedback. It does not independently validate neural control or retained learning. A retention test must remove tutor information while keeping the necessary movement bridge and comparable task conditions, then compare performance with the original controls. No participant recording, stimulation, actuation or such retention experiment is reported here. Physical writers and actuators remain outside the existing acquisition/virtual-work budget scope. The retained LINK failures remain the current computational result; this sequence specifies prospective comparisons.

23 Measure both ends. Test the translation.

The following prospective protocol tests the paired learner described in “Two pairs. One conditional learner”: can permitted observations on one side of a sensorimotor task predict withheld measurements on the other, on recordings the model has not seen? The apparatus remains non-invasive EEG, non-invasive surface EMG, a calibrated

motion-capture relay-composite, an open musculoskeletal-simulation physics-engine with a lifelike explanatory view, and an open multimodal model. Its acquisition and validation are unfinished.

The earlier brain–muscle and measured-movement–virtual-body comparisons define the two pairs. Here the protocol specifies their records, targets, calibration and independent tests. Anatomical form constrains possible mechanisms; measured function must test the proposed connections. A conditional learner's combined prediction must answer to measurements outside itself.

23.1 Keep the observations distinct

Keep raw EEG, raw EMG, derived features and movement estimates in separate records. Identify the participant and session in each record, then declare the sensor sites, references, coordinate conventions, units, sampling and timestamp semantics, and clocks. Trace every processing step and preserve missing intervals and uncertainty. Retain task instructions, stimulus events, observed behaviour and any separately obtained participant report. Independent participants or recordings cannot supply a simultaneous brain–muscle–movement observation.

Surface EMG records muscle electrical activity. A sensory-loop claim also needs known stimuli and appropriate measured feedback, contact or force channels; muscle recordings do not provide incoming sensory-nerve traffic. Preserve the actual spatial resolution of EEG measurements and source estimates. Templates and explanatory cutaways do not register participants or localize individual neurons.

23.2 The spine shapes the loop

The spinal cord changes how sensory input becomes muscle output. Account for local coordination, sensory gating, reflex gain and some lasting adaptations, rather than reducing it to a transmission delay. Separate a response to the current posture or task from a change retained after training. Keep both apart from adaptation in sensory receptors, peripheral pathways, neuromuscular junctions and mechanical tissues. These processes interact; changed output alone does not locate a memory.

Human H-reflex conditioning and the constrained mouse adaptation studied by Lavaud and colleagues ask different questions.[38, 46, 47] The human experiments do not establish brain-independent skill learning. The mouse study does not establish unrestricted walking or reaching. Circuits required for acquisition or recall are not thereby exclusive memory-storage sites. Our human comparison uses the abstracts, selected results and limited methods passages. The mouse comparison also draws on relevant author-manuscript passages; equivalence to the final published edition and the complete methods and statistics have not been established here. Passive proprioceptive training motivates a peripheral observation model; its psychophysical and spindle-afferent experiments were separate.[48] Those observations do not establish complex memory stored in peripheral axons.

Define the function before choosing its computational substitute. Candidate functions include gain and inhibition, phase- and load-dependent sensory responses, coordination across supported muscles or joints, relevant rhythms, and acquisition, reten-

tion and expression of a declared adaptation. Give each function its own task, species, anatomical scope, measurement and reference. Naming a mechanism does not show that the model implements it. A hand task cannot inherit an unmeasured locomotor circuit.

Record task context and history. Distinguish the state within a trial from a learned parameter change; retain elapsed time, intervening activity, resets, drift and uncertainty. Recorded spinal or peripheral activity needs supported channels of its own. Where those observations are absent, label internal state as estimated or as a simulated controller. EEG, EMG and movement cannot be relabeled as recorded spinal activity or identified interneurons.

Declare controls and endpoints before fitting or experimentation. Test immediate context effects separately from retention after a declared interval and from peripheral or tissue changes. Preserve the original held-out subjects and tasks, eligible independent measurements, matched baselines and ablations. Check alternative explanations, time shifts, missing inputs and artifact leakage. Lower movement error alone cannot locate a biological adaptation, establish its cause or certify useful plasticity.

To externalize a specified function, propose a substitute that can be tested through declared peripheral and cerebral interfaces. Compare latency, state dependence, stability, uncertainty and measured outcomes with the specified biological function. Computational feedback, peripheral stimulation and restored biological function require different evidence. A matched virtual trajectory does not demonstrate spinal replacement, relocated learning or clinical repair. Keep the spinal-cord site, measured function and proposed computational substitute distinct. Physical spinal scope remains read-only.

23.3 Give the two directions different targets

For brain-to-limb prediction, give the model EEG and only the declared context available at inference. Test its predicted muscle and movement observables against withheld measurements. A brain-only condition receives no measured EMG or movement input. If those inputs help, evaluate them in a separately labeled hybrid condition.

For limb-to-brain prediction, give the model muscle activity, movement and declared context. Its answer must allow for different patterns in the supported recorded EEG features. A conditional distribution assigns probabilities to specified feature values or ranges, given those inputs. Compare that prediction with independently retained EEG, then report uncertainty and alternatives. Movement can narrow the possibilities without identifying a unique hidden brain state or subjective experience. Agreement between learned models is a consistency check; it cannot supply the measured target.

23.4 Reading a signal and writing an output

The protocol records EEG and EMG; its two prediction directions do not themselves deliver sensation or enact a movement. “Write sensation. Enact action” defines the computer and physiological boundaries. Testing a writer would additionally require a separately qualified sensory writer or transducer, a specified actual response and distinct device, measurement, safety and clinical evidence. Indirect sensory feedback and

direct neural writing remain different interfaces; a predicted EEG feature is neither. Enaction needs an appropriate virtual, robotic or biological output interface. A robot command has an action role, not the anatomical role of an efferent nerve.

Physical sensory writers, transducers and body actuators remain excluded from the acquisition and virtual-work plan, unqualified and unpriced. This proposal authorizes no stimulation, participant acquisition, biological actuation, new equipment purchase or training.

23.5 Align before learning

Calibrate the acquisition views. Verify their clocks against recorded events, offsets and drift, while preserving occlusion and dropout. Keep a missing observation missing: a predicted finger position is not a measurement. Declare frames, scale, joint conventions and acceptable spatial, temporal and end-to-end errors before selecting equipment or models. Position, joint, contact and force claims each need an appropriate independent reference.

Specify how the virtual body represents subject scale, muscles, tendons, joint constraints and contact. Declare its integration settings and observation models. Biomechanical simulation predicts mechanics; VFX rendering makes those predictions visible. Judge individual parts and their assembly against the original anatomical, physical, functional and real-reference-series criteria.

23.6 Build an open corpus without inventing pairs

Inventory neural structure, neural activity, muscle structure and muscle activity as separate domains. Use measured movement and task context as constraints. For each source, record rights, version, access conditions, species, participants, tasks, montages, units, clocks, preprocessing and missing paired modalities. Broad open-data coverage remains a continuing objective, not an assertion that every dataset is accessible, licensed, compatible or already used. Check code, pretrained weights and training-data rights separately.

The candidate uses an encoder for each modality, explicit input-availability masks, a shared conditional representation and separate decoders for the two directions. Keep simple EEG-only, EMG-only, movement-only and hybrid baselines. Define pretraining, post-training and fine-tuning tasks before fitting. An existing encoder supplies a candidate component, not an already established brain–body model.

23.7 Make the reward answer to measurements

First test paired supervised prediction in each direction. Compare any later reinforcement-learning or other closed-loop refinement with that baseline. Evaluate independent movement measurements, task outcomes, relevant contact or force, latency and physical validity. Freeze reward units, terms, weights and decision thresholds before evaluation. The candidate cannot invent the target against which it earns a reward.

Backpropagation updates model parameters. Reverse prediction estimates another observable. Measured biological feedback comes from the body. Identify which process the study actually implements. A supervised prediction error alone does not establish reinforcement learning: a proposed control study must also declare its state, action, transition and reward.

Partition eligible participants, sessions and novel tasks before preprocessing or model selection. Preserve TRAIN-only normalization, frozen seeds and checkpoint selection, untouched held-out evaluation, uncertainty and original negative controls. Include informative missing-modality, time-shift, artifact, drift and alternative-inverse comparisons. Keep reserved recordings closed and retain completed fits and failed benefit conjunctions. Evaluate frozen inference separately from any proposed adaptive update. This prospective protocol admits no new fitting, optimizer update or model/backend promotion.

23.8 Show what was measured and what was inferred

Show the selected source clock and the inputs available to each prediction. Place recorded EEG, muscle activity and movement beside their predictions, with uncertainty, errors, gaps and stimulus or contact context. Identify recordings, inferred features and simulation as separate layers. Preserve site/channel inspection and source-time play, pause, seek and step. Unsupported peripheral or spinal pathways remain visible gaps. An animated cord cannot substitute for a spinal recording or supported anatomical localization.

24 Results and Remaining Questions

The LINK comparison shows why an improved mean cannot stand in for a better decoder. Neither model meets the frozen requirements, and all five benefit claims fail. The earlier studies retain their narrower results: specified within-recording estimates, a separately calibrated different-day specialist, bounded simulated muscle correction and traceable geometry checks. They answer different questions and cannot reverse the LINK outcome.

The hand studies make the difference visible. The classifier identifies 116/150 attempted-finger labels. In the linked simulation, 54/54 issued commands are executed, yet only 41/60 source intentions become correct contacts and 13 unintended presses remain. A separate goal-driven policy completes 32/32 constrained simulated phrases without recorded brain activity driving that skill. Executing a decision, making the right decision and learning a phrase are different achievements. None establishes biological teaching, a neural policy for the phrases, voluntary whole-chain control, retained mastery or universal translation.

The next paired comparison would ask whether permitted neural observations predict withheld muscle or movement measurements better than the specified simpler controls. Unknown units, upstream clocks, contact identities, participant registration, joint and force measurements and missing nerve or spinal observations must be resolved or left explicitly unavailable. More computation cannot supply an absent measurement by assertion. Source support, independent implementation, final-checkpoint/update

verification and reserved confirmation remain separate requirements; completing one cannot stand in for the others.

That experiment must keep the original physical, statistical, held-out, uncertainty and failure thresholds, with independent acceptance for the claimed domain and real-reference series for the applicable anatomy and function. Biological function and a complete working GUI require their own evidence. Publication and edition-review limits are collected in “Evidence, scope and edition status”. The test returns to the opening distinction: does measuring the intermediary improve the declared intent-to-action mapping? If it yields useful assistance, does any learning remain after tutor information is removed under controlled conditions? Neither result is established here.

Evidence, scope and edition status

The computational studies in these unpublished working manuscripts lead to a separate proposed paired experiment. No new fitting, reserved-data evaluation, model promotion or biological teaching is reported here. The negative results constrain the next test; they do not finish it. The common optimizer still needs independent implementation checks, including final-checkpoint and update verification. Reserved confirmation, complete GUI and offline checks, and applicable anatomical and functional reference comparisons also remain open.

The original tolerances and reserved recordings remain unchanged. A failed comparison cannot become a successful one through revised wording. Scientific, clinical, physical, real-reference, rights and runtime claims still require their original validation. Manuscript sources, citations, figures, rendered outputs and publication have not completed their checks. Source support for a negative report establishes neither full scientific validity nor a qualified interface or release.

This edition uses the 194-entry shared glossary, revision 29, SHA-256 4189722b c823c1634084c4a28b93dcc198e583f2b11443e315f0cb593fd9b038. Its 178-entry definition supplement and 16-entry addendum retain their scientific meanings. Four component entries now use the October 19 names and explain the proposed roles beside their measurement limits. Lookup labels can refer to related or contrasting ideas; the entries explain how the terms are used here. Earlier definitions remain in their historical editions. Updating the vocabulary does not revalidate the earlier studies or establish a proposed capability.

A generated illustration can explain a mechanism; it cannot replace the working model or measured evidence.

Reproducibility record. The retained packaging record describes where files are stored. Moving opaque resources outside the candidate HTML, the browser page’s textual entry file, reduced that entry from 113,298,857 to 34,295,597 bytes. The complete candidate directory was 93,831,812 bytes and online delivery required its assets and whole directory. The separate static offline export compiled from the original source was 113,581,899 bytes. Views dependent on the application’s service interface (API) were unavailable offline. The recorded review could not use the candidate’s browser

route and did not promote it; these receipts establish no newer browser qualification. The counts measure neither interaction nor loading latency, memory, crashes or GPU performance. Whole-directory online delivery and original-source offline access remain separate claims.

The requested public byline is `@eliminatenoise`. Author identity, applicable affiliation and repository submission requirements must be resolved before publication. No affiliation or publication acceptance is implied.

The retained metadata and bounded passages for Logothetis and colleagues, Tang and colleagues, the quantum-measurement scientific background, and the NIST measurement-uncertainty guidance support only their stated uses here. Complete manuscript citation and claim verification remains pending.

25 Frozen LINK Development Methods

A predictor can lower its average error while making one finger group worse on one recording date. The LINK comparison therefore keeps each date and output visible, then asks which comparisons survive all their stated conditions. The prospective protocol states the intended comparison. The retained implementation specifies the operations; the completed-study report describes the execution and its outcomes. Agreement among these records remains to be verified.

25.1 Observations and Common Support

The development design uses one macaque's four training dates, January 27 and 30 and February 4 and 5, 2020, and two validation dates, February 6 and 11, 2020. The two measured targets are index-finger and combined middle/ring/small-finger normalized positions. They have source-normalized position units, without a conversion to radians, force or 23 separate joint angles. The input comprises 96 nonnegative threshold-crossing counts transformed by $\log(1 + x)$ and 96 archived spiking-band-power features. Physical electrode contacts identify columns; matching a contact across recordings does not identify a persistent single neuron. These data do not supply simultaneous peripheral-nerve, spinal or EMG recordings.

For a scored endpoint t, the input contains source frames $t - 10, \ldots, t$ and the auxiliary target contains the neural features at $t + 1$. The position target is at t. All twelve frames must belong to one source trial, and each adjacent source-clock interval must lie in $[0.0185, 0.0215]$ seconds. The nominal bin interval is 20 milliseconds; original timestamp jitter is retained. No window crosses a trial boundary, no interpolation fills a gap, and recurrent state does not continue between windows. Every candidate, including the constant and both ridge baselines, uses this same endpoint support and requires the following neural frame. A position-only model consequently receives no extra trial-end examples unavailable to the auxiliary model.

The first training date defines the canonical order of 96 physical contacts. Both feature blocks are reordered together. Feature means and population standard deviations are fitted in float64 on each unique trial-owned frame once across the four training dates, rather than counting a frame again for every overlapping window. Exactly zero standard deviations become one; there is no other floor, clipping or validation-derived normalization. The transform is calculated in float64 and cast to float32. The constant baseline is the mean of unique trial-owned training positions. Position targets remain source float32 values. Nonfinite measurements, unknown contacts or incompatible schemas cause a failure rather than silent row removal or imputation.

The archived threshold-crossing unit attribute says volts despite count-valued semantics; the comparison makes no voltage conversion. Archived power values are retained without the author loader's unexplained factor of four. The documented 32-millisecond interval also differs from the retained nominal 20-millisecond clock. Full author trial counts must agree with the trial stop timestamps; uniform one-short time-series references are identified separately. These distinctions matter because changing units or trial support would change the experiment.

25.2 Candidates, Initialization and Training Order

The two recurrent candidates have 192 inputs, one gated recurrent-unit layer with 128 hidden values, and two position outputs. The position head combines the final hidden values with the mean neural feature vector over the eleven input bins. A separate linear head predicts the 192 next-neural features. Both candidates contain 169,090 parameters. The position-only objective uses 144,322; the 24,768-parameter neural head receives zero objective weight in that candidate. The historical software name of the position head does not change its target into velocity.

For position target y, position prediction $\hat{y}$, next-neural target z and prediction $\hat{z}$, training minimizes

$$L = \operatorname{mean}\left((\hat{y} - y)^2\right) + \lambda \operatorname{mean}\left((\hat{z} - z)^2\right), \qquad \lambda \in \{0, 0.1\}.$$

Each residual is squared before the mean is taken over the corresponding minibatch and output elements. Otherwise, residuals of opposite signs could cancel before the square and conceal prediction error. Training uses Adam with learning rate 0.001, averaging constants $(0.9, 0.999)$, stabilizer 10^{-8}, zero weight decay, and disabled fused and foreach routes. There is no dropout, autocast, TF32, gradient clipping or target clipping. Minibatches contain 256 windows, including the final partial minibatch. Nonfinite losses, gradients, updated parameters, optimizer states or predictions fail the run. The reported study route uses four CPU mathematical threads and one framework interoperation thread; that historical configuration is separate from manuscript preparation and does not establish a speed improvement.

The fixed seeds are 20261007, 20261008 and 20261009. Each run resets Torch's seed, constructs the same architecture, checks its initialized float32 tensors against the seed's frozen initialization, and loads that initialization. The intended pairing therefore includes both recurrent candidates and all three input conditions for a given seed. A separate NumPy PCG64 generator starts once per run from that seed. Training dates are concatenated in the declared order above, with ascending scored source endpoints within each date. One permutation of all concatenated training rows is drawn per epoch; every row appears once. The implementation obtains matching schedules by resetting this generator identically in each paired run. It saves each epoch's order hash and generator state, rather than reading a shared bank of saved order arrays.

The run order is measured inputs, association control, then order control; within a condition it is position-only, then position plus next-neural; within a candidate it is the three seeds in their declared order. Each run retains all 40 epochs. Restoring a completed epoch restores model, Adam, NumPy and Torch state; an unacknowledged partial epoch is not permission to repeat updates. The source specifies these safeguards. This documentary account does not independently verify their use throughout the completed execution.

25.3 What Each Control Changes

The association control breaks the training alignment between neural evidence and position. For each seed, the implementation creates one PCG64 generator from `seed+1000000`, then advances that same stream across the four training dates. It draws one permutation separately within each date. The shuffle unit is an eligible whole-window row: its complete 11 × 192 neural window and associated 192-element

next-neural target move together. Position targets and the destination rows' date, trial, source-endpoint and scoring identities stay fixed. This is neither a channel shuffle nor a within-window bin shuffle, and it does not move whole trials as blocks. The permutation is fixed for the run, shared by the two candidate types for that seed, and is not redrawn each epoch. Its hash and accidental fixed-point fraction are recorded.

Association-control models are fitted on these transformed training rows and selected on unchanged measured validation windows and positions. Their final comparison also uses unchanged validation support. Training losses refer to the transformed training pairing; the main candidate and control scores refer to the two validation dates. The control retains within-window history, within-date distributions and possible common task structure. Fixed points can retain some original pairings. It does not create a causal null.

The temporal-order control applies the same fixed permutation to the bins of every training and validation window. In zero-based bin indices it reads

$$(9, 8, 7, 6, 5, 4, 3, 2, 1, 0, 10) \quad \text{in place of} \quad (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10).$$

Thus the ten earlier bins reverse and the current bin stays last. Position, next-neural targets, feature identities and scoring rows remain unchanged. The transformation uses no random draw. Each order-control run retains the same seed initialization, epoch-order rule, optimizer and selection policy as its measured-input counterpart. Reversal preserves the window's multiset of values and is invertible; a recurrent model can adapt to it. The comparison tests benefit from this declared order, not an assurance that all temporal information has been removed. Both control conditions are excluded from the best-model competition.

25.4 Selection and Uncertainty

Let $E_{dg}(A)$ be model A's frame-mean squared position error for validation date d and finger group g. The primary score is

$$\overline{E}(A) = \frac{1}{4} \sum_{d=1}^{2} \sum_{g=1}^{2} E_{dg}(A).$$

Frames are weighted equally within a cell; the four date/group cells then have equal weight. Each seed retains the epoch with the lowest primary score; an exact tie takes the earliest epoch. Predictions from all three selected seeds are averaged arithmetically with equal weights. Seed performance is also retained separately, including its mean, sample standard deviation with one degree-of-freedom correction, minimum and maximum. No favourable seed replaces the ensemble.

The current-bin ridge uses only the final 192 features; history ridge flattens all eleven bins oldest to newest in row-major order. Both fit float64 ridge regression with unpenalized intercept and objective $\sum_i \|y_i - X_i W - b\|^2 + \alpha \|W\|_F^2$. The frozen penalties are $0.01, 0.1, 1, 10, 100$. The lowest validation primary score selects the penalty, with the smallest penalty winning an exact tie. An exact tie between ridge types selects current-bin ridge. An exact tie between measured recurrent ensembles selects position-only. Control models do not enter either candidate-selection contest.

For every comparison, candidate-minus-reference squared errors are paired on the same validation endpoints. The bootstrap initializes PCG64 with 20261007, uses validation dates in declared order and sorted unique source trial IDs, and draws 1,000 samples. Within each date, each sample draws the original number of eligible trials with replacement. Every supported frame of a drawn trial contributes with its multiplicity; variable trial lengths retain their frame weighting. The same trial draws serve both models and both groups. The generator advances from the first validation date to the second, and each comparison restarts the same seed. The primary difference averages the four cell differences. The 95-percent interval uses the 0.025 and 0.975 quantiles with linear interpolation. Fewer than two supported trials per date prevents this uncertainty calculation. The intervals are descriptive and conditional on these dates and validation-selected models.

Ordinary R^2 uses the evaluation cell's observed mean:

$$R^2_{dg}(A) = 1 - \frac{\sum_i (\hat{y}_{i,g} - y_{i,g})^2}{\sum_i (y_{i,g} - \overline{y}_{d,g})^2}.$$

The separately labelled training-mean skill score substitutes the frozen training mean in the denominator. A zero denominator produces an undefined score, without an added epsilon, and cannot pass a rule requiring defined R^2.

25.5 Model Admission and the Five Benefit Comparisons

Define the comparison conjunction

$$C(A, B) = [\overline{E}(A) < \overline{E}(B)] \ \wedge \ [E_{dg}(A) \le E_{dg}(B) \text{ for every } d, g] \ \wedge \ [U_{.95}(A - B) < 0],$$

where $U_{.95}$ is the upper endpoint of the paired two-sided 95-percent interval just defined. Equality passes the per-cell no-harm condition, but does not pass either strict improvement condition. Let P and N denote the measured position-only and auxiliary ensembles, R_c current-bin ridge, $R_\star$ selected best ridge, and P_o, N_o their matched order controls. The retained independent postfit checker evaluates these five comparisons separately:

1. Position-only history benefit: $C(P, R_c)$.
2. Position-only order-use benefit: $C(P, P_o)$.
3. Auxiliary-model history benefit: $C(N, R_c)$.
4. Auxiliary-model order-use benefit: $C(N, N_o)$.
5. Auxiliary-objective benefit: $C(N, P)$.

Each displayed conjunction includes its own lower primary error, no worse date/group and negative paired upper bound. The history comparison asks whether the recurrent candidate improves on the specified current-bin linear reference. It does not isolate recurrent ordering from the mean-history input or every other architectural difference. An order-use claim additionally needs its matched reversal comparison; a history comparison alone does not establish that claim.

The distinct implemented model-admission gate requires $C(A, R_*)$ and, in every cell, defined ordinary $R^2(A) > 0$ and $R^2(A) > R^2(A_s)$, where A_s is the matched association-control ensemble. A missing or nonfinite cell cannot pass. This model gate must not be substituted for the five comparison rules.

The retained independent postfit checker implements the three-part conjunction separately for every comparison above and records each Boolean condition. All five decisions fail. Each has a lower average error, but each fails its per-date/group no-harm condition; three have a paired upper bound below zero. The separately recorded model-admission decisions also fail. This account reuses those existing decisions without recomputing an interval or prediction.

That support is limited to the correctly negative report. The retained order-control ledgers have empty control-provenance fields: source identity and earlier accepted control evidence do not supply a new dynamic reversal check. Nor does postfit agreement independently certify the shared optimizer, every new checkpoint forward calculation, restored random state or complete update trajectory. Those requirements remain open. A favourable average cannot replace a failed condition.

25.6 Completion and Causal Limits

The inventory records completion of 18 runs with 40 saved epochs each, for 720 saved epochs. That completion and the later postfit checks support the unchanged negative report; they do not verify every optimizer calculation, checkpoint or update, or establish whole-study scientific acceptance. The prospective protocol's `execution_started=false` records its state when frozen, before the later study ran. Completion admits neither a model nor reserved confirmation.

The saved inventory records that the service was terminal and inactive, with exit status zero and no automatic restart. That record establishes custody of the files; it does not, by itself, bind them to the complete execution command, loaded modules and later decision records. Those missing links limit what can be reconstructed about execution. They do not explain away the failed development comparisons.

February 22 and 24, 2020, remain closed later-day dates; June 21 and 22, 2023, remain closed drift dates. The later task difference also limits a pure drift interpretation. No reserved result is inferred from development validation.

The history is causal only with respect to the supplied source indices: position prediction uses indices no later than t, while next-neural learning uses $t + 1$ as a training target. Full archive extraction and filtering support has not established causal online acquisition or physiological delay. The study concerns predictive associations between recorded features and two position coordinates. It does not demonstrate nerve transmission, spinal externalization, sensory delivery, biological teaching, retained skill or full-body neural translation.

NeuroTranslation Definitions and Lookup Index

Canonical registry revision 26; `fc5dbd61462ecf98ada2629e84659ab2d05c0005 8df44efa80e3b397e20ca042`.

A model, a measurement and a claim about biology need different words. These 178 entries state what each term means in this programme. Naming a capability does not establish that it has been demonstrated.

Lookup labels help find an entry. Some name related or contrasting concepts, so they are not all interchangeable. The portable JSON retains the selected fields and complete lookup index. Original definition histories remain in the canonical registry.

Organizational level describes which part or signal is considered. Physical ranges and units describe dimensions. Computational granularity describes the detail represented in a model; statistical aggregation describes how observations are grouped. A missing range or unit remains unspecified.

Some technical notes describe a named source edition or prototype. Its measurements, numerical results and implementation limits belong to that context; they are not timeless claims about every model. The entries distinguish recorded observations, calculated quantities, simulations and proposed capabilities. This supplement reports no new experimental result.

A checked message between software parts

A structured message and interface specify what the values mean and how software parts exchange them.

The checks include recording, ordered channels, quantity, timing where supplied, units and body axes. Passing a check does not prove that the caller's claimed physical source is true.

Lookup labels: Translation packet, typed port and API; packet; translation port; typed interface; API; application programming interface

A group of recorded neural features

Several recorded neural features considered together when predicting an outcome.

Their joint pattern can help predict movement. That does not establish that the same pattern has the same meaning in another animal, task or day.

Lookup labels: Neural population; population decoding; population model

A map of physical connections

A structural map records physical connections. A functional recording measures activity over time, such as which recorded features produced events during a task.

The structure can guide a model, but it cannot supply missing time-recorded activity. A drawing, connection map and synchronized functional recording have different evidential roles.

Lookup labels: Structural connectome and anatomical prior; connectome; connectomic connectivity; anatomical prior; structural connectivity; structural reference; recorded function; structural versus functional measurement

A model spanning the body

A future model connecting relevant observations and actions across the brain, spinal cord, nerves, muscles and movement.

The project does not yet have a matched voluntary recording across that whole chain. Separate datasets cannot fill the missing measurements.

Lookup labels: Body-wide functional model; body-wide neural fusion; whole-body neural translator

A specified human-related feature

Studying a defined human-related biological component or feature in a model organism.

This can overlap with adding donor cells. One selected immune or neural feature does not make an animal a miniature human or establish human-like cognition.

Lookup labels: Humanization; humanized; human-related feature

Ability that remains after guidance is removed

A learned ability that remains without the training guidance and can be tested again later.

Remove tutor commands while keeping any bridge needed to express the pupil's own control. Switching off that required route can prevent movement regardless of what the pupil learned.

Lookup labels: Retained learning; retention; independent pupil control; retained skill; pupil-alone; necessary bridge held constant; pupil-alone with necessary bridge constant

Action2World

The hand reaches; the cup moves or stays still. Contact, load and the measured outcome tell us whether the action did what we intended, and that feedback shapes the next attempt.

The relationship between an action and its consequences for the body and surrounding environment, including the sensory information those consequences produce.

A predicted consequence is not an observed outcome. Specify the object, reference frame, task, units and uncertainty. Footage supplies movement examples without supplying the performer's neural, muscle or force measurements.

Organizational level: whole system; network; signal/feature

Lookup labels: action2world; action to world; action-to-world

Activity state inside the muscle model

The model's internal state that responds to a command and affects its force.

This state can change more slowly than the command. The displayed activation comes from the computer model, not recovered physiological activation from the macaque recording.

Lookup labels: Muscle activation; activation; activation dynamics

Adjustment using evidence from the new setting

Changing or checking a model using examples from where it will be used.

Adjusting a decoder, checking probability scores and measuring a body's physical properties are different kinds of calibration. The new-day model uses labeled examples.

Lookup labels: Calibration; recalibration; calibrated transfer; uncalibrated; physiological calibration; physiological fidelity; subject calibration

Agent2Agent

One software agent passes declared information to another.

A proposed link carrying specified information between software agents.

Name the agents, message or representation, direction and tested outcome. Software communication does not by itself supply a biological recording, sensory writer, shared experience or demonstrated skill transfer.

Lookup labels: agent2agent; agent-to-agent; agent to agent

Agent2Brain

An agent would give a person feedback during a stated task. Tests would have to distinguish help during that task from learning that remains afterwards.

A proposed path from a software agent to a person through specified feedback or a separately qualified neural output interface.

Feedback would reach a person through an ordinary sensory route or a separately qualified direct neural interface. Those routes need different devices, evidence and admission: feedback on a screen is not direct neural delivery. EEG and EMG record; they do not write. The proposed agent-to-person path leaves the reverse observation and feedback direction intact. Help during a task must be tested separately from learning retained after it ends. Agent2Brain establishes no stimulation or physical-action authority. Physical spinal scope remains read-only.

Lookup labels: Agent2Brain?; agent2brain; agent2brain?; agent-to-brain; agent to brain

All outputs inside their ranges together

How often every output of interest falls within its stated range in the same case.

A 90% range for each separate output does not guarantee 90% coverage of all outputs at once or of a whole body.

Lookup labels: Simultaneous coverage; all-output coverage; whole-body coverage

An independent way the model can move

One independent model movement, such as rotation at a joint. The formal shorthand is DOF.

The native arm has seven coordinates and no separate finger movements. The human replay controls two goals while bracing other independent joints.

Lookup labels: Degree of freedom; DOF; DOFs; independent coordinate; braced joint

Analysis timing and biological travel time

An analysis offset aligns two recorded quantities. A conduction delay concerns how long a signal travels through biology.

The 40 ms sensory offset estimates an earlier limb state. The motor target's 4 ms offset comes from sampling. Neither measures neural conduction time.

Lookup labels: S1 alignment offset and conduction delay; 40 ms sensory lag; S1 lag; 4 ms sampling offset; conduction delay; physiological latency

Animal drawing and measured anatomy

An illustration explains a proposed animal or body. An empirical anatomical atlas is built from verified observations of anatomy; it cannot be established by a convincing drawing.

The antechinus illustration is a proposal, separate from acquired macaque and human sources. The implementation has no matched validated antechinus neural/body atlas and makes no living-animal learning claim from the image.

Lookup labels: Antechinus proposal; antechinus; marsupial proposal; hypothetical organism; Proposed marsupial setting; anatomy illustration; illustrative model; empirical atlas; anatomical atlas

Another brain area used in the finger study

A brain area recorded alongside the movement area in the human finger study.

The formal name is posterior parietal cortex, or PPC. The study provides attempted-finger labels, not measured trajectories of executed finger movement.

Lookup labels: Posterior parietal cortex; PPC; Parietal brain area

Assumed repair to the native muscle model

A separately named model version replaces unusable zero-rate properties with the software's default values.

The source uses a named Schutte muscle model. Restoring 117 default properties permits numerical tests but does not measure the subject's muscle response rates.

Lookup labels: Schutte muscle properties and default-rate fork; Schutte; zero-rate; assumed-rate fork; class defaults

Attachment order and the chosen route segment

The ordered connection points and the numbered segment to which a wrapping rule applies.

The rule uses the source model's attachment order. The SDK counts from one and includes both range endpoints; the correction does not fit arbitrary points to an error frame.

Lookup labels: Attachment topology and SDK range; attachment topology; one-based inclusive range; humerus-to-ulna segment

Axes, pushes and turns

A coordinate frame states the axes used for movement. A handle wrench combines forces and turning moments measured in that frame.

The handle record has three force and three moment components. An unknown mapping into the computer-body axes prevents treating those forces as verified simulator inputs.

Lookup labels: Coordinate frame and handle wrench; body frame; XY; physical frame; handle wrench; forces and moments

Biological tissue repair

Restoration or replacement of biological tissue or its function, supported by direct biological evidence.

Better simulated tracking, assisted movement and retained behavior are different outcomes. The prototype does not demonstrate neural tissue repair.

Lookup labels: Regeneration; tissue regeneration; repair

Biological virtualization

For a movement model, matching the path is one question; matching the living system's response to an intervention is another.

A claim that a digital model reproduces specified biological behaviour and causal dynamics within an independently tested domain and stated error bounds.

Independent ground truth must establish measured error and repeatability within declared conditions and tolerances. For causal dynamics, the comparison also needs independently measured responses to appropriate causal interventions; a similar-looking trajectory alone cannot establish them. Approximation can make a model plausible and may contribute to a validated model when its assumptions, errors and domain are tested and disclosed. A musculoskeletal simulation is itself a model to test, not independent empirical ground truth. Numerical detail and repeated verifier results guarantee neither biological accuracy nor complete physiology, subjective continuity, consciousness transfer or implemented Brain2Body capability. This biological claim is distinct from operating-system virtualization.

Scope: validation: Specified observed behaviours and causal dynamical responses under declared conditions.

Organizational level: whole system; network; organ; local circuit; cell; signal/feature

Computational granularity: Declared model and brain-body interfaces

Statistical aggregation: Declared validation aggregation; separate from physical scale

Lookup labels: virtualization; virtualisation; biological virtualisation

Body dimensions and resistance to motion

Mass is the amount of matter modeled. Inertia describes resistance to changes in motion; anthropometry means body dimensions.

The scaled model keeps assumed source masses and applies geometry-related inertia updates. Its length fit does not identify the animal's actual masses, strengths or individual bone dimensions.

Lookup labels: Mass, inertia and anthropometry; inertia; mass; anthropometry; subject scaling; specimen scale

Body position stored by the simulator

The model's stored configuration, such as its joint positions.

The formal variable name is qpos. Its entries depend on the model and are not automatically measured joint angles. The matching velocity variable, qvel, is explained separately.

Lookup labels: Generalized position and velocity; qpos; joint state

Body-model software and its programming tools

OpenSim provides tools for body models, geometry, scaling and simulation. Its software development kit, or SDK, lets code work with those models.

Checks on these tools establish the stated model's numerical behavior. They do not certify the exact upstream model version or the recorded animal's anatomy.

Lookup labels: OpenSim and its SDK; OpenSim; SDK; software development kit; native SDK; OpenSim 4.5.2

Body-position feedback

Proprioception means sensing body position and movement. For example, it helps a body monitor where a limb is. Biofeedback uses bodily information to guide action or learning.

The implemented correction uses simulated body position. It does not add measured peripheral nerve signals or show that a biological feedback process taught a retained skill.

Lookup labels: Proprioception and biofeedback; proprioception; simulated proprioception; biofeedback; Body-state sensing and feedback; proprioceptive feedback

Body2Action

The proposed link from body mechanics and sensory interaction to a specified, independently measured task action or outcome.

A task outcome makes this proposed link testable: the body's forces, contacts and timing have to be measured in the relevant context. Body2Action connects body mechanics and sensory interaction to that specified action or outcome; it establishes no working controller. World labels are historical conceptual lookup references, not a strict synonym for every environmental interaction.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Lookup labels: Body to action; Body2World; Body to world

Brain imaging related to blood oxygen

An imaging method, called fMRI, that measures signals associated with changes in blood oxygenation.

It does not record individual electrical spikes. The paper discusses a published model for this kind of image, not a project result showing that brain scans and spikes are interchangeable.

Lookup labels: Functional magnetic resonance imaging; fMRI

Brain stimulation

An intervention that applies a specified physical stimulus to alter brain activity.

The method must be named, such as transcranial magnetic/electrical stimulation or intracortical microstimulation. These differ in access, risks and capabilities. EEG records; a scalp EEG headset does not provide general sensory writing or safe skill upload. Broad neuromodulation is not identical to precise task-content encoding.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Lookup labels: Neural stimulation; Neurostimulation

Source: A brain-computer interface that evokes tactile sensations improves robotic arm control

Source identifier: flesher2021

DOI: 10.1126/science.abd0380

https://pubmed.ncbi.nlm.nih.gov/34016775/

Research precedent for motor decoding plus sensory feedback through intracortical microstimulation; invasive and not a consumer EEG capability

Brain2Agent?

A recording supplies observations that an agent may use for a stated task.

A proposed path from specified brain observations to a software agent through declared recording and interpretation interfaces.

Recorded brain observations would reach an agent through a declared recording and interpretation interface. Where applicable, feedback toward the agent remains distinct from Agent2Brain's proposed agent-to-person path. Neither direction identifies unique private intent or experience. Recording supplies observations; inference and simulation interpret or model them, while actual sensory delivery requires its own interface. Brain2Agent? remains an open proposal.

Lookup labels: Brain2Agent; brain2agent; brain2agent?; brain-to-agent; brain to agent

Brain2Body

Connecting brain and body function; the 2 means to.

The numeral 2 is the preferred spelling. The programme includes body-to-brain feedback as well as outward pathways; the name does not establish complete biological translation or novelty of body–brain research.

Organizational level: whole system; network; local circuit; signal/feature

Lookup labels: brain2body; Brain2body; brain 2 body; brain-to-body; brain to body

Brain2Brain

Information links between two brain or neural systems.

The project's label for studying specified information links between brain or neural systems, including tutor–pupil comparisons.

Brain2Brain is the numeral-style public label for the project's existing brain-to-brain research direction. Each experiment must specify its signal, task and controls; the name itself establishes no shared thoughts, general skill transfer or successful biological teaching.

Organizational level: whole system; network; local circuit; signal/feature

Lookup labels: brain2brain; brain 2 brain; brain-to-brain; brain to brain

Brain2Motion

The proposed movement pathway connecting Brain2Body with Body2Action and the resulting measured motion, including the returning sensory feedback.

The label concerns the movement part of the wider programme, not every sensory/cognitive function. Brain/spinal/peripheral observations, muscle activity, body mechanics and external outcomes stay distinct. A decoded motion signal, simulation or lifelike animation is not a complete physiological explanation or validated skill transfer.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Lookup labels: Brain to motion

Brain2Spine

A movement command meets a spinal cord that already regulates reflexes, coordinates muscles and adapts through experience. Understanding that middle step helps explain how the same intention can produce different actions.

The pathway from brain activity into spinal processing: descending signals influence circuits that combine them with sensory input and the body's current state.

A scalp EEG recording does not identify every descending signal or measure the spinal circuit's hidden state. The physical spine remains read only. This label does not name an admitted stimulation interface.

Organizational level: whole system; network; signal/feature

Lookup labels: brain2spine; brain to spine; brain-to-spine

BrainNet

A network of specified links between brain or neural systems.

A working network metaphor for connected brain or neural systems within the programme.

BrainNet pictures specified links between brain or neural systems as a network. It is a programme metaphor; it establishes no validated universal neural network, uniquely coined field or available trademark.

Organizational level: whole system; network; local circuit; signal/feature

Lookup labels: brainnet; Brain Net

Choosing a model and deciding if it passes

Choosing settings and applying a stated pass rule using the validation data.

The frozen-transfer rule requires a prediction score at least 0.01 above every specified control. A failing candidate stays rejected rather than being given an easier success rule.

Lookup labels: Validation selection and admission; validation-selected; acceptance criterion; admission; failure criterion

Command sent to a modeled muscle

The input command that starts the modeled muscle's activation process.

The command is bounded and nonnegative here. It differs from the resulting activity state, recorded muscle electricity and instructions for a biological intervention.

Lookup labels: Muscle excitation; excitation; actuator command; muscle control

Communication route around an interruption

An interface carrying selected information around a modeled break or between stated recording and control channels.

A bridge can compensate without repairing tissue or supplying muscle energy. The virtual interruption does not establish a biological injury model or treatment.

Lookup labels: Digital bridge; compensatory bridge; brain–spinal bridge; BCI; brain-computer interface; virtual pathway interruption

Comparable roles in different organisms

Comparing roles such as reaching or support while keeping each organism's anatomy explicit.

Related roles do not imply identical neurons or muscles. A combined fictional body cannot replace the measured sources and checks needed for each real organism.

Lookup labels: Organism role and homology; organism adapter; limb role; functional coordinate; homology; biped/quadruped

Comparisons that test whether guidance matters

Matched replay uses the relevant recorded activity under matched task conditions. Stale or unmatched replay uses activity from a different time or condition. No-link and inactive or sham comparisons test whether the proposed information route adds benefit.

These are proposed comparison types, not new animal results. Shuffling disrupts timing; a yoked comparison gives matched help from another case. Guidance amount, timing and later pupil control need their own stated comparisons.

Lookup labels: Shuffled and yoked controls; shuffled-input control; time-shuffled; yoked signal comparison; Matched comparisons with altered information; matched replay; matched own-session activity; stale replay; unmatched replay; no-link; sham; inactive comparison

Completed time interval for counting events

A stated time interval groups events into one count. A half-open interval includes its start and excludes its end.

Inputs use completed intervals. The exact boundary rule, interval duration and timing must remain explicit rather than being guessed from the display.

Lookup labels: Completed time bin and half-open binning; bin; 20 ms bin; 50 ms bin; half-open; bin end; cadence

Computer processors, memory and file size

A CPU is a general processor, a GPU is another kind of parallel processor, and RAM is working memory. Bytes and GiB measure digital data size.

The training described in this source edition used four CPU threads; GPU acceleration was neither verified nor used. AI-assisted coding does not establish independent biological validation or an authorship credential.

Lookup labels: Computing hardware and data size; CPU; GPU; RAM; AMD; AI; Ryzen AI Max+395; GiB; byte; bytes

Correct-label rate and chance comparison

Accuracy is the proportion of labels chosen correctly. A confusion table shows which labels were mistaken for others.

A uniform guess among ten balanced classes has a 10% reference rate. Accuracy and label scores do not prove executed movement or confidence in biological truth.

Lookup labels: Classification accuracy and chance; accuracy; uniform chance; ten-class confusion; confusion matrix

Count-prediction improvement per recorded event

Bits per spike describes how much a count-prediction score improves over a defined baseline, divided by the number of recorded spikes.

The baseline uses the training mean rate. A positive result concerns estimated average counts, not newly recorded activity or verified stimulation instructions.

Lookup labels: Bits per spike; BPS; ΔBPS; bits/spike; likelihood gain

Data for learning, choosing and checking

Separate portions used to fit the model, choose its settings, and check the selected result.

The paper uses local time-ordered splits. If test results have already been inspected, later reuse cannot be described as another independent test.

Lookup labels: Training, validation and test partitions; train/validation/test; train/val/test; split; chronological 70/15/15

Data kept out of a stated step

Data reserved from training or from another specified development step.

Independence depends on what was actually excluded. A previously inspected case does not become a fresh test just because it is called held-out.

Lookup labels: Holdout; held-out; reserved window; independent evaluation

Describing motion and calculating what causes it

Describing a body's position and movement is one task; calculating how modeled forces change that movement is another.

The paper calls these kinematics and dynamics. A replayed pose and a muscle-driven trajectory are different results; neither alone proves physiological accuracy.

Lookup labels: Kinematics and dynamics; kinematics; dynamics; forward dynamics; numerical plant

Electrical recording in outer brain tissue

Electrodes in the brain's outer layer, called the cortex, record electrical signals outside the cells.

This differs from a scalp recording. It also does not observe every cell in a circuit; the source file defines which events and channels were recorded.

Lookup labels: Intracortical extracellular recording; intracortical; extracellular cortical events; Electrical recording inside the cortex; cortex

Error range based on validation examples

A range placed around a prediction using the errors seen on the data used for model selection.

The ranges have constant width for each output. They do not guarantee future coverage when examples overlap in time or their errors are related.

Lookup labels: Residual band; empirical residual bounds; validation residual quantile; 90% per-output residual band

Events grouped as one neural feature

The source recording process groups events into features that the model can count separately.

A group is not a verified lifelong identity for one neuron. It must stay separate from unassigned events and from the channel labels used in other recordings.

Lookup labels: Sorted neural unit; sorted unit; sorted cell; sorted feature

Expected event count and its fit score

A model estimates the average event count for given inputs, then a score checks how that estimate fits the recorded counts.

The paper uses Poisson negative log likelihood for that check. A predicted average may be fractional; a good count-fit score does not establish a complete biological spike model.

Lookup labels: Conditional count mean and Poisson likelihood; expected count mean; conditional count; Poisson; negative log likelihood; NLL

External movement model

The external part of the body model, constrained by independently measured surface movement and pose in a declared task and coordinate frame.

The motion-capture relay-composite retains the source views, clocks, calibration and coordinate frame for the declared task. The earlier upper/lower hand concept and the later synchronized left/right tripod arm proposal are different arrangements; each needs its own visibility, synchronization and error checks. The composite is derived. Hidden internal states, forces and intent are not uniquely measured. LiDAR and optical infrared tracking require distinct hardware, calibration and uncertainty qualification.

Extra correction and its scoring rule

A model can add a correction to an existing controller. A reward scores the result; command-change rate describes how quickly the input changes.

The formal names include residual policy, reinforcement learning and actuator slew. A simulator can score tracking or task completion without proving biological learning or correct hidden neural activity.

Lookup labels: Residual policy, reward and actuator slew; residual policy; reinforcement learning; RL; verifiable reward; actuator slew; task completion

File fingerprints

A checksum or hash is calculated from file bytes to detect changes or match a stated reference.

The named methods here include MD5 and SHA-256. A match checks bytes against that reference; it does not prove scientific truth or verify an entire file when only a range was acquired.

Lookup labels: Checksums and hashes; checksum; hash; MD5; SHA-256; source checksum; archive hash

Force in the modeled muscle and tendon

The force produced by the computer model for a muscle-tendon unit, or a force measured by a stated physical method.

The native test reads forces from the running simulator rather than an old lookup table. Those values are still simulated rather than isolated forces recorded in the source animal.

Lookup labels: Tendon force; muscle force; isolated muscle force

Form → function → action

The programme narrative follows structural constraints, the context-dependent processes they support, and independently measured actions or outcomes.

The three words organize the explanation; every molecular and neural level has its own structure and dynamics. Form informs and constrains function but does not determine the full process alone. Feedback returns outcomes to the system; this is not a one-way causal chain or a guaranteed one-to-one mapping.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Lookup labels: Form to function to action

Functional electrical stimulation

Electrical stimulation coordinated to produce a specified functional movement or task.

FES is the appropriate functional-actuation label for the proposed later assisted-body study. It requires separately qualified devices, control/safety limits and specialist-supervised research approval. This registry contains no stimulation settings or self-experiment instructions.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Lookup labels: FES

Source: Restoring cortical control of functional movement in a human with quadriplegia

Source identifier: bouton2016

DOI: 10.1038/nature17435

https://www.nature.com/articles/nature17435

Research precedent linking implanted cortical recording to a custom neuromuscular stimulation sleeve; not an EEG/OpenBCI or elite-skill result

Further training for the target task

Continuing model training so it predicts the desired target, using the relevant inputs and labels.

The paper calls this fine-tuning. Changing software weights is not evidence that an animal or person learned a biological skill.

Lookup labels: Fine-tuning; paired-output fine-tuning

Future testing and analysis of existing data

A future test checks later unseen outcomes under a fixed plan; an existing-data test analyzes outcomes already recorded.

The second source file was chosen before inspection, but its recording day is earlier than the original model's day. Choosing a file in advance does not make it future biological deployment.

Lookup labels: Prospective and retrospective evaluation; prospective; retrospective; prospective biological deployment

Gap between data portions

Leaving out an interval around a split boundary so overlapping windows do not cross into another portion.

The continuous pipeline uses a one-second gap and checks the input and target support. This protects the software split, not every assumption about the original measurement.

Lookup labels: Partition embargo; embargo; boundary embargo

Guide and learner

The tutor supplies guidance; the pupil is the learner whose later control is tested. An experienced guide helping a novice asks a different question from two novices practicing together.

The proposed comparisons include experienced-to-novice, novice-to-novice, individual practice and matched no-link conditions. Later pupil-alone testing removes tutor commands while retaining any bridge required to express movement.

Lookup labels: Tutor and pupil; biological tutor; artificial tutor; live tutoring; tutor withdrawal; experienced-to-novice tutor; novice-to-novice; individual practice

Hallucination

A model can predict a movement correctly while inventing the biological mechanism it claims produced it.

Invented or ungrounded model content presented as if it corresponded to reality. In programme claims, this includes fabricated biological structure, dynamics or ability presented as established biological fidelity.

Research on AlphaFold3 uses the term for generative-model errors; this programme also applies it to unsupported biological attributions. A disclosed approximation, hypothesis, counterfactual or unvalidated candidate is a different kind of statement, and missing evidence alone does not prove it false. Numerical error concerns the calculation; objective misspecification concerns what is being optimized; causal attribution error concerns the mechanism being credited. Those differences matter when deciding what a result supports. An ungrounded generated answer can happen to be correct without validating its claimed biological mechanism or reliability elsewhere. Informed predictions therefore still require independent performance evaluation. The rolling-dice analogy describes reliance on unverified claims, not literal randomness or equal odds. Calling every approximation a hallucination, or every connectome or simulation fictitious, would erase the distinction.

Scope: usage: Model-generated biological content and claims of biological fidelity.

Computational granularity: Model output and its interpretation

Lookup labels: hallucination; model hallucination

Help during current performance

Help from a controller, replay, tutor or interface that improves performance while it is present.

For example, a pupil can move successfully while following tutor commands. This does not show independent learning or tissue regeneration.

Lookup labels: Assistance; compensation; assisted performance

How a muscle’s pull turns a joint

To relate a muscle’s pulling force to joint turning, the model uses its geometry. Consistent work calculations also relate those forces to small coordinate changes.

The formal terms are moment arm and virtual work. Their numerical checks do not measure biological muscle strength; the human controller uses them to account for dependent shoulder movements.

Lookup labels: Moment arm and virtual work; moment arm; virtual-work projection; dependent shoulder-girdle load

How many labeled examples were supplied

The labeled examples given to train or choose a model, including examples from prior recordings.

The first 5% counts new training labels only. Validation labels, earlier-source labels and extra neural samples are additional information.

Lookup labels: Label budget and supervision; label budget; paired supervision; adaptation labels; validation labels; source-task supervision

How scores from several outputs are combined

Scores can be averaged across individual outputs or first combined within defined groups. In this entry, macro describes the averaging and weighting rule, not the size of a body part, the scope of a nervous system, or the level of detail in a computer model.

The sensory result gives equal weight to five physical output groups. This avoids letting a larger group dominate but does not make their different physical units interchangeable.

Lookup labels: Per-output and group-macro scoring; per-output; group-macro; macro R^2

How uncertain the result remains

The remaining doubt about a prediction, including why it could be wrong or change.

Noise in the instrument, errors in the model, changing conditions and missing observations are different causes. The current ranges do not separate all of them.

Lookup labels: Prediction uncertainty; uncertainty; model uncertainty; measurement noise; drift

Human arm model used for playback

MyoArm is the human arm model used for the displayed muscle-driven replay.

The replay uses 63 modeled muscles and two mapped joint goals. Those quantities are simulated, and the macaque velocity mapping does not prove neural or skill transfer into a human body.

Lookup labels: MyoArm; human MyoArm; MyoArm right

Internal musculoskeletal model

The internal anatomy part of the body model: specified bones, joints and soft tissues, assembled from appropriate registered anatomical measurements.

Proposed complementary X-ray/CT and MRI inform different structures. Anatomy does not directly measure activation, force, material properties or neural function. Wētā FX provides visual/simulation inspiration, not a verified identical acquisition pipeline.

Interoception

Information about the body's internal condition, such as signals related to heartbeat or breathing.

Scope: Sensing the body's internal condition differs from sensing a limb's position and movement. Internal sensing remains a research objective; this prototype does not provide a fully measured body-wide internal signal.

Lookup labels: interoception; interoceptive signal; internal sensations

Interoceptomics

How does the body sense a heartbeat or its breathing, and how are the nerve connections involved arranged? Interoceptomics brings those questions together in a proposed study, relating observations of internal-body sensing to maps of the connections carrying or processing that information.

A proposed field combining interoception and connectomics to study internal-body sensing and its relationships with neural connectivity across scales.

The programme uses Interoceptomics as an umbrella for this proposed research. Giving the work a name does not establish a validated comprehensive map or a recognized external standard.

Scope: macro: Whole-body, organ and network relationships relevant to internal-body sensing; micro: Local signals, cells, circuits and interfaces relevant to those relationships.

Organizational level: whole system; network; organ; local circuit; cell; signal/feature

Lookup labels: interoceptomics

Keeping each source separate

Keeping track of which recording, channel, time and physical setting a value belongs to.

A channel numbered 12 in one day is not automatically the same cell as channel 12 on another day. A software label also cannot prove where supplied data physically came from.

Lookup labels: Source conditioning; source-conditioned; source-specific; recording-specific

Keeping each trial together

Assigning an entire trial to one data portion instead of dividing it among portions.

This prevents pieces of the same trial from being shared between the relevant splits. It does not remove prior inspection or other source limitations.

Lookup labels: Whole-trial separation; whole-trial-separated; successful trial; trial partition

Keeping up now and computing for later

Real-time computation keeps pace with incoming data; offline computation can take longer and be played back later.

Native planning takes about 4.52 seconds for a 20 ms simulated step and is offline. Smooth cached playback does not prove live planning or biological control.

Lookup labels: Real-time and offline execution; real-time; offline; paced playback; cached reference

Kind of measurement

What a method records, such as electrical spikes, scalp voltage, muscle electrical activity or a brain-scan signal.

These signals measure different things. They cannot be treated as interchangeable numbers merely because all of them relate to the nervous system.

Lookup labels: Signal modality; modality; observation process

Large Movement Model

The shorthand “NeuroTranslate with MoCap MyoSim = Large Movement Model” names the proposed combination of NeuroTranslate, MoCap ReComp and MyoSim Replay. It brings a learned mapping, measured movement and virtual mechanics into one architecture. The equals sign is not a mathematical identity or evidence of a trained model. EEG, EMG, measured movement, virtual mechanics and the learner retain their distinct roles.

The proposed movement-focused learned model that brings neural structure and functional observations together with internal musculoskeletal anatomy and independently measured external movement in a declared task.

The programme model-family name, analogous in ambition to large language model. It does not replace accurate technical architecture descriptions or mean that every DNN is a movement model. Scans and capture alone do not train a model. Parameter count, scale, generalization and nervous-system fidelity require evidence. System 3 thinking remains the partnership framing.

Lookup labels: large movement models; LMM; Large movement model

Lasting change in the nervous system

Changes in the nervous system that can contribute to learning or adaptation.

Updating software weights or controller gains is a different event. The practice interface in the computer model does not measure synaptic change in a living organism.

Lookup labels: Biological plasticity; synaptic plasticity; biological learning

Learning before the final task

Training first on another objective, such as predicting a later neural count, before training for a paired movement target.

The paper calls this pretraining. Earlier source-task labels remain part of the information the model received, even when only new-setting labels are counted in a small budget.

Lookup labels: Pretraining; neural next-bin pretraining

Mapping movement onto another body model

Turn a specified movement quantity, such as fingertip velocity components, into movement goals for a particular computer body. The mapping depends on that body and the chosen task.

Mapping specified movement quantities into goals for a particular body model, with the source quantities, target coordinates and transformation declared. This does not by itself reconstruct full posture or demonstrate biological transfer.

The retained source-specific adapter takes two macaque fingertip velocity components as its source and produces human-model shoulder and elbow goals. Their signs carry direction; scalar speed would discard it. These are goals for that model and task, so the mapping establishes neither cross-species skill transfer nor verified biological function.

Lookup labels: Retargeting; body adapter; organism-specific body adapter; joint goal

Meme-jacking

Borrow the recognition; earn the explanation. A familiar image should help the audience see the scientific point before the script develops it.

Use a recognizable meme or cultural reference to make a scientific point immediately legible, drawing on familiarity already earned by others.

Memification hijacking is the user's expansion. References may come from any online culture, not only films. Apply one purposeful reference per teaching concept without crowding formal methods or labels. Recognition does not grant reuse rights, endorsement, proven comprehension or scientific validity. Preserve the original Meme-ification history.

Lookup labels: meme-ification; memeification; mimification; Meme-ification; meme-jacking; meme jacking; memification hijacking

Meme-ification/memeification are spelling forms; mimification is retained only as the corrected historical transcription.

Mixed-source animal and added tissue

A chimera has biological contributions from different sources. A graft is added tissue; when it crosses species, it is a xenotransplant. An organoid is an organized tissue model.

Survival, physical connection, activity and useful function are separate questions. The proposed comparisons do not establish a humanized whole brain or an implemented treatment.

Lookup labels: Chimera, graft and organoid; chimera; chimerism; graft; organoid; host–graft; xenotransplant; xenograft

Model that chooses a label

A statistical model that assigns scores to categories, such as the ten attempted-finger labels.

The selected human model uses logistic classification, but its scores do not establish validated confidence in biological truth. Its reported accuracy comes from development data that have already been seen.

Lookup labels: Logistic classifier; logistic regression; classifier; classification; logistic

Model weights kept fixed

Keeping the fitted numbers in a model unchanged during a defined comparison.

The tested transfer trains a connector while retaining the older main model. Fixed weights make the comparison clear; they do not prove biological sameness across days.

Lookup labels: Frozen parameters; frozen trunk; frozen readout; frozen checkpoint

Modeled route from one muscle attachment to another

The route and length assigned to a muscle and tendon as the model's joints move.

The native arm retains 39 named routes. Their modeled lengths differ from direct measurements of living tissue, muscle commands and force.

Lookup labels: Muscle-tendon path; musculotendon length; modeled muscle length; geometric path

Models that convert inputs or predict outputs

One model can turn an input into a representation; another can use available inputs to predict a named result.

The paper calls these encoders and decoders. Predicting counts from mechanics or mechanics from counts does not by itself show the biological cause of either.

Lookup labels: Encoder and decoder; encoding; decoding; readout; decode; encoder; decoder

MoCap ReComp

MoCap ReComp is Motion-Capture Relay-Composite: the proposed measurement component that brings views of the same movement into a common physical frame. It should preserve which view supplied each point, so the joined map can be compared with its original observations.

The proposal starts with two dot maps of the same moving body: one from above, one from below. Align their time and position in a common physical coordinate frame, then join them into one motion map while retaining the source of each point.

A proposed method that creates two dot maps (3D point clouds) of the same moving body, one captured from above and one from below, aligns them in time and a common physical coordinate frame, then stitches them into one composite motion map while retaining each point's source.

To compare the same movement from above and below, the proposal must align both streams in time and a common physical coordinate frame while retaining each observation's source. Bidirectional denotes those opposing measurements, not neural communication or causal direction. LiDAR ranging and infrared optical motion capture are proposed alternatives; marker triangulation and LiDAR point clouds are different measurements, so the actual modality and camera/ranging geometry must be qualified. The maze geometry calls for measurements perpendicular to the floor, with viewing paths from both sides through material intended to be clear. Transmission and material effects need testing; concentrated and distributed arrays remain alternatives. Hardware, coverage, timing, calibration, spatial alignment, occlusion, motion error and latency require validation against real references. The joined map is a derived record, not a direct observation of hidden joints, tissue mechanics, muscle commands or neural function. Neither successful implementation, complete body reconstruction nor material transparency has been established by the name. The single vertical camera illustration remains schematic.

Geometry: Declare the top/bottom sensor positions, measured object, coordinate frame, viewing/illumination paths and coverage. Preserve the requested

perpendicular-to-maze-floor geometry; document deviations rather than inferring them from an illustration.

Modality: Declare LiDAR ranging or infrared optical motion capture, including marker-based/markerless method, camera count, triangulation geometry, material path and actual calibration. No complete reconstruction or physiological writing follows from opposing views.

Timing: Record source timestamps, clocks, acquisition cadence, synchronization error and any interpolation, offline processing or end-to-end latency. No real-time capability is assumed.

Calibration and registration: Preserve calibration data and transformations placing both streams into a common physical frame; measure registration error and keep original observations and reconstruction steps traceable.

Limits: Record overlap, occlusion, missing observations, material effects, uncertainty and failure cases. Opposing measurements alone do not prove complete 3D pose, forces or biological function.

Physical scale: Declare measured distance ranges and units for an actual acquisition; no supported numeric range or physical unit assignment is presently available for this proposal.

Statistical aggregation: Declare any chosen evaluation grouping/averaging rule separately from physical scale and computational granularity.

References and acceptance: Use the Reference Comparison Protocol and unfilled record to identify the real-reference series needed for each acquisition component and the assembled map. Its provenance, rights, organism-stage, physical-unit, modality, clock, calibration, occlusion, contact, independent held-out, error, uncertainty and acceptance fields remain unfilled and HOLD. It supplies no measured series or apparatus validation. The unchanged programme reference criteria and every stronger original requirement govern their respective claims; no acceptance threshold is introduced or relaxed.

Outputs: Retain the top dot map, bottom dot map and joined map as separate traceable outputs. Their timestamps and common-frame transformations must be explicit; do not fill unobserved regions as though they were measured.

Scope: proposed application: Complementary top/bottom motion measurements of a body moving through the maze; concentrated coordinated platforms and distributed arrays remain alternatives; measurement boundary: Observed distances or geometry remain distinct from registered/stitched composites and any model-estimated pose or musculoskeletal quantity.

Organizational level: whole system; signal/feature

Computational granularity: top dot map (3D point cloud); bottom dot map (3D point cloud); time alignment and spatial registration; joined composite motion sequence

Lookup labels: motion capture composite relay; motion-capture composite relay; motion capture composite; bidirectional composite motion capturing using LiDAR; Motion capture composite relay; motion capture relay composite; motion-capture relay composite; bidirectional LiDAR; opposed-view infrared motion capture; bidirectional infrared; motion-capture relay-composite; Motion-Capture Relay-Composite

Historical spellings name this same proposed capture method. Relay here means passing captured measurements onward; the separate agent-coordination Relay protocol does not describe motion capture.

Movement brain area

A brain area involved in movement. The paper uses the names motor cortex, MC and M1 for relevant source recordings.

The signals help predict recorded movement-related targets. Knowing the brain area does not identify matching cells across recordings or reveal a whole-body command pathway.

Lookup labels: Motor cortex; M1; MC; primary motor cortex

Muscle electrical recording

Electrical activity recorded from muscles; the method is called EMG.

Muscle electrical activity differs from force and from the computer model's command and activity states. The trained targets here are rescaled signal summaries.

Lookup labels: Electromyography; EMG

MyoSim Replay

MyoSim Replay is the proposed MusculoSkeletal Simulation Physics-Engine. Declared anatomy, mechanical parameters and control inputs would produce movement predictions to compare with independent measurements. The external MyoSim body-model collection is a separate resource. Simulation replay is not a measured biological replay.

The proposed digital model would calculate how muscles and the skeleton move and exert forces, with real measurements used to check its predictions.

The proposed physics-based simulation component that models specified muscles, bones, joints and their mechanical interactions, using declared anatomy, parameters, forces and contact rules, with predictions tested against independent movement and other applicable physical measurements.

In the proposed five-component arrangement, declared anatomy, muscle/control inputs and physical context would drive a simulation of movement and forces. The motion-capture relay-composite would provide traceable measured movement to compare with those predictions. That comparison needs matching timing, coordinates, units and calibration, and tests that preserve anatomical and physical constraints, uncertainty, held-out cases, baseline controls, a series of real references and independent acceptance. Movement geometry alone does not measure muscle activation, internal forces, hidden neural state or subjective experience; force, activation and causal claims need additional independent references. Visual effects (VFX) rendering shows the model but cannot validate its mechanics. This remains a proposed simulation component, not a newly demonstrated implementation or external scientific field. Approximate simulation and biological virtualization verified within declared error

bounds remain distinct; this component establishes neither universal precision nor whole-body completeness.

Scope: programme role: Virtual-body mechanics component; initially constrained hand and foot tasks; inputs and outputs: Declared anatomical parameters, muscle/control inputs and physical context produce simulated movement and mechanical quantities. Independently acquired motion and other necessary observables remain references, not simulator ground truth.

Organizational level: whole system; organ

Computational granularity: musculoskeletal model; mechanical solver; measurement-to-prediction comparison

Lookup labels: musculoskeletal simulation physics engine; musculoskeletal-simulation physics engine; musculoskeletal simulation; musculoskeletal physics model; musculoskeletal-simulation physics-engine; MusculoSkeletal Simulation Physics-Engine

Names used for source subjects

Indy, Han and Monkey L label macaque sources; N1 labels the human finger source.

These labels keep different subjects separate. Similar brain-region or task names do not make them one synchronized recording.

Lookup labels: Source subject labels; Indy; Han; Monkey L; N1

Nerves bringing body information in

Afferent paths carry body-related information toward the brain and spinal cord, together called the central nervous system.

For example, a limb-related sensory signal can travel inward. The source-specific cortical predictions do not measure this whole path or make missing nerve activity observed.

Lookup labels: afferent; afferent pathway; body-to-CNS; CNS; central nervous system

Nerves carrying commands out

Efferent paths carry outputs from the brain and spinal cord toward the body, including muscles and organs.

A prediction of movement is not a recording of that outward pathway. The prototype does not supply a synchronized voluntary sequence through spinal cord, nerves and muscles.

Lookup labels: efferent; efferent pathway; CNS-to-body

Neural decoding

Estimating declared information such as intended action from recorded neural observations using a specified model.

Decode accuracy, channels, source identity, task and held-out domain must be demonstrated. Brain-only control requires exclusion of covert muscle, eye and motion signals; decoding does not write signals into the brain.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: Restoring cortical control of functional movement in a human with quadriplegia

Source identifier: bouton2016

DOI: 10.1038/nature17435

https://www.nature.com/articles/nature17435

Research precedent linking implanted cortical recording to a custom neuromuscular stimulation sleeve; not an EEG/OpenBCI or elite-skill result

Source: A brain-computer interface that evokes tactile sensations improves robotic arm control

Source identifier: flesher2021

DOI: 10.1126/science.abd0380

https://pubmed.ncbi.nlm.nih.gov/34016775/

Research precedent for motor decoding plus sensory feedback through intracortical microstimulation; invasive and not a consumer EEG capability

Neural encoding

Mapping declared information to a representation in neural activity; in a sensory-interface design, constructing candidate stimulation patterns for specified feedback.

A computational encoder is distinct from a qualified stimulation device and evidence that a participant receives the intended sensation. The term does not mean arbitrary thought writing or skill upload.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: A brain-computer interface that evokes tactile sensations improves robotic arm control

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Neural Replay

Revisiting a stored sequence of neural samples with its stated timing and channel order.

This playback is not a live recording or instructions for stimulating a nervous system. A displayed arm movement does not show what the source animal's muscles actually did.

Lookup labels: Neural replay; recorded replay; archived replay; cached playback; Playing back a recording

NeuroArcher

A proposed archery game that studies coordinated use of both hands and arms through physical motion control, brain-signal control of a digital body, and a separately admitted later assisted-body experiment.

The near-term prop is a toy bow with small suction-cup tips and an over-arm arrow pouch. Target/creature choice is unresolved. Simulated performance does not establish muscle stimulation or retained biological archery skill.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Lookup labels: Neuro Archer

NeuroBoxer

A proposed boxing game that compares arm-motion control, brain-signal control of a digital body, and a separately admitted later assisted-body experiment.

Prime-era Mike Tyson is the requested virtual opponent and meme reference. A game, recorded prediction, stimulation-assisted motion and retained biological skill learning need different evidence. No actual opponent, head contact, finished game or stimulation capability is claimed.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Lookup labels: Neuro Boxer

NeuroConnectome

A map of specified neural connections used to constrain a model of the nervous system.

Connectome is the established term. The user's NeuroConnectome label does not establish complete coverage or identify functional dynamics, state, plasticity, sensory context or unique thought from wiring alone. Declare species, anatomical coverage, resolution and provenance.

Lookup labels: Neuroconnectome

Neuromuscular electrical stimulation

Electrical stimulation intended to elicit muscle contraction through the relevant neuromuscular pathways.

NMES is a writing/actuation intervention, whereas EMG records muscle activity. EMG electrodes or an OpenBCI recording board do not by themselves provide qualified stimulation. Neither contraction nor imposed movement demonstrates retained skill.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Lookup labels: NMES; Electrical muscle stimulation; EMS

Source: Restoring cortical control of functional movement in a human with quadriplegia

Source identifier: bouton2016

DOI: 10.1038/nature17435

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Research precedent linking implanted cortical recording to a custom neuromuscular stimulation sleeve; not an EEG/OpenBCI or elite-skill result

Neuroscience recording files

NWB stores neuroscience data and descriptions. In the paired source files here, SET holds recording descriptions and FDT holds binary samples.

A stale unit in the NWB muscle source must not override verified processing. SET headers match source hashes; FDT previews have checked local byte ranges rather than full verification of each upstream file.

Lookup labels: Neuroscience recording files; Neurodata Without Borders file; NWB; Recording header and sample file; SET and FDT files; SET; FDT; EEGLAB header; byte-range preview

NeuroTransfer

Specify what would move between which systems, then test what the recipient can do with and without assistance.

The proposed transfer of specified task-relevant information or a mapping between declared systems, with assistance and retained learning evaluated separately.

The label does not establish a successful biological transfer or retained skill. A weight copy, recorded replay, guided movement and independently retained learning answer different questions. Each proposed interface needs its own source, target, task, controls, rights and qualified outcome.

Lookup labels: neurotransfer

NeuroTranslate

Brain2Body Neural Activity is the long-form component label paired with NeuroTranslate in this proposed architecture. Recorded activity is an observation. NeuroTranslate names the mapping operation that could relate specified observations to movement, or reconstruct specified recorded features in the reverse direction. The observation, learned relationship and predicted value remain different things.

State the observations, the target and the task, then test how well the mapping works.

The proposed operation of mapping specified neural and body observations between declared inputs and targets in a stated task.

A model could predict movement from specified observations, or reconstruct specified recorded features in the reverse direction. Each direction needs its own test in the stated task. Neither establishes a unique hidden state, sensory delivery or biological skill transfer. NeuroTranslate names this proposed mapping operation; NeuroTranslation names the field and NeuroTranslator the implemented software.

Lookup labels: neurotranslate; NeuroTranslate; Brain2Body Neural Activity

NeuroTranslation

How could a particular neural recording be related to a defined movement task? NeuroTranslation is the proposed study of mapping nerve signals and the patterns that describe them between clearly defined situations like these. Every mapping must state where its inputs come from, what its outputs mean, and how it is tested.

A proposed field for translating or mapping neural signals and representations between explicitly defined source and target domains, with stated context, scale, tests and limitations.

Each connection must be assessed against its named source and stated test. The project has not demonstrated a universal brain language or an uploaded skill.

Scope: macro: Relationships between neural systems, body systems and defined tasks or organisms; micro: Mappings involving identified local signals, channels, features and circuit-level representations.

Organizational level: whole system; network; organ; local circuit; signal/feature

Lookup labels: neurotranslation; neural translation; Connecting neural recordings to defined tasks; Neurotranslation

NeuroTranslation Network

A plan for connecting more neural and body components while keeping the origin and meaning of each value clear.

The proposed network would connect neural and body components while carrying the origin and meaning of each value through the mapping. Each link could then be examined in its own recording context. Body-wide translation and biological teaching remain unproved.

Lookup labels: Source-Conditioned Neurotranslation Architecture; SCNA; Neurotranslation Network

NeuroTranslator

The software that joins particular recordings, prediction models, playback and computer-body checks.

Predictions and playback can be checked against the particular recordings they use. That is the scope of the implemented NeuroTranslator; neither a universal brain language nor skill upload has been demonstrated.

Lookup labels: Neurotranslation Engine; NTE; Neurotranslator

Numerical arrays, saved arrays and prediction

NumPy provides tools for numerical arrays. NPZ saves arrays; inference means running a trained model to produce an output.

Portable execution still needs the correct processing and source identity. Matching saved-model outputs does not establish success on new organisms or biological skills.

Lookup labels: NumPy inference and NPZ archive; NumPy; NPZ; portable inference; inference

Numerical checks on allowed movement

Checks that the computer model stays defined and within its stated joint limits.

Joint-limit penetration means going past an angular bound, not entering tissue. The 0.001 rad tolerance is a numerical rule, not a certificate of correct neural commands.

Lookup labels: Mechanical feasibility and anatomical tolerance; anatomical limit; limit penetration; finite state; joint range; numerical tolerance

Numerical software and its data file

MATLAB is numerical-computing software, and MAT is a format for its structured data.

The second Indy recording uses the original MAT source. The format does not prove matching cells, past-only source processing or complete observation.

Lookup labels: MATLAB and MAT file; MATLAB; MAT

Observe it, connect it, keep it useful

Observation burden, prediction on excluded cases and lasting task benefit need separate measurements. A clear picture does not answer those experimental questions.

Lookup labels: Access it, Ground it, Keep it; access; grounding; stability; signal stability

One multiplier for modeled lengths

A single number chosen to reduce differences between the model's muscle lengths and the source-derived length references.

The number is fitted on training poses and checked on distinct observed poses. It is not complete body calibration; no movement-control experiment is run on the scaled candidate.

Lookup labels: Scalar length fit; global length factor; uniform factor; a*; scalar bridge; recording length-fit candidate; SDK-scaled candidate

One recording period

One experiment or recording period with its own subject, channels, timing and task.

Two recording days, or recordings from different animals, do not form one observed nervous system simply because the files are loaded together.

Lookup labels: Recording and session; source recording; session; recording family

Open body-model collection

MyoSim supplies computer-body models with muscles and bones, including models used in this project.

The collection is earlier work, not a new project invention. Availability of an arm model does not establish matching macaque and human muscles or neural commands.

Lookup labels: MyoSim, MyoSuite and MyoArm; MyoSim

Open muscle-control task tools

MyoSuite provides tasks and tools for experiments with muscle-driven computer-body control.

It is earlier software reused here. Success on a simulated task is not proof of biological learning or a recovered neural command.

Lookup labels: MyoSuite

Other published models of neural recordings

NDT2 learns patterns across recordings made inside the cortex, the brain's outer layer. POYO handles differing sets of recorded neural features by representing events and learning relationships between them.

The paper cites these as earlier work. It does not claim to have implemented them or use their existence as proof of universal transfer or matching cells.

Lookup labels: Other published models of neural recordings; NDT2; large-scale neural pretraining; Published model for varying recorded populations; POYO; neural-event tokenization; cross-attention bottleneck

Passing information and gaining a lasting skill

Receiving useful task information differs from acquiring a durable ability that works independently in new conditions.

A short software label budget is not measured biological practice time. Sewing, cube solving and breakdancing remain future examples without matched skill recordings here.

Lookup labels: Task information transfer and skill acquisition; task transfer; skill transfer; skill upload; compressed acquisition; learning speed; learning; physical skill learning; skill acquisition

Past-only inputs and processing that uses later data

The local model uses completed past inputs. Some earlier source processing may still have used later samples.

Past-only indexing cannot reverse earlier filtering or rescaling that used the whole recording. It does not certify the full measurement-to-output delay in a living system.

Lookup labels: Causal indexing and acausal preprocessing; causal; algorithmic support; acausal filtering; offline interpolation; upstream filtering delay

Peripheral nerve electrical recording

A method for recording electrical signals associated with peripheral nerves, called ENG.

The human preview contains aggregate electrical observations. It does not provide separately identified nerve-cell spike trains or a complete voluntary command sequence.

Lookup labels: Electroneurography; ENG; surface ENG; peripheral recording

Placing body parts from joint positions

Calculating where body parts and muscle routes lie when joint positions are given.

The formal name is forward kinematics, or FK. It places geometry; it does not calculate muscle-driven motion through time or recover muscle commands.

Lookup labels: Forward kinematics; FK; kinematic replay

Planned commands, corrections and constant input

A planned command is feedforward; adjusting it from current model information is feedback; a constant muscle input is tonic control.

The human correction can change the bounded command goal. Recorded movement stays a scoring reference, and the native correction candidates remain rejected against constant-input comparisons.

Lookup labels: Feedforward, feedback and tonic control; feedforward; feedback; corrected controller; tonic excitation; constant tonic

Prediction score compared with a mean-value guess

R^2 compares squared prediction errors with a constant mean-value reference. Higher values usually indicate better prediction on the stated data.

The score can be negative. Different recordings, target processing and ways of combining outputs prevent treating all R^2 values as directly comparable.

Lookup labels: Coefficient of determination; R^2; R2; R^2; variance explained

Processing within spinal and peripheral pathways

Body-related information can be processed locally in spinal and peripheral structures rather than passing through one simple unchanged wire.

A cortex-only prediction does not observe those local processes. The separate recordings and computer-body checks do not establish a complete measured brain-to-body causal chain.

Lookup labels: local spinal processing; local peripheral processing; spinal/peripheral processing

Programmable neurobiology

The proposed ability to specify a bounded nervous-system or sensorimotor outcome, select a qualified intervention, predict its effects, measure what happens and adapt within a validated closed loop.

Scope: Brain, spinal, peripheral, muscle, mechanical and sensory feedback roles need separate evidence. Bidirectional prediction is different from safe sensory/action writing. An intermediary may improve a mapping without making it one-to-one. Arbitrary skill upload or universal disease/disability repair is not demonstrated.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Programmable synthetic biology

Engineering specified biological behaviours by combining design, context-dependent molecular prediction, experimental measurement and iterative correction.

DNA/regulation, RNA/cell state and protein/interactions are explanatory layers, not a sufficient literal three-model recipe. Metabolism, feedback, environment and mea-

sured function matter. AlphaGenome and AlphaFold do not jointly certify whole-cell programmability; AlphaTranscriptome is a desired capability placeholder here.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Lookup labels: Programmable molecular biology

Public reaching recordings

Public files pairing cortical electrical events with reaching behavior. The sources use names such as RTT and MC_RTT.

The original model's release and the separately acquired Indy day have different processing. Their dates and target definitions must remain explicit.

Lookup labels: Reaching source archive; RTT; MC_RTT; MC_RTT; O'Doherty reaching archive; Sabes reaching corpus

Public source records and document links

Repositories such as DANDI and OpenNeuro identify source versions. PDF is a document format, and a URL gives a resource location.

Source licenses and credit must be retained. A local paper draft or public input dataset does not mean the project's new code is published, licensed or submitted to arXiv or a journal.

Lookup labels: Public records, document files and links; DANDI; OpenNeuro; PDF; URL; arXiv; public deposition; license; attribution

Published models for brain scans and scalp recordings

BrainLM learns from fMRI brain-scan data. Neuro-GPT combines an EEG input model with a model of changes over time and is discussed for motor imagery.

These are earlier methods for different measurements. The project does not claim to have trained them, make scans and electrical spikes interchangeable, or upload skills.

Lookup labels: Published models for brain scans and scalp recordings; BrainLM; Published model of scalp signals; Neuro-GPT; NeuroGPT

Published neural-model comparison tasks

The Neural Latents Benchmark, or NLB, provides shared tasks for comparing models of neural recordings.

It informs the choice of comparisons. The paper's own local scores are not official NLB benchmark scores.

Lookup labels: Neural Latents Benchmark; NLB

Published tests of changing recordings

FALCON is a published benchmark concerned with neural decoding when recordings change and recalibration is limited.

The paper uses related source recordings and its own data splits. Its local muscle-signal scores are not official FALCON scores.

Lookup labels: FALCON benchmark; FALCON; FALCON M1; FALCON M1-A

Ranges that need statistical assumptions

A confidence interval gives a range for a quantity being estimated. A conformal prediction range tries to contain a future actual result at a stated frequency when its required assumptions hold.

The transfer figure shows one fixed run. The paper's ranges are based on earlier prediction errors, with related errors across time; they do not guarantee how often future true values will fall inside.

Lookup labels: Ranges that need statistical assumptions; Confidence interval and conformal coverage; confidence interval; repeated-seed interval; Prediction range with tested coverage; conformal; conformal coverage; prospective-conformal; distribution-free coverage

Recent-signal model

A model component that keeps a learned memory of recent inputs as it predicts the next result.

Its formal name is gated recurrent unit, or GRU. This established method uses completed input history; its memory does not establish a universal neural language.

Lookup labels: Gated recurrent unit; GRU; compact recurrent model; recurrent network

Record of what was observed

A record showing which channels or intervals were actually measured and usable.

Missing observations stay missing; they do not become zeros. The second Indy source does not supply separate observation intervals for every recorded unit.

Lookup labels: Observation mask; observation interval; invalid support; missing observation

Recorded neural events

A spike is an electrical event identified in a recording. A spike train is the sequence of those events over time.

A recorded event count is a whole number tied to one channel and time interval. A model's average expected count can be a fraction and is a different kind of value.

Lookup labels: Spike and spike train; neural event; spike count; spike train

Records linking files, checks and claims

A manifest lists specified files and identities; a receipt records checks; an evidence matrix links a claim to its supporting record.

These records also keep failures and withdrawn results visible. A catalog listing does not show that every dataset was acquired or used to train a model.

Lookup labels: Manifest, receipt and evidence matrix; manifest; source manifest; audit receipt; processing receipt; evidence matrix; withdrawal record

Relay

Pass focused work between agents, then check their results before accepting it.

A manual coordination protocol that passes bounded work and verified evidence between specialist sessions, with one integration owner and an independent acceptance reviewer. Each session follows the currently authorized model, effort and speed routing.

A specialist can work on a bounded question while an independent reviewer checks its evidence before the integration owner accepts it. This is a project coordination workflow, distinct from an established external standard or native Ultra mode. Efficiency and quality claims apply only to tested tasks; universal performance equivalence and a 50 percent weekly saving remain undemonstrated.

Computational granularity: finite delegated tasks and evidence artifacts

Lookup labels: relay; manual Relay protocol; Relay coordination

Rescaled muscle signal

A processed summary of muscle electrical activity put onto a relative scale.

The 16 targets have no physical voltage unit. A stale millivolt label in the source does not make them measured voltage, force or muscle activation.

Lookup labels: Normalized EMG envelope; EMG envelope; quantile normalization; dimensionless EMG

Rescaling using training data only

Learning a rescaling rule from the training portion, then applying the same rule to later values.

This avoids using local test values to fit that rule. It cannot undo source processing that already used future samples or the whole recording.

Lookup labels: Training-only normalization; scaler; target statistics; normalization

Response after an electrical stimulus

An electrical recording made while observing the response to a stimulus.

The human preview is a stimulus-response experiment, not voluntary movement. The stimulus itself can also leave an unwanted signal in the recording; that is an artifact.

Lookup labels: Evoked potential and stimulus artifact; evoked stimulation; evoked recording

Results already seen during development

Results that have been inspected and can therefore influence later decisions.

The human 90% result and local motor test windows have this limitation. They can be reported, but their later reuse is not a new independent test.

Lookup labels: Development evidence; development holdout; pilot; descriptive local test; inspected test

Reusing learning in another recording

Trying to use information learned in one setting to improve performance in another.

The comparison must state the new-setting labels and the controls. The reported frozen motor transfer and unadjusted human cross-day result do not establish successful biological or cross-species transfer.

Lookup labels: Cross-recording transfer; transfer; cross-day transfer; cross-species transfer

Routing a muscle around a modeled obstacle

A rule bends a modeled muscle route around an obstacle. One obstacle shape, a torus, is shaped like a doughnut.

The correction restricts three elbow wrapping rules to a stated attachment segment. It removes reproduced route errors without deleting a muscle or claiming a new anatomical measurement.

Lookup labels: Muscle wrapping and torus; wrap; wrapping route; torus; ELBOW_ULNA; ELBOW_ULNA; off-torus contact

Saved model and matching-output check

A checkpoint saves a model's state. A parity check asks whether another execution produces the specified matching output.

Repeating a result needs the same support, ordering, conversions, data splits, settings and body-model version. Passing software checks does not certify biological validity.

Lookup labels: Checkpoint and parity; checkpoint; replay/checkpoint parity; model parity; reproduction

Scalp electrical recording

Electrodes on the scalp record electrical potential; the method is called EEG.

This is an aggregate electrical signal, not a list of individual cortical spikes. The project's paired preview follows electrical stimulation rather than voluntary motor training.

Lookup labels: Electroencephalography; EEG; scalp potentials

Settings needed to repeat a run

The numbers and choices fixing one training run's starting point and procedure.

One fixed run can be reproduced. It does not show how results vary across repeated starts, animals or future recording days.

Lookup labels: Random seed and configuration; seed; single-seed; fixed protocol

Shared internal model

A model can learn an internal pattern and reuse that pattern when making several predictions.

The paper calls the pattern a latent representation and the reusable model a backbone or trunk. The tested shared model and frozen-transfer candidates were rejected.

Lookup labels: Latent representation and backbone; latent model; shared latent; backbone; trunk; shared parameters

Simple weighted prediction model

A linear model that combines inputs using fitted weights while discouraging excessively large weights.

The formal method is ridge regression. It provides a meaningful comparison: a more complex model must add value rather than merely produce a plausible display.

Lookup labels: Ridge regression; ridge; linear control; linear model

Simplicity principle

Introduce one concept or element at a time, using separate slides or successive reveals according to what makes the relationship easiest to understand.

Introduce each new element on its own, then show how the elements connect. Use separate slides or successive reveals to make the logical relationship visible. Simplicity cannot remove scientific qualifications, evidence, source rights or required methods. Actual comprehension and exported readability still need review.

Small connector model

A smaller fitted mapping that connects one recording to an existing model or interface.

The connector can change while the older model stays fixed. That does not establish matching neurons or useful transfer without new-setting calibration.

Lookup labels: Adapter; linear adapter; recording adapter

Software that calculates a computer body's motion

MuJoCo calculates how a modeled body moves under forces and control inputs.

It is established software reused here. Its availability does not prove that the chosen body model or mapping matches a particular animal's physiology.

Lookup labels: MuJoCo; dynamics engine

Spine2Body

A muscle signal becomes a movement through tendons, joints, contact and load. Measure those steps together, and a failed reach becomes a specific question about activation, coordination or mechanics.

The pathway through which spinal output reaches peripheral motor nerves and muscles, producing action through the body's mechanics, while sensory signals return from the body.

Surface EMG measures muscle electrical activity, not peripheral nerve firing or force. Spinal plasticity, momentary reflex modulation and peripheral adaptation remain distinct. Simulated movement is a prediction to compare with independent measurements.

Organizational level: whole system; network; signal/feature

Lookup labels: spine2body; spine to body; spine-to-body

Stopping training under a stated rule

Ending training when the chosen development rule says to stop, rather than always using every allowed round.

Compared methods have equal round limits but can perform different numbers of actual updates. This is not measured biological learning time.

Lookup labels: Early stopping; epoch cap; training update

Structured task text and supported instructions

JSON stores structured text values. The task compiler converts a supported instruction into a defined program.

The compiler chooses available recorded building blocks and limited computer-controller practice. It does not program arbitrary biological skills or supply missing task evidence.

Lookup labels: JSON and task compiler; JSON; task JSON; natural-language compiler; supported primitive

Surface electrical recording over the spine

Surface electrodes placed over spinal regions record electrical potentials; the method is called ESG.

These measurements combine electrical contributions. They are not separately identified spinal-neuron spike trains.

Lookup labels: Electrospinography; ESG; surface spinal recording; surface ESG

System 3 thinking

System 3 thinking is the partnership; a large movement model is the movement-focused model we aim to build within it.

The programme's human–AI partnership framing in which an external supervisory model helps interpret, train or calibrate task-specific neural/body mappings; the movement-focused model being built is called a large movement model.

Expert footage provides movement examples, but it does not supply the depicted experts' neural or muscle recordings. An external supervisor would help interpret, train or calibrate task-specific neural/body mappings while the large movement model remains the movement-focused model being built within that partnership. Shaw and Nave use System3 for artificial cognition outside the brain in a different research account. This programme's task-specific framing establishes no new anatomical brain layer, biological neo-neocortex, third anatomical thinking system or achieved universal nervous-system model.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Lookup labels: System three thinking; System3 thinking

Source: Thinking—Fast, Slow, and Artificial

Source identifier: shaw-nave2026

DOI: 10.2139/ssrn.6097646

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6097646

Prior proposed System3 usage for artificial cognition outside the brain; not an established third anatomical brain system

Source: Thinking Fast, Slow, Artificially: AI and Your Brain

Source identifier: wharton-authors2026

https://executiveeducation.wharton.upenn.edu/thought-leadership/wharton-at-work/2026/05/thinking-fast-slow-and-artificially/

Primary institutional account of Shaw and Nave System3 research

Testing on an excluded day, animal or species

Checking a model on complete identities that were kept out of the relevant development steps.

The paper uses formal names such as leave-session-out or leave-species-out. A good score within one session does not answer these broader tests.

Lookup labels: Leave-out generalization; leave-session-out; leave-animal-out; leave-species-out; unseen-organism support

Testing possible next actions in the model

Simulating near-future actions before choosing the next command.

The copied test states must not change the live model. The native result remains rejected and too slow for real-time use; cached playback is a separate function.

Lookup labels: Short-horizon planning and lookahead; horizon controller; lookahead; copied-state rollout; model predictive planning

The comparison needed to judge improvement

A defined comparison that shows whether the proposed method adds useful information or performance.

A simple predictor, random fixed model and constant muscle command answer different questions. Producing a plausible animation is not a substitute for beating the relevant comparison.

Lookup labels: Baseline and control; simple baseline; tonic baseline; random frozen control; behavior-only predictor

Thought2Action

Begin with one intention and one action: reach for the cup. Measure the neural, muscular and mechanical steps between them. The aim is to make that intent-to-action relationship as close to one-to-one as possible within the declared task, person, state, context and tolerated error.

A task-conditioned mapping from a participant's stated or inferred intent to a specified action, evaluated through independently measured outcomes.

Intent inferred from EEG is not direct access to private thought. Movement constrains competing explanations; it does not certify an achieved universal bijection or the best possible mapping.

Organizational level: whole system; network; signal/feature

Lookup labels: thought2action; thought to action; thought-to-action

Thought2World

Follow the intention all the way to its consequence, then follow sensation and perception back to the next intention. Measuring the middle makes errors easier to locate and the next experiment easier to define.

The composed pathway Thought2Action followed by Action2World: intent is translated into an action, and that action produces an outcome that can be independently checked.

This is a proposed programme pathway, not a claim of unrestricted thought control. It is not silently equivalent to the existing Body2World lookup label, Body2Action or Brain2Motion. External touch and movement tasks alone do not solve internal-body sensing or the full Interoceptomics programme.

Organizational level: whole system; network; signal/feature

Lookup labels: thought2world; thought to world; thought-to-world

Timing relationships between spikes

How closely spikes can repeat and how spikes from different recorded features occur together.

The project checks predicted average counts. It does not establish these finer timing relationships or a complete account of biological spike noise.

Lookup labels: Refractoriness and spike synchrony; refractoriness; joint spike synchrony; biological count-noise model

Touch brain area

A brain area that processes body-related sensory information. The source study records from somatosensory cortex, including Area 2.

The formal shorthand is S1. The selected model estimates an earlier limb state; it does not recover voluntary motor commands or spinal and nerve recordings.

Lookup labels: Somatosensory cortex and Area 2; S1; Area 2; Area2_Bump; somatosensory

Training a fresh model

Training a new model for the current recording without importing an older model's learned main weights.

This is the scratch specialist. It performs better than the frozen-transfer candidate in both reported label budgets.

Lookup labels: Scratch training; scratch specialist; new-recording scratch specialist

Training with neural samples without movement labels

Using neural samples for a training objective that does not require paired movement labels.

These samples still provide information and require computation. The first 5% label condition permits full training-neural support, so it is not only 28.5 seconds of total observation.

Lookup labels: Neural-only training support; neural-only pretraining; unlabeled neural support

Trying to move and actually moving

Trying or being cued to move is different from measuring the movement that occurred.

The human labels describe attempted finger movement. They do not independently separate cue from intent or provide executed finger trajectories.

Lookup labels: Attempted versus executed movement; attempted finger identity; cued attempted movement; executed kinematics

Typical error on the original scale

RMS or RMSE squares the deviations, averages them, and takes a square root so the result has the original unit.

An error per joint differs from a combined seven-joint error. An error against a changed command goal also differs from an error against the same recorded reference.

Lookup labels: Root mean square error; RMSE; RMS; root-mean-square; tracking error

Unassigned event channel

A channel collecting events not assigned to the sorted features used by the model.

These events are excluded from the 171 sorted features in the second Indy recording. They cannot silently become another identified cell.

Lookup labels: Residual or hash cell; residual/hash; hash slot; threshold crossing

Units for distance, speed and angle

Metres, centimetres, millimetres and micrometres measure distance. Metres per second measures speed; degrees and radians measure angle. Radians per second (rad/s) measures the rate of change of angle.

One metre is 100 cm or 1000 mm; one mm is 1000 µm. Right-angle and varied-angle maze layouts differ in geometry, not in a validated difficulty multiplier. Unit conversion or a small model error does not prove anatomical or decoded-pose accuracy.

Lookup labels: Length, velocity and angle units; m; metre; cm; centimetre; mm; millimetre; µm; micrometre; um; m/s; rad; radian; degrees; °; rad/s; orthogonal; nonorthogonal; varied-angle maze; maze layout

Units for pushes, turns and electrical potential

Newtons measure force; newton metres measure turning moment; millivolts measure electrical potential. A dimensionless value has no physical unit.

The handle record mixes force and moment units. The rescaled muscle targets are dimensionless despite a stale mV label and cannot be read as force or activation.

Lookup labels: Force, moment and voltage units; N; newton; N m; N·m; newton metre; mV; millivolt; dimensionless

Units for time and sample rate

Seconds measure time; milliseconds are thousandths of a second. Hertz counts repetitions or samples per second.

A 250 Hz source sample interval is 4 ms; a 500 Hz simulation interval is 2 ms. These rates differ from playback pace, biological travel time and time spent computing.

Lookup labels: Time and frequency units; s; second; seconds; ms; millisecond; milliseconds; Hz; hertz; sample rate; 250 Hz; 500 Hz

Unwanted stimulus signal in a recording

Part of a recording caused by the stimulus or measuring setup rather than the intended response alone.

Stimulus artifacts remain in the human previews. Their presence and the evoked task limit interpretation as voluntary motor evidence.

Lookup labels: stimulus artifact; artifact

Use in a new setting without labeled adjustment

Applying a learned model to a new setting without using its labels to adjust the model.

The second M1 model uses explicit new-recording labels. Its test score is a result after calibration, not zero-shot success.

Lookup labels: Zero-shot transfer; zero-shot

Velocity estimated from earlier positions

Two recorded fingertip positions can give an estimate of how fast the fingertip moved and in which direction between the samples. The signed components describe velocity; speed is its non-negative magnitude.

A position-derived velocity estimate formed from the difference between current and earlier recorded positions divided by their elapsed time; individual components retain sign. Speed is the magnitude of the velocity vector.

In the separate calibrated motor recording, each interval uses the last available fingertip sample and the actual elapsed time to form a velocity target in m/s. This makes the calculation causal with respect to available position samples, not proof that the tracker supplied them without filtering or delay. Upstream filtering and latency remain unestablished; the derived target is not an original, directly supplied velocity trace.

Lookup labels: Backward finite difference; backward difference; finite difference; causally derived velocity; 250 Hz position support; Speed estimated from earlier positions

Velocity of each model movement

The model's stored velocity for its movement degrees of freedom, including direction.

The formal variable name is qvel. Its signed generalized velocities depend on the model's degrees of freedom; speed is a magnitude. Component units and frames depend on the declared coordinate and model. In MuJoCo, quaternion orientations can make qpos and qvel different lengths, so qvel is not an entry-by-entry derivative of qpos.

Lookup labels: qvel; generalized velocity; joint speed; Speed of each model movement

What a model estimated

A model's estimate of a defined result, such as speed, from the inputs it was given.

An estimate remains an estimate even when it matches the test data well. It does not become a new measurement of the spinal cord, nerves or muscles.

Lookup labels: Decoded or inferred prediction; decoded; inferred; reconstructed; prediction

What an instrument recorded

A value produced by a stated measuring method, such as an electrode or a movement tracker.

Recorded data can contain noise, filtering or artifacts. Calculating muscle length from tracked markers does not turn that length into a direct tissue measurement.

Lookup labels: Recorded measurement; measured; observed; recorded signal

What the computer body produced

A value produced while running a computer model, such as a modeled muscle's activity.

Computer-body movement and forces can be checked for numerical errors. They remain separate from measurements of a living body and from evidence that an organism learned.

Lookup labels: Simulated quantity; simulated; numerical model state

What was calculated from tracking

A value calculated from measurements together with a body model, such as an estimated joint angle.

The study's joint angles and muscle lengths use tracked movement and a model. They are not direct recordings of muscle commands, tissue length or peripheral nerve spikes.

Lookup labels: Model-derived quantity; tracking-derived; source-derived; biomechanical covariate

Where a value came from

The record of a value's source and the steps used to produce it.

For example, a measured hand position, an angle calculated from that position and a computer-generated muscle state have different origins. The software must preserve those differences.

Lookup labels: Provenance; lineage; source integrity

Which recorded channel is which

The labels and order used to identify neural features within one recording.

The labels do not identify the same neurons across days. The project did not accept unverified matches between cells in different recordings.

Lookup labels: Source-local unit identity; ordered channel identity; cell correspondence; unit order; neuron index

Which samples a value depends on

The input history and exact time interval needed to produce a value or target.

A single timestamp does not show all of this support. Overlapping histories and invalid samples near data splits need explicit checks.

Lookup labels: Temporal support and context; support; history window; context; next-target support; overlapping histories

Composed pathways

The proposed intermediary has a positive purpose: understand and measure the processes between intent and action so the task-specific mapping can be made as close to one-to-one as possible. The task, participant, state, context, target and tolerated error must be specified and independently tested. This is an aim for identifiability and control, not an achieved universal bijection or a demonstrated optimum.

Form → function → action and intent → action describe connected questions. The following proposed pathways separate the links that would need independent tests before their combined prediction could be trusted.

Biological pathway: Brain2Spine → Spine2Body; resulting in Brain2Body

Relate brain activity to spinal processing, and spinal/peripheral processing to the body; compose the tested links into an intent-to-body-action pathway.

Functional pathway: Thought2Action → Action2World; resulting in Thought2World

Relate a declared intention to a measured action, and that action to its measured consequence in the task environment.

Molecular analogy: DNA → RNA; RNA → protein; resulting in DNA → protein

The intermediate processes explain how an initial specification reaches its downstream product. This analogy illustrates the need for a measured middle; it does not assert identical biology or a context-free one-to-one relationship.

System 3 engineering route: Neural activity → Musculoskeletal simulation → Programmable model

The proposed engineering route to NeuroTranslation, coordinated by the System3 human–AI supervisory model.

Paired source-bound brain/spinal/peripheral/muscle and sensory/outcome observations constrain mechanics and a task-specific learner. A musculoskeletal simulation represents the biomechanical middle. Model read/write capabilities must be qualified individually.

Intent → action → outcome returns through sensory feedback to perception and subsequent intent; actual observed/inferred/simulated links remain distinguishable.

Interoceptomics combines interoception and connectomics. Touch, proprioception and external sensorimotor tasks can support this programme; they do not by themselves demonstrate internal-body sensing. Combining the proposed pathways advances a research aim rather than establishing a comprehensive solution.

Brain2Body names the wider programme. Body2World remains a contextual lookup label for Body2Action; Brain2Motion and Body2Action retain their own scopes. These related names do not make their measurement targets equivalent to those of the five composed paths.

The physical spinal pathway remains read-only in this proposal. Each claimed recording or writing capability needs its own independent qualification.

Additional programme terms and proposed names

This addendum uses glossary revision 28 and programme specification 30. The preceding glossary supplement preserves its historical revision 26.

Glossary revision 28 (SHA-256):

`c253cdfa96c3ca68877ae2f5af05322d9876952bef8bed5eb646928d3ec90041`

Programme specification 30 (SHA-256):

`deacac166f7a00021e84516c640279fa59ade74a18dd6aa3daf568503716d147`

A connection map, an individual profile and a proposed feedback route answer different questions. The terms below keep those questions apart.

Body2Body

A measured movement can constrain a digital body; a digital body can suggest a movement to test. Body2Body asks which quantities can travel between those settings and whether the resulting movement matches an independent reference.

A proposed research direction relating specified physical-body measurements to digital body models and testing mappings or feedback from those models to declared target bodies.

The user's physical-to-digital and digital-to-physical wording names a proposed direction, not a proven symmetric inverse. Observation-to-model fitting, retargeting and a separately qualified physical output have different requirements. Homologous and mechanically analogous features can support unpaired pretraining or transfer when their mechanics, scale, contacts, timing and missing features are declared. That learned relationship is not an actually measured neural pairing. Keep inferred and simulated quantities distinct from measurements. Biological virtualization still requires independent biological outcomes in a stated domain; no real body actuation, retained skill or whole-organism equivalence follows here.

Lookup labels: Body to body

Brain–Body Interface

A brain recording and a measured hand movement answer different questions about the same reach. A Brain–Body Interface would connect the stated observations and any separately qualified outputs so that the proposed relationship can be tested.

A proposed interface linking specified brain observations with body observations or control outputs in a declared task, with each direction and physical route identified.

This names an interface role, rather than the Brain2Body programme or a model spanning the body. A Digital Bridge is the existing selected-information route around an interruption or between specified recording and control channels; it is related, not an automatic synonym for every Brain–Body Interface. Recording, inference, simulated control and physical output require distinct evidence. EEG and EMG record; a feedback or stimulation output needs its own qualified interface. The label establishes neither complete coverage nor a bidirectional biological loop. No new acronym is introduced.

Lookup labels: Brain-Body Interface

Ecotype

A population adapted to a coastal habitat is an example of the ecological distinction at issue. Ecotype names the adapted population or form; the surroundings themselves are its environment.

A locally adapted population or form associated with a particular ecological setting.

Ecotype is an established population/ecological concept, not an individual's surroundings or an obligatory third stage after genotype and phenotype. State the organism and evidence for the claimed local differentiation or adaptation. The retained Turesson source is a historical plant example [49], not a complete modern evolutionary theory or evidence about the proposed nervous-system experiments.

Enactome

A reach has a path, a time and a result. An Enactome would organize observations like these so that the action and what followed can be compared across stated conditions.

A proposed organization of specified action observations and the recorded relationships between actions and their outcomes in declared tasks.

Declare which actions, body parts, tasks, units, frames and times the data cover. Keep commands, measured movement, sensory consequences and simulated outcomes separate. An observed association does not establish causation, a complete repertoire or the person's unique intent. This proposed data organization differs from the tentative Enactotype individual-profile label.

EndoHumanisation

Choose a particular human-related feature, then ask what introducing or studying that feature would change in the model organism. EndoHumanisation names that proposed line of work.

A proposed research stage studying specified human-related genetic features within a model organism and testing their relevance to a declared biological question.

The roadmap places XenoTransgenesis at step 1.1. The working interpretation concerns defined genetic features; the actual manipulation and its established scientific name remain to be specified. A selected feature does not make an animal human or establish human cognition, disease fidelity or an appropriate intervention age. Biological relevance, function and animal/participant, ethical, safety and rights requirements

need their own evidence and admission. This entry supplies no experimental procedure.

ExoHumanisation

Added tissue can survive without doing the job expected of it. ExoHumanisation names a proposed study of a specified human-related graft: what survives, what connects, and what function can actually be measured?

A proposed research stage studying specified human-related donor cells or tissue in a model organism and testing their survival, integration and function.

The roadmap associates step 2.1 with XenoTransplantation. A cross-species graft retains that established meaning; cell survival, anatomical connection, activity and useful functional integration are distinct outcomes. A graft does not establish a human nervous system, a shared mind or whole-organism human equivalence. Species, tissue identity, biological purpose, ethics, safety and rights remain explicit dependencies. This label provides no transplantation procedure or intervention-age window.

Genotype

A genetic description can record which versions of a DNA sequence an organism carries. Genotype names that genetic make-up, whether the description covers a selected part of the DNA or a wider set. The scope must be stated; the genetic description alone does not tell us every observable trait.

The genetic constitution being described for an organism, with the genetic scope of the description stated.

Genotype is an established scientific term. Its description is not a complete prediction of the organism's observed traits; phenotype also depends on environment and development. The three labels in the user's organism-level example are not three obligatory sequential stages or three fixed individual types.

Interoceptome

A heartbeat signal and a breathing signal describe different parts of the body's internal condition. An Interoceptome would organize the selected observations and their relationships, retaining how, where and when each was recorded.

A proposed organization of specified observations of internal-body sensing and the relationships being studied among them.

This is a data-organization proposal within Interoceptomics. Its coverage must name the organism, body systems, measurements, timing and task. Measured signals, calculated quantities and model estimates remain distinguishable. Touch and limb-position sensing are not relabelled as interoception, and the suffix does not establish a complete internal-body map or an individual biological type.

Mind-Meld

What information could one system give another, and would it help with the task? Mind-Meld names the proposed first Brain2Brain encounter, whose signal and effects would have to be measured.

A proposed first Brain2Brain study of a specified information link between neural systems during a declared task.

The roadmap places this at step 3.1 before later NeuroTransfer studies. The name does not describe fused minds, shared private experience, arbitrary thought reading or demonstrated skill transfer. Specify the source, recipient, recording and feedback interfaces, task, controls and independent outcome. Assistance during a link and learning retained without tutor information need separate tests, with any necessary compensatory bridge held constant. A reported natural connection is a motivating case, not evidence that this engineered link exists.

Mind-Merge

An agent could guide practice by giving feedback about a stated task. Mind-Merge names the proposed study of that interaction: what help does the person receive, and what can they do later without it?

A proposed Agent2Brain study of task-specific feedback from a software agent to a person and its measurable effects.

The roadmap places this at step 4.1. Ordinary sensory feedback and direct neural delivery remain different routes with different interfaces, evidence and admission. EEG and EMG record; neither supplies a neural writer. No merged identity, shared mind, retained skill or physical intervention follows from the name. Preserve the reverse observation path, matched practice and delayed/new-task retention tests, removing tutor information while keeping a necessary bridge constant. Physical spinal work remains read-only.

NeuroHandler

How well can a hand carry out the chosen task, and what information helps it do so? NeuroHandler is the proposed manual-dexterity experiment, with the hand's measured action providing the result to explain.

A proposed manual-dexterity experiment studying specified neural/body observations, task control and independently measured outcomes.

The earlier proposed manual tasks remain: rapid hand rubbing, the requested knuckle gesture and the interlaced-hand stretch. NeuroHandler names the manual-dexterity experiment; it does not establish a finished apparatus or successful model. Define the action, available inputs, physical or simulated setting, comparison and outcome before evaluation. Keep motion control, neural prediction, assistance and retained learning separate; any physical output requires separate qualification and admission. No dexterity gain, clinical benefit or intervention procedure is established.

NeuroJuggler

Moving each limb successfully does not settle whether the limbs work together. NeuroJuggler is the proposed coordination experiment: define the joint task, then measure the relationship between the limb actions and the result.

A proposed multi-limb coordination experiment studying how specified limb actions are coordinated in a declared task.

The earlier coordination proposal remains a hot-potato transfer between the hands, followed preferably by two-ball juggling. Those are proposed tasks; the final apparatus and qualified measurement setup remain unestablished. The label supplies no successful performance or implemented neural interface. Declare the task, limbs, inputs, timing, body model or physical setting and independent outcomes. Separate coordination from isolated-limb performance, simulated success from biological function, and current assistance from retained learning. Physical output, safety, rights and held-out tests keep their own admission requirements.

NeuroReplay

What changes when a learner receives replay-based guidance? NeuroReplay names that proposed study: compare performance while guidance is available, then test what remains when the guidance ends.

A proposed Stage 0 research direction for studying replay-guided sensorimotor feedback and its effects on current performance and later learning.

Proposed replay-guided feedback and learning studies.

This is the new roadmap label for steps 0.2–0.5, with a derived working scope; it does not rename Neural Replay, the existing stored-sample playback entry playing-back-a-recording. A stored sequence can supply a comparison, but playback alone establishes neither delivery to a nervous system nor biological learning. Fix the task, signal, feedback route, timing and equal-practice comparisons; distinguish matched, stale or unmatched replay, no-link and inactive or sham controls. Test delayed retention and new tasks after removing tutor information while keeping any necessary compensatory bridge constant. The questioned antechinus age ranges are not established developmental or intervention windows. No physical writer, procedure or present result is supplied by this label.

Phenotype

The trait is what must be observed: for example, a specified body characteristic or response in the setting being studied. Phenotype names these observable traits, rather than the genetic description alone.

The observable traits of an organism arising through genotype, environment and development.

State the actual trait, measurement method, developmental and environmental context. A phenotype is not a universal function that can be read uniquely from structure or sequence. Retain the distinction between a measured trait and a model's estimate.

Whole Body Interface

The hand is one part of the question. Whole Body Interface names the wider ambition to connect declared body systems, while showing which pathways have actually been observed and which remain missing.

A proposed extension of brain–body interfacing toward specified observations and outputs across the brain, spinal, peripheral and body systems.

Whole body states an ambition, not achieved coverage. An interface exchanges specified observations or outputs; the existing model spanning the body is a related modelling ambition, not the same object. State the recorded systems, missing pathways, units, clocks, frames and output routes. Separate datasets can support qualified learning and transfer, but they do not become simultaneous measurements of one body. Physiological input/output direction, software input/output and ordinary sensory feedback remain distinct; no spinal writer or physical actuation is authorized.

XenoTransgenesis

Which genetic feature would be studied, in which organism, and what would count as the relevant change? XenoTransgenesis is the user's working label for that proposed question within EndoHumanisation.

A working roadmap name for proposed research involving specified genetic components from another species in a model organism.

XenoTransgenesis is the preferred spelling; XenoTransgenicism remains a historical lookup label. This is not an established replacement name for every form of transgenesis, genome editing or humanisation; use the established term appropriate to any actual manipulation. Neither a genetic change nor its name establishes the intended functional effect or human equivalence. No procedure, approved organism, developmental schedule or experimental permission is defined here.

Lookup labels: XenoTransgenicism

A cross-species graft

A chimera has biological contributions from different sources. A graft is added tissue; when it crosses species, it is a xenotransplant. An organoid is an organized tissue model.

Survival, physical connection, activity and useful function are separate questions. The proposed comparisons do not establish a humanized whole brain or an implemented treatment.

Cross-species-graft component only; not a strict synonym for every chimera, graft or organoid, or achieved ExoHumanisation.

Additional lookup label: XenoTransplantation

Proposed names and explanatory comparisons

Neo-neocortex

A metaphor for an external machine contribution to human cognition; no anatomical addition is claimed.

AlphaTranscriptome

AlphaTranscriptome is a proposed name for a desired RNA/cell-state prediction capability. It is not an adopted general-purpose model. AlphaGenome already predicts some expression and splicing [36]; other RNA/cell-state models have bounded domains. These precedents leave whole-cell programmability to be independently demonstrated.

AlphaCell

A label for integrating cellular layers appears in the supplied passage; no implementation, novelty or product entitlement follows.

Connetype

A connection profile would describe the selected observations for one individual. Its anatomical coverage, resolution, method and uncertainty must remain visible. The profile does not identify that person's unique function, thoughts or a fixed biological type.

An individual's specified neural-connection profile under a declared measurement and representation.

The original three-name line ends in a question mark; all three remain tentative. No formal empirical definition or repeatability threshold is fixed.

The tentative name is Connetype. Connectype remains an unapproved editorial alternative; it is neither a lookup alias nor an adopted spelling.

Interotype

A heartbeat-related observation can vary with state and measurement conditions. A proposed individual profile must state what was observed and how repeatability would be assessed; it cannot treat touch or proprioception as interchangeable with interoception.

An individual's profile of specified internal-body sensing observations under declared conditions.

The original three-name line ends in a question mark; all three remain tentative. No formal empirical definition or repeatability threshold is fixed.

Interotype is a tentative name, rather than an established scientific type.

Enactotype

A profile of selected actions is bounded by the tasks, observations and conditions used to construct it. Repeatability, state dependence and generalization remain empirical questions; a stable innate action type or inferred unique intent is not established.

An individual's profile of specified actions and action–outcome observations in declared tasks.

The original three-name line ends in a question mark; all three remain tentative. No formal empirical definition or repeatability threshold is fixed.

Enactotype is a tentative name, rather than an established scientific type.

Form, Function, Feature

Start with a structure, ask what it does in the stated setting, then name the observable feature being compared. A hand's form, its grasping function and a measured property of a particular grasp answer related but different questions.

Feature must identify the actual observed or defined trait in its context; it is not a new universal third biological layer or a substitute for function or action. The existing Form → function → action account and intent–action distinction remain unchanged. Structure constrains function; context, history, selection and co-option prevent a guaranteed one-to-one or teleological chain.

Semiosis as a sensorimotor analogy

A drawing can resemble a hand, a pointing finger can indicate an object, and an instruction can follow a learned convention. Such signs can guide action through different relations; they do not assign perception, sensation and action to three exclusive boxes.

Icon, index and symbol describe sign–object relations, while sensation, perception and action name processes. A sign can combine relations. No one-to-one biological correspondence, neural code, private-experience identity or strict isomorphism is established. Outside-in and inside-out perspectives must state whether they concern physiology, modelling or interpretation.

Agent vs Neurology

The roadmap ends by asking what an agent could contribute to a defined neurological problem. Agent vs Neurology retains that proposed destination; the specific problem, task and comparison still have to be chosen.

An unresolved Stage 5 research question, not a selected disorder, intervention, diagnostic system or clinical trial.

No specific disease, patient population, endpoint, treatment mechanism, universal cure or successful clinical benefit is inferred. Any future study must define those matters and its ordinary feedback or separately qualified physical interface under the

original scientific, participant, rights, safety and independent gates. A prior software, metadata or model result cannot supply this evidence.

Agent vs Neurology names Stage 5 of the proposed roadmap.

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A Reference Comparison Protocol: Unfilled Record

A moving hand gives the instruments something in common to answer to. Scalp EEG would observe electrical activity at declared scalp sites; surface EMG would observe muscle electrical activity at declared arm sites. Neither tells us, by itself, where the hand went. The proposed LEFT and RIGHT tripod views would provide complementary movement observations. Each acquisition needs its own identifiable series of real references, followed by an independent comparison of the assembly. Two streams can agree because they share the same mistaken calibration.

This is an unfilled protocol and record. No observations, admitted reference series, measured errors or acceptance decisions have been entered. The apparatus and its anatomical and functional claims remain unvalidated. The older above/below maze illustration retains its separate schematic context, perpendicular-to-floor geometry and material-path questions. Its directions are not synonyms for the current LEFT/RIGHT tripod arrangement. Actual positions, viewing paths, coverage and transmission would need evidence.

Identify the body and the observation

A photograph can constrain a visible proportion; it cannot supply an EEG channel or a calibrated trajectory. Record each reference's provider, stable identifier, version or acquisition context, checksum if materialized, license, access terms and supporting claim. Identify the organism, developmental stage, participant or specimen, session, body part, task and view or time interval. Record physical dimensions and units, posture, support and contact, so that the comparison concerns the same body and movement. Keep those dimensions separate from computational detail and from the rule used to average errors. Existing datasets, atlases and photographs retain their separate study identities; none is silently admitted for this apparatus.

EEG needs sites, reference, channel order and electrical units. Surface EMG needs named muscles, sleeve and contact identities, reference and electrical units; it supplies neither incoming sensory-nerve traffic nor direct force. Each tripod view needs its selected modality and camera or ranging geometry. Optical triangulation and LiDAR ranging measure different things. Preserve raw observations, derived features and model estimates separately. These recordings do not identify hidden spinal activity, individual neurons or a unique brain state.

Make time and position comparable

A reach can follow the measured path and still arrive late. Record source timestamps, clock identities, cadence and units, synchronization events, offsets, drift, dropped samples and interpolation. Retain calibration observations, frame transformations, their uncertainty and checks against an independent reference. Report offline processing and end-to-end latency explicitly. An alignment error must not disappear inside a movement score, and no real-time capability follows from this protocol.

Compare scalp EEG, each arm's surface EMG, each tripod acquisition, and the time/frame calibration and any actual material path against their respective series

of real references. Keep overlap, occlusion, dropout and failed views alongside anatomy, proportion, support and contact. Preserve both motion observations and the joined map. A missing finger remains missing unless its replacement is labelled as an estimate. The assembled map then needs its own independent movement comparison; agreement among its inputs cannot establish the accuracy of their common reconstruction.

Test what the assembly claims

Independently measured movement can test a virtual body's path. Contact and force claims need appropriate measurements of contact and force; a lifelike image cannot establish valid mechanics. A learner's prediction needs a withheld measured target, declared input availability, baselines, negative controls, independent held-out participants, sessions, tasks or species where applicable, and leakage checks. Keep the reserved recordings closed and the original tolerances unchanged. This record admits no new fitting, biological teaching, participant acquisition, stimulation, physical intervention or model promotion. Physical spinal scope remains read-only.

For each component and the assembled apparatus, retain matched reference views or time intervals, the target observable, result, error with units, repeatability, uncertainty and failures. Declare the applicable original acceptance rule, its source, domain, comparator and tolerances before evaluation. The programme's unchanged real-reference contract and every stronger original scientific, anatomical, functional, physical, held-out, rights, visual, runtime and export criterion still govern their respective claims. No universal numeric threshold is introduced or relaxed. Biological function and clinical readiness need evidence beyond a matched motion record.

Unfilled comparison record

Use one record per reference and comparison, retaining a series for every component and for the assembled map. All values below are `null`; reference-series lists are empty and every acceptance state is `HOLD`.

Source and rights. Record ID, component or assembly, provider, source identifier, version or acquisition context, local checksum if materialized, license, access and supporting claim.

Body and task. Organism, developmental stage, participant or specimen, session, body part, task, view or time interval, physical dimensions and units, posture, support, contact, actual sensor geometry and material path.

Acquisition. Modality, measured quantity and units, sites, montage, reference, channel and contact identities, source frame and clock, timestamps, cadence, synchronization events, offsets, drift, calibration observations and transforms, calibration uncertainty, interpolation, missingness, occlusion, offline processing and end-to-end latency.

Independent comparison. Component-series and assembly-series identifiers, independent reference and its acquisition, aligned interval, raw/derived/simulated provenance, input availability, held-out split and custodian, baseline, negative controls, leakage checks and target observable.

Result and decision. Result, error and units, repeatability, uncertainty, failure cases, applicable original acceptance rule and source, domain, comparator, tolerances, independent reviewer, evidence and verdict.

No measured series or scientific validation is supplied by this empty record. Rights, physical, clinical, runtime, native, render, export and publication acceptance remain separate and open; no hosted release is claimed.